

B.Tech. Course Structure (R23UG)

B.Tech. I-Semester

S.No.	Category	Course Code	Course Name	L	T	P	Credits
1	BS&H	2321101	Linear Algebra and Calculus	3	0	0	3
2	BS&H	2323102	Chemistry	3	0	0	3
3	Engg Science	23CM105	Basic Civil and Mechanical Engineering	3	0	0	3
4	Engg Science	2305107	Introduction to Programming	3	0	0	3
5	Engg Science	2305109	IT Workshop	0	0	2	1
6	BS&H	2323111	Chemistry Lab	0	0	2	1
7	Engg Science	2303112	Engineering Workshop	0	0	3	1.5
8	Engg Science	2305113	Computer Programming Lab	0	0	3	1.5
9	BS&H	2306117	Health And Wellness, Yoga and Sports	-	-	1	0.5
Total				12	0	11	17.5

B.Tech. II-Semester

S. No.	Category	CourseCode	Course Name	L	T	P	Credits
1	BS&H	2321201	Differential Equations & Vector Calculus	3	0	0	3
2	BS&H	2324201	Communicative English	2	0	0	2
3	PC	2305202	Data Structures	3	0	0	3
4	BS&H	2322204	Engineering Physics	3	0	0	3
5	Engg Science	23EE206	Basic Electrical & Electronics Engineering	3	0	0	3
6	Engg Science	2303208	Engineering Graphics	1	0	4	3
7	PC	2305203	Data Structures Lab	0	0	3	1.5
8	BS&H	2324210	Communicative EnglishLab	0	0	2	1
9	Engg Science	23EE214	Electrical & ElectronicsEngineering Workshop	0	0	3	1.5
10	BS&H	2322215	Engineering Physics Lab	0	0	2	1
11	BS&H	2306216	NSS / NCC / Scouts & Guides / Community Service	-	-	1	0.5
Total				15	0	15	22.5

B.Tech. III-Semester

S. No.	Category	Course Code	Course Name	Hours per Week			IM	EM	Credits
				L	T	P	30	70	
1	BS&H	23HS302	Discrete Mathematics and Graph Theory	3	0	0	30	70	3
2	HSMC	23HS311	Universal Human Values – Understanding Harmony and Ethical Human Conduct	2	1	0	30	70	3
3	Professional Core	2305301	Digital Logic and Computer Organization	3	0	0	30	70	3
4	Professional Core	2305302	Advanced Data Structures and Algorithms Analysis	3	0	0	30	70	3
5	Professional Core	2305303	Object Oriented Programming through Java	3	0	0	30	70	3
6	Professional Core	2305351	Advanced Data Structures and Algorithms Analysis Lab	0	0	3	30	70	1.5
7	Professional Core	2305352	Object Oriented Programming through Java Lab	0	0	3	30	70	1.5
8	SEC	2305353	Python Programming	0	1	2	30	70	2
9	AC	23HS331	Environmental Science	2	0	0	30	--	--
Total				16	2	8	270	560	20

B.Tech. IV-Semester

S. No.	Category	Course Code	Course Name	Hours per Week			IM	EM	Credits
				L	T	P	30	70	
1	Management Elective	23HS421	Managerial Economics and Financial Analysis	2	0	0	30	70	2
2	BS&H	23HS402	Probability and Statistics	3	0	0	30	70	3
3	Professional Core	2305401	Operating Systems	3	0	0	30	70	3
4	Professional Core	2339402	Database Management Systems	3	0	0	30	70	3
5	Professional Core	2305402	Software Engineering	3	0	0	30	70	3
6	Professional Core	2305451	Operating Systems Lab	0	0	3	30	70	1.5
7	Professional Core	2339452	Database Management Systems Lab	0	0	3	30	70	1.5
8	SEC	2305452	Full Stack Development-1	0	1	2	30	70	2
9	Engg Science	2304453	Design Thinking and Innovation	1	0	2	30	70	2
Total				15	1	10	270	630	21

COMPUTER SCIENCE AND ENGINEERING**B.Tech. (R23UG) Course Structure****V-SEMESTER**

S. No.	COURSE CODE	COURSE TITLE	SC	L	T	P	IM	EM	CR
1	2305501	Artificial Intelligence	PCC	3	0	0	30	70	3
2	2305502	Computer Networks	PCC	3	0	0	30	70	3
3	2305503	Automata Theory and Compiler Design	PCC	3	0	0	30	70	3
4	2305504	Object Oriented Analysis and Design	PEC-1	3	0	0	30	70	3
	2305505	Soft Computing							
	2304503	Microprocessors and Microcontrollers							
	2305506	Augmented Reality and Virtual Reality							
5		Open Elective Course–I	OEC-1	3	0	0	30	70	3
6	2305507	Introduction to Quantum Technologies and Applications	ESC	3	0	0	30	70	3
7	2305551	Artificial Intelligence Lab	PCC	0	0	3	30	70	1.5
8	2305552	Computer Networks Lab	PCC	0	0	3	30	70	1.5
9	2305553	Tinkering Lab	ESC	0	0	2	30	70	1
10	2305554	Community Service Project	PROJ	-	-	-	-	50	2
11	23HS451	Soft Skills	SC	0	1	2	30	70	2
Total				18	01	10	300	750	26

B.Tech. (R23UG) Course Structure

VI-SEMESTER

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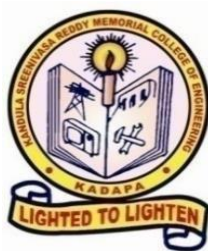
COMPUTER SCIENCE AND ENGINEERING

B.Tech. (R23UG) Course Structure**VII-SEMESTER**

S.No.	COURSE CODE	COURSE TITLE	CC	L	T	P	IM	EM	CR
1	2339609	Deep Learning	PCC	3	0	0	30	70	3
2	2332604	Cloud Computing	PCC	3	0	0	30	70	3
3	2332605	Software Architecture and Design Patterns	PEC-IV	3	0	0	30	70	3
	2305701	Cyber Security							
	2305702	Mobile Adhoc Networks							
	2332603	Computer Vision							
4	2305703	Agile methodologies	PEC-V	3	0	0	30	70	3
	2305704	Block chain Technology							
	2305705	Cyber Physical Systems							
	2305706	DevOps							
5		Open Elective Course –III	OEC-III	3	0	0	30	70	3
6		Open Elective Course –IV	OEC-IV	3	0	0	30	70	3
7	2339751	Prompt Engineering	SC	0	1	2	30	70	2
8	2305752	Industry Internship	PROJ	-	-	-	-	50	2
9	23HS701	Gender Sensitization	MC	2	0	0	30	-	-
Total				20	01	02	240	540	22

VIII-SEMESTER

S.No.	COURSE CODE	COURSE TITLE	CC	L	T	P	IM	EM	CR
1	2305851	Internship	INTERN	-	-	-	-	100	4
2	2305852	Project Work	PROJ	-	-	-	60	140	8
Total				-	-	-	60	240	12



**Kandula Sreenivasa Reddy Memorial College of Engineering
(Autonomous)**

Kadapa – 516005, A.P.

**(Approved by AICTE, Affiliated to JNTUA, Ananthapuram, Accredited by NAAC)
(An ISO 9001-2008, ISO 14001:2015 Certified Institution)**

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B. Tech (Regular)

**(Effective for the students admitted into I year from the
Academic Year 2023-24 onwards and Lateral Entry
students admitted into II year from Academic Year:
2024-25 onwards)**

**Academic Regulations (R23UG)
Course Structure and Syllabus
(I, II, III and IV Sem.)**

VISION AND MISSION OF THE INSTITUTE

VISION

To evolve as center of repute for providing quality academic programs amalgamated with creative learning and research excellence to produce graduates with leadership qualities, ethical and human values to serve the nation.

MISSION

- M1:** To provide high quality education with enriched curriculum blended with impactful teaching-learning practices.
- M2:** To promote research, entrepreneurship and innovation through industry collaborations.
- M3:** To produce highly competent professional leaders for contributing to Socio-economic development of region and the nation.

VISION AND MISSION OF THE DEPARTMENT

VISION

To evolve as a recognized centre of excellence in the area of Computer Science and Engineering and other related inter-disciplinary fields.

MISSION

M1: To produce competent and industry ready professionals through well balanced curriculum and innovative pedagogy.

M2: To provide conducive environment for research by establishing centre of excellence and industry collaborations.

M3: To instil leadership qualities, ethical values among students through various co-curricular and extracurricular activities.

PROGRAM EDUCATIONAL OBJECTIVES

A graduate of the K.S.R.M.C.E, C.S.E should have a successful career in CSE or a related field, and within three to five years, should

PEO1: To excel in their career as competent software engineer in IT and allied organizations.

PEO2: To pursue higher education and to demonstrate research temper for providing solutions to engineering problems.

PEO3: To contribute for the societal development by exhibiting leadership, through professional, social and ethical values.

PROGRAM OUTCOMES

PO1 - Engineering Knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2 - Problem Analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3 - Design/Development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4 - Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5 - Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6 - The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues, and the consequent responsibilities relevant to the professional engineering practice.

PO7 - Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8 - Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.

PO9 - Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10 - Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11 - Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12 - Life-long learning: Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OBJECTIVES

PSOs are statements that describe what the graduates of a specific engineering program should be able to do:

PSO1 - Professional Skills: The ability to understand, analyze and develop computer programs in the areas related to algorithms, system software, multimedia, web design, big data analytics, and networking for efficient design of computer-based systems of varying complexity.

PSO2 - Problem-Solving Skills: The ability to apply standard practices and strategies in software project development using open-ended programming environments to deliver a quality product for business success.

PSO3 - Successful Career and Entrepreneurship: The ability to employ modern computer languages, environments, and platforms in creating innovative career paths to be an entrepreneur, and a zest for higher studies.

Academic Regulations (R23UG) for B. Tech (Regular-Full time)

(Effective for the students admitted into I year from the Academic Year 2023-24 onwards)

1. Award of the Degree

- (a) Award of the B.Tech. Degree / B.Tech. Degree with a Minor if he/she fulfils the following:
 - (i) Pursues a course of study for not less than four academic years and not more than eight academic years. However, for the students availing Gap year facility this period shall be extended by two years at the most and these two years would in addition to the maximum period permitted for graduation (Eight years).
 - (ii) Registers for 160 credits and secures all 160 credits.
- (b) Award of B.Tech. degree with Honors if he/she fulfils the following:
 - (i) Student secures additional 15 credits fulfilling all the requisites of a B.Tech. program i.e., 160 credits.
 - (ii) Registering for Honors is optional.
 - (iii) Honors is to be completed simultaneously with B.Tech. program.

- 2. Students, who fail to fulfil all the academic requirements for the award of the degree within eight academic years from the year of their admission, shall forfeit their seat in B.Tech. course and their admission stands cancelled. This clause shall be read along with clause 1(a) (i).

3. Admissions

Admission to the B. Tech Program shall be made subject to the eligibility, qualifications and specialization prescribed by the A.P. State Government/University from time to time. Admissions shall be made either based on the merit rank obtained by the student in the common entrance examination conducted by the A.P. Government/University or any other order of merit approved by the A.P. Government/University, subject to reservations as prescribed by the Government/University from time to time.

4. Program related terms

Credit: A unit by which the course work is measured. It determines the number of hours of instruction required per week. One credit is equivalent to one hour of teaching (Lecture/Tutorial) or two hours of practical work/field work per week.

Credit Definition:

1 Hr. Lecture (L) per week	1 credit
1 Hr. Tutorial (T) per week	1 credit
1 Hr. Practical (P) per week	0.5 credit
2 Hrs. Practical (Lab) per week	1 credit

a) **Academic Year:** Two consecutive (one odd + one even) Semesters constitute one academic year.

b) **Choice Based Credit System (CBCS):** The CBCS provides a choice for students to select from the prescribed courses.

5. Semester/Credits:

i) A Semester comprises 90 working days and an academic year is divided into two Semesters.

ii) The summer term is for eight weeks during summer vacation. Internship/ apprenticeship / work-based vocational education and training can be carried out during the summer term, especially by students who wish to exit after two Semesters or four Semesters of study.

iii) Regular courses may also be completed well in advance through MOOCs satisfying prerequisites.

6. Structure of the Undergraduate Program

All courses offered for the undergraduate program (B. Tech.) are broadly classified as follows:

S. No.	Category	Breakup of Credits (Total 160)	Percentage of total credits	AICTE Recommendation (%)
1.	Humanities and Social Science including Management (HM)	13	8 %	8 – 9%
2.	Basic Sciences (BS)	20	13 %	12 - 16%
3.	Engineering Sciences (ES)	23.5	14%	10 – 18%
4.	Professional Core (PC)	54.5	34 %	30 – 36%
5.	Electives – Professional (PE) & Open (OE); Domain Specific Skill Enhancement Courses (SEC)	33	21 %	19 - 23%
6.	Internships & Project work(PR)	16	10 %	8 – 11%
7.	Mandatory Courses (MC)	Non-credit	Non-credit	-

7. Course Classification:

All subjects / courses offered for the undergraduate program in Engineering & Technology (B.Tech degree programs) are broadly classified as follows:

S. No.	Broad Course Classification	Course Category	Description
1.	Foundation Courses	Foundation courses	Includes Mathematics, Physics and Chemistry; fundamental engineering courses; humanities, social sciences and management courses
2.	Core Courses	Professional Core Courses (PC)	Includes subjects related to the parent discipline / department / branch of Engineering
		Professional Elective Courses (PE)	Includes elective subjects related to the parent discipline / department / branch of Engineering
		Open Elective Courses (OE)	Elective subjects which include inter disciplinary subjects or subjects in an area outside the parent discipline/ department/ branch of Engineering
		Domain specific skill enhancement courses (SEC)	Interdisciplinary / job-oriented / domain courses which are relevant to the industry
4.	Project & Internships	Project	B.Tech. Project or Major Project
		Internships	Summer Internships – Community based and Industry Internships; Industry oriented Full Semester Internship
5.	Audit Courses	Mandatory non-credit courses	Covering subjects of developing desired attitude among the learners

8. Program Pattern

- Total duration of the of B. Tech (Regular) Program is four academic years.
- Each academic year of study is divided into two Semesters.
- Minimum number of instruction days in each Semester is 90 days.
- There shall be mandatory student induction program for fresher's, with a three-week duration before the commencement of first Semester. Physical activity, Creative Arts, Universal Human Values, Literary, Proficiency Modules, Lectures by Eminent People, Visits to local Areas, Familiarization to Dept. / Branch & Innovations etc., are included as per the guidelines issued by AICTE.
- Health/wellness/yoga/sports and NSS /Scouts & Guides / Community service activities are made mandatory as credit courses for all the under graduate students.
- Courses like Environmental Sciences, Indian Constitution, Technical Paper Writing & IPR are offered as non-credit mandatory courses for all the undergraduate students.
- Design Thinking for Innovation & Tinkering Labs are made mandatory as credit courses for all the undergraduate students.
- Increased flexibility for students through an increase in the elective component of the curriculum, with 05 Professional Elective courses and 04 Open Elective courses.

- ix. Professional Elective Courses, include the elective courses relevant to the chosen specialization/branch. Proper choice of professional elective courses can lead to students specializing in emerging areas within the chosen field of study.
- x. A total of 04 Open Electives are offered in the curriculum. A student can complete the requirement for B.Tech. Degree with a Minor within the 160 credits by opting for the courses offered through various verticals/tracks under Open Electives.
- xi. While choosing the electives, students shall ensure that they do not opt for the courses with syllabus contents similar to courses already pursued.
- xii. A pool of interdisciplinary/job-oriented/domain skill courses which are relevant to the industry are integrated into the curriculum of all disciplines. There shall be 05 skill-oriented courses offered during III to VII Semesters. Among the five skill courses, four courses shall focus on the basic and advanced skills related to the domain/interdisciplinary courses and the other shall be a soft skills course.
- xiii. Students shall undergo mandatory summer internships, for a minimum of eight weeks duration at the end of second and third year of the program. The internship at the end of second year shall be community oriented and industry internship at the end of third year.
- xiv. There shall also be mandatory full internship in the final Semester of the program along with the project work.
- xv. Undergraduate degree with Honors is introduced for the students having good academic record.
- xvi. The college shall take measures to implement Virtual Labs (<https://www.vlab.co.in>) which provide remote access to labs in various disciplines of Engineering and will help student in learning basic and advanced concept through remote experimentation. Student shall be made to work on virtual lab experiments during the regular labs.
- xvii. The Principal shall assign a faculty advisor/mentor after admission to a group of students from same department to provide guidance in courses registration /career growth/placements/opportunities for higher studies/GATE/other competitive exams etc.
- xviii. Preferably 25% of course work for the theory courses in every Semester shall be conducted in the blended mode of learning.

9. Evaluation Process

The performance of a student in each Semester shall be evaluated subject wise with a maximum of 100 marks for theory and 100 marks for practical subject. Summer Internships shall be evaluated for 50 marks, Full Internship & Project work in final Semester shall be evaluated for 200 marks, mandatory courses with no credits shall be evaluated for 30 mid Semester marks.

A student has to secure not less than 35% of marks in the end examination and a minimum of 40% of marks in the sum total of the mid Semester and end examination marks taken together for the theory, practical, design, drawing subject or project etc. In case of a mandatory course, he/she should secure 40% of the total marks.

Theory Courses

Assessment Method	Marks
Continuous Internal Assessment	30
Semester End Examination	70
Total	100

- i) For theory subject, the distribution shall be 30 marks for Internal Evaluation and 70 marks for the End-Examination.
- ii) For practical subject, the distribution shall be 30 marks for Internal Evaluation and 70 marks for the End- Examination.
- iii) If any course contains two different branch subjects, the syllabus shall be written in two parts with 3 units each (Part-A and Part-B) and external examination question paper shall be set with two parts each for 35 marks.
- iv) If any subject is having both theory and practical components, they will be evaluated separately as theory subject and practical subject. However, they will be given same subject code with an extension of 'T' for theory subject and 'P' for practical subject.

a) Continuous Internal Evaluation

- i) For theory subjects, during the Semester, there shall be two midterm examinations. Each midterm examination shall be evaluated for 30 marks of which 5 marks for objective paper (20 minutes duration), 20 marks for subjective paper (100 minutes duration) and 5 marks for assignment.
- ii) Objective paper shall contain 05 short answer questions with 2 marks each or maximum of 20 bits for 10 marks. Subjective paper shall contain 4 either or type questions (totally eight questions from 1 to 8) of which student has to answer one from each either-or type questions. Each question carries 10 marks. The marks obtained in the subjective paper are condensed to 20 marks, the marks obtained in the objective paper is condensed to 5 marks.

Note:

- The objective paper shall be prepared in line with the quality of competitive examinations questions.
- The subjective paper shall contain 4 either or type questions of equal weightage of 10 marks. Any fraction shall be rounded off to the next higher mark.
- The objective paper shall be conducted by the institution on the day of subjective paper test.
- Assignments shall be in the form of problems, mini projects, design problems, slip tests, quizzes etc., depending on the course content. It should be continuous assessment throughout the Semester and the average marks shall be considered.

Assignments:

The assignments shall aid and hone the daily routine of students.

Assignments shall be stimulating and thought provoking to the student. While some questions may test student's understanding of the subject, there shall be questions that imply connect to real world applications. A variety of questions can posed in assignments.

- i. *Number*: A minimum of four assignments shall be given in each subject with one assignment from Unit I to IV of syllabus of that subject.
 - ii. *Quantum of work*: An assignment shall take about four to six hours of study / work per week. Assignments shall not be overloaded nor under loaded. As a guideline, each assignment may contain five questions, each question taking an hour to answer.
 - iii. *Marks*: Each assignment must be evaluated for fifty marks. Final marks are obtained by averaging all the assignment marks and reducing it to five marks.
 - iv. *Deadlines*: Students shall be given at least one-week time to complete and submit assignments. Assignments shall be submitted within deadline. Late submissions should be awarded zero marks.
 - v. *General*: It is advised to administer assignments using Google Classroom.
- iii) If the student is absent for the mid Semester examination, no re-exam shall be conducted and mid Semester marks for that examination shall be considered as zero.
- iv) First midterm examination shall be conducted for I, II and III units of syllabus with one / two either or type questions from each unit. The second midterm examination shall be conducted for III, IV and V units with one/two either or type questions from each unit. (Each midterm test shall cover 50% of the syllabus approximately).
- v) Final mid Semester marks shall be arrived at by considering the marks secured by the student in both the mid examinations with 80% weightage given to the better mid exam and 20% to the other.

For Example:

Marks obtained in first mid: 25 Marks obtained in second mid: 20
Final mid Semester Marks: $(25 \times 0.8) + (20 \times 0.2) = 24$

If the student is absent for any one midterm examination, the final mid Semester marks shall be arrived at by considering 80% weightage to the marks secured by the student in the appeared examination and zero to the other. For Example:

Marks obtained in first mid: Absent Marks obtained in second mid: 25
Final mid Semester Marks: $(25 \times 0.8) + (0 \times 0.2) = 20$

- vi) After the course work is over, the student is permitted to improve his / her internal marks of any 3 theory subjects in the entire course. However he/she will have to attend the course work.

b) Semester-End Examination Evaluation:

End examination of theory subjects shall have the following pattern:

- i) There shall be 6 questions and all questions are compulsory.
- ii) Question '1' shall contain 10 compulsory short answer questions for a total of 20 marks such that each question carries 2 marks. There shall be 2 short answer questions from each unit.
- iii) The questions from '2' to '6' shall be set by covering one unit of the syllabus for each question. In each of the questions from '2' to '6', there shall be either/or type questions of 10 marks each. Student shall answer any one of them.

End examination of theory subjects consisting of two parts of different subjects, for Example: Basic Electrical & Electronics Engineering shall have the following pattern:

- i) Question paper shall be in two parts viz., Part A and Part B with equal weightage of 35 marks each.
- ii) In each part, question 1 shall contain 5 compulsory short answer questions for a total of 5 marks such that each question carries 1 mark.
- iii) In each part, questions from 2 to 4, there shall be either/or type questions of 10 marks each. Student shall answer any one of them.
- iv) The questions from 2 to 4 shall be set by covering one unit of the syllabus for each question.

Practical Courses

- a) Practical courses shall be evaluated as tabulated below

Assessment Method	Marks
Continuous Internal Assessment	30
Semester End Examination	70
Total	100

- a) For practical courses, there shall be a continuous evaluation during the Semester for 30 sessional marks and end examination shall be for 70 marks.
- b) For Continuous internal assessment, Day-to-day work in the laboratory shall be evaluated for 30 marks by the concerned laboratory teacher based on the lab observation, record and performance of student in the laboratory.
- c) The semester-end examination shall be evaluated for 70 marks, conducted by the concerned laboratory teacher and a senior expert in the subject from the same department.
 - Procedure: 20 marks
 - Experimental work & Results: 30 marks
 - Viva-voce: 20 marks.

In a practical subject consisting of two parts (Eg. Basic Electrical & Electronics Engineering Lab), the end examination shall be conducted for 70 marks as a single laboratory in 3 hours. Mid Semester examination shall be evaluated as above for 30 marks in each part and final mid Semester marks shall be arrived by considering the average of marks obtained in two parts.

- d) For the subject having design and/or drawing, such as Engineering Drawing, the distribution of marks shall be 30 for mid Semester evaluation and 70 for end examination.

Assessment Method	Marks
Continuous Internal Assessment	30
Semester End Examination	70
Total	100

Day-to-day work shall be evaluated for 15 marks by the concerned subject teacher based on the reports/submissions prepared in the class. And there shall be two midterm examinations in a Semester for duration of 2 hours each for 15 marks with weightage of 80% to better mid marks and 20% for the other. The subjective paper shall contain 3 either or type questions of equal weightage of 5 marks. There shall be no objective paper in mid Semester examination. The sum of day- to-day evaluation and the mid Semester marks will be the final sessional marks for the subject. The end examination pattern for Engineering Graphics, shall consist of 5 questions, either/or type, of 14 marks each. There shall be no objective type questions in the end examination. However, the end examination pattern for other subjects related to design/drawing, multiple branches, etc is mentioned along with the syllabus.

- e) There shall be no external examination for mandatory courses with zero credits. However, attendance shall be considered while calculating aggregate attendance and student shall be declared to have passed the mandatory course only when he/she secures 40% or more in the internal examinations. In case, the student fails, a re-examination shall be conducted for failed candidates for 30 marks satisfying the conditions mentioned in item 1 & 2 of the regulations.
- f) Three batches complete record of laboratory records and Internal and End Semester test papers shall be preserved in the college and shall be produced to the Committees of the University as and when the same are asked for.
- g) Revaluation of End Examination Scripts**
- Revaluation of End Examination scripts is allowed for theory subjects only by paying requisite fee.
 - Procedure for Revaluation: The script will be revaluated by an examiner appointed by the principal. The maximum of revaluation and regular end examination marks will be awarded for that subject. If the deviation is more than 12 marks, the script is reevaluated second time.
 - A student can apply for revaluation in a subject only once
- h) **Curriculum Delivery:** The curriculum will also be delivered by the industry expert or adjunct faculties, who have industry experience if some MOU is signed between industry and institution.
- i) Subjects taught in collaboration with the industry, the assessment will be done by either the industry or the college as decided by the concerned Chairman of the Board

of Studies.

- j) The question papers for theory subjects shall be set by faculty members outside of the Institute. The external faculty members for question paper setting shall be appointed by the Principal.
- k) Evaluation of answer scripts shall be done by either Internal or External examiners appointed by the Principal. A minimum of 50% of subjects will be evaluated by external examiners.
- l) For laboratory subjects, end examination shall be conducted by a committee consisting of two internal examiners. One examiner shall be appointed by Head of Department of concerned Major, and the other examiner shall be appointed by the Principal.
- m) For project work viva-voce, end examination shall be conducted by a committee consisting of one internal examiner, one external examiner, and the concerned guide of the student. Internal examiner shall be appointed by Head of Department of concerned Major, and the external examiner shall be appointed by the Principal.

10. Skill oriented Courses

- i) There shall be five skill-oriented courses offered during III to VII Semesters.
- ii) Out of the five skill courses two shall be skill-oriented courses from the same domain. Of the remaining three skill courses, one shall be a soft skill course and the remaining two shall be skill-advanced courses from the same domain / Interdisciplinary / Job oriented.
- iii) The course shall carry 100 marks and shall be evaluated through continuous assessments during the Semester for 30 sessional marks and end examination shall be for 70 marks. Day-to-day work in the class / laboratory shall be evaluated for 30 marks by the concerned teacher based on the regularity/assignments/viva/mid Semester test. The end examination similar to practical examination pattern shall be conducted by the concerned teacher and an expert in the subject nominated by the principal.
- iv) The Head of the Department shall identify a faculty member as coordinator for the course. A committee consisting of the Head of the Department, coordinator and a senior Faculty member nominated by the Head of the Department shall monitor the evaluation process. The marks / grades shall be assigned to the students by the above committee based on their performance.
- v) The student shall be given an option to choose either the skill courses being offered by the college or to choose a certificate course being offered by industries / Professional bodies or any other accredited bodies. If a student chooses to take a Certificate Course offered by external agencies, the credits shall be awarded to the student upon producing the Course Completion Certificate from the agency. A committee shall be formed by the principal of the college to evaluate the grades / marks given for a course by external agencies and convert to the equivalent marks /

grades.

- vi) The recommended courses offered by external agencies, conversions and appropriate grades/marks are to be approved by the respective BoS chairman at the beginning of the Semester.
- vii) If a student prefers to take a certificate course offered by external agency, the department shall mark attendance of the student for the remaining courses in that Semester excluding the skill course in all the calculations of mandatory attendance requirements upon producing a valid certificate as approved by the University.

11. Massive Open Online Courses (MOOCs):

A Student has to pursue and complete one course compulsorily through MOOCs approved by the University. A student can pursue courses other than core through MOOCs and it is mandatory to complete one course successfully through MOOCs for awarding the degree. A student is not permitted to register and pursue core courses through MOOCs.

A student shall register for the course (Minimum of either 8 weeks or 12 weeks) offered through MOOCs with the approval of Head of the Department. The Head of the Department shall appoint one mentor to monitor the student's progression. The student needs to earn a certificate by passing the exam. The student shall be awarded the credits assigned in the curriculum only by submission of the certificate. Examination fee, if any, will be borne by the student.

Students who have qualified in the proctored examinations conducted through MOOCs platform can apply for credit transfer as specified and are exempted from appearing internal as well as external examination (for the specified equivalent credit course only).

Necessary amendments in rules and regulations regarding adoption of MOOC courses would be proposed from time to time.

12. Credit Transfer Policy

Adoption of MOOCs is mandatory, to enable Blended model of teaching-learning as also envisaged in the NEP 2020. As per University Grants Commission (Credit Framework for Online Learning Courses through SWAYAM) Regulation, 2016, a maximum of 20% of the total courses being offered in a particular program are allowed i.e., maximum of 32 credits through MOOCs platform.

- i) The College shall offer credit mobility for MOOCs and give the equivalent credit weightage to the students for the credits earned through online learning courses.
- ii) Student registration for the MOOCs shall be only through the respective department of the institution, it is mandatory for the student to share necessary information with the department.
- iii) Credit transfer policy will be applicable to the Professional & Open Elective and skill courses only.
- iv) The concerned department shall identify the courses permitted for credit transfer.
- v) The College shall notify at the beginning of Semester the list of the online learning courses eligible for credit transfer.
- vi) The institution shall designate a faculty member as a Mentor for each course to

- guide the students from registration till completion of the credit course.
- vii) The college shall ensure no overlap of MOOC exams with that of the end Semester examination schedule. In case of delay in results, the college will re-issue the marks sheet for such students.
 - viii) Student pursuing courses under MOOCs shall acquire the required credits only after successful completion of the course and submitting a certificate issued by the competent authority along with the percentage of marks and grades.
 - ix) The institution shall submit the following to the examination section of the college:
 - a) List of students who have passed MOOC courses in the current Semester along with the certificate of completion.
 - b) Undertaking form filled by the students for credit transfer.
 - x) The College academic council shall resolve any issues that may arise in the implementation of this policy from time to time and shall review its credit transfer policy in the light of periodic changes brought by UGC, SWAYAM, NPTEL and state government.

Note: Students shall be permitted to register for MOOCs offered through online platforms approved by the Head of the department from time to time.

13. Academic Bank of Credits (ABC)

The institution has implemented Academic Bank of Credits (ABC) to promote flexibility in curriculum as per NEP 2020 to

- i. provide option of mobility for learners across the universities of their choice
- ii. provide option to gain the credits through MOOCs from approved digital platforms.
- iii. facilitate award of certificate/diploma/degree in line with the accumulated credits in ABC
- iv. execute Multiple Entry and Exit system with credit count, credit transfer and credit acceptance from students' account.

14. Mandatory Internships

Summer Internships: Two summer internships either onsite or virtual each with a minimum of 08 weeks' duration, done at the end of second and third years, respectively are mandatory. It shall be completed in collaboration with local industries, Govt. Organizations, construction agencies, Power projects, software MNCs or any industries in the areas of concerned specialization of the Undergraduate program. One of the two summer internships at the end of second year (Community Service Project) shall be society oriented and shall be completed in collaboration with government organizations / NGOs & others. The other internship at the end of third year is Industry Internship and shall be completed in collaboration with Industries. The student shall register for the internship as per course structure after commencement of academic year. The guidelines issued by the APSCHE / University shall be followed for carrying out and evaluation of Community Service Project and Industry Internship.

Evaluation of the summer internships shall be through the departmental committee. A student will be required to submit a summer internship report to the concerned department and appear for an oral presentation before the departmental committee comprising of Head of the Department, supervisor of the internship and a senior

faculty member of the department. A certificate of successful completion from industry shall be included in the report. The report and the oral presentation shall carry 50% weightage each. It shall be evaluated for 50 external marks. There shall be no internal marks for Summer Internship. A student shall secure minimum 40% of marks for successful completion. In case, if a student fails, he/she shall reappear as and when Semester supplementary examinations are conducted by the institution.

Full Semester Internship and Project work: In the final Semester, the student should mandatorily register and undergo internship (onsite/virtual) and in parallel he/she should work on a project with well-defined objectives. At the end of the Semester the candidate shall submit an internship completion certificate and a project report. A student shall also be permitted to submit project report on the work carried out during the internship.

The project report shall be evaluated with an external examiner. The total marks for project work 200 marks and distribution shall be 60 marks for internal and 140 marks for external evaluation. The supervisor assesses the student for 30 marks (Report: 15 marks, Seminar: 15 marks). At the end of the Semester, all projects shall be showcased at the department for the benefit of all students and staff and the same is to be evaluated by the departmental Project Review Committee consisting of supervisor, a senior faculty and HOD for 30 marks. The external evaluation of Project Work is a Viva-Voce Examination conducted in the presence of internal examiner and external examiner appointed by the principal and is evaluated for 140 marks.

The college shall facilitate and monitor the student internship programs. Completion of internships is mandatory, if any student fails to complete internship, he/she will not be eligible for the award of degree. In such cases, the student shall repeat and complete the internship.

15. Guidelines for offering a Minor

To promote interdisciplinary knowledge among the students, the students admitted into B.Tech. in a major stream/branch are eligible to obtain degree in Minor in another stream.

- i) The Minor program requires the completion of 12 credits in Minor stream chosen.
- ii) Two courses for 06 credits related to a Minor are to be pursued compulsorily for the minor degree, but maybe waived for students who have done similar/equivalent courses. If waived for a student, then the student must take an extra elective course in its place. It is recommended that students should complete the compulsory courses (or equivalents) before registering for the electives.
- iii) Electives (minimum of 2 courses) to complete a total of 12 credits.

Note: A total of 04 Open Electives are offered in the curriculum. A student can complete the requirement for Minor by opting for the courses offered through various verticals/tracks under Open Electives.

16. Guidelines for offering Honors

The objective of introducing B.Tech. (Hons.) is to facilitate the students to choose additionally the specialized courses of their choice and build their competence in a specialized area in the UG level. The program is a best choice for academically excellent

students having good academic record and interest towards higher studies and research.

- i) Honors is introduced in the curriculum of all B. Tech. programs offering a major degree and is applicable to all B. Tech (Regular and Lateral Entry) students admitted in Engineering & Technology.
- ii) A student shall earn additional 15 credits for award of B.Tech.(Honors) degree from same branch/department/discipline registered for major degree. This is in addition to the credits essential for obtaining the Undergraduate degree in Major Discipline (i.e., 160 credits).
- iii) A student is permitted to register for Honors in IV Semester after the results of III Semester are declared and students may be allowed to take maximum two subjects per Semester pertaining to the Honors from V Semester onwards.
- iv) The concerned Principal of the college shall arrange separate class work and timetable of the courses offered under Honors program.
- v) Courses that are used to fulfil the student's primary major may not be double counted towards the Honors. Courses with content substantially equivalent to courses in the student's primary Major may not be counted towards the Honors.
- vi) The attendance for the registered courses under Honors and regular courses offered for Major degree in a Semester are to be considered separately.
- vii) A student shall maintain an attendance of 75% in all registered courses under Honors to be eligible for attending Semester end examinations.
- viii) A student registered for Honors shall pass in all subjects that constitute the requirement for the Honors degree program. No class/division (i.e., second class, first class and distinction, etc.) shall be awarded for Honors degree program.
- ix) If a student drops or is terminated from the Honors program, the additional credits so far earned cannot be converted into open or core electives; they will remain extra. However, such students will receive a separate grade sheet mentioning the additional courses completed by them.
- x) The Honors will be mentioned in the degree certificate as Bachelor of Technology (Honors) in XYZ. For example, B.Tech. (Honors) in Mechanical Engineering

Enrolment into Honors:

- i) Students of a Department/Discipline are eligible to opt for Honors program offered by the same Department/Discipline
- ii) The enrolment of student into Honors is based on the CGPA obtained in the major degree program. CGPA shall be taken up to III Semester in case of regular entry students and only III Semester in case of lateral entry students. Students having 7 CGPA without any backlog subjects will be permitted to register for Honors.
- iii) If a student is detained due to lack of attendance either in Major or in Honors, registration shall be cancelled.
- iv) Transfer of credits from Honors to regular B. Tech degree and vice-versa shall

not be permitted.

- v) Honors is to be completed simultaneously with a Major degree program.

Registration for Honors:

- i) The eligible and interested students shall apply through the HOD of his/her parent department. The whole process should be completed within one week before the start of every Semester. Selected students shall be permitted to register the courses under Honors.
- ii) The selected students shall submit their willingness to the principal through his/her parent department offering Honors. The parent department shall maintain the record of student pursuing the Honors.
- iii) The students enrolled in the Honors courses will be monitored continuously. An advisor/mentor from parent department shall be assigned to a group of students to monitor the progress.
- iv) There is no fee for registration of subjects for Honors program offered in offline at the respective institutions.

17. Attendance Requirements:

- i) A student shall be eligible to appear for the end Semester examinations if he/she acquires a minimum of 40% attendance in each subject and 75% of attendance in aggregate of all the subjects. b) Condonation of shortage of attendance in aggregate up to 10% (65% and above and below 75%) in each Semester may be granted by the College Academic Committee.
- ii) Shortage of Attendance below 65% in aggregate shall in NO CASE be condoned.
- iii) A stipulated fee shall be payable towards condonation of shortage of attendance to the institution.
- iv) Students whose shortage of attendance is not condoned in any Semester are not eligible to take their end examination of that class and their registration shall stand cancelled.
- v) A student will not be promoted to the next Semester unless he satisfies the attendance requirements of the present Semester. They may seek readmission for that Semester from the date of commencement of class work.
- vi) If any candidate fulfils the attendance requirement in the present Semester, he shall not be eligible for readmission into the same class.
- vii) If the learning is carried out in blended mode (both offline & online), then the total attendance of the student shall be calculated considering the offline and online attendance of the student.
- viii) For induction program attendance shall be maintained as per AICTE norms.

18. Promotion Rules:

The following academic requirements must be satisfied in addition to the attendance

requirements mentioned in section 16.

- i) A student shall be promoted from first year to second year if he/she fulfils the minimum attendance requirement as per university norms.
- ii) A student will be promoted from II to III year if he/she fulfils the academic requirement of securing 40% of the credits (any **decimal** fraction should be **rounded off to lower** digit) in the subjects that have been studied up to III Semester.
- iii) A student shall be promoted from III year to IV year if he/she fulfils the academic requirements of securing 40% of the credits (any **decimal** fraction should be **rounded off to lower** digit) in the subjects that have been studied up to V Semester. And in case a student is detained for want of credits for a particular academic year by
 - ii) & iii) above, the student may make up the credits through supplementary examinations and only after securing the required credits he/she shall be permitted to join in the V Semester or VII Semester respectively as the case may be.
- iv) When a student is detained due to lack of credits/shortage of attendance he/she may be re-admitted when the Semester is offered after fulfilment of academic regulations. In such case, he/she shall be in the academic regulations into which he/she is readmitted.

19. Grading:

As a measure of the student's performance, a 10-point Absolute Grading System using the following Letter Grades and corresponding percentage of marks shall be followed:

After each course is evaluated for 100 marks, the marks obtained in each course will be converted to a corresponding letter grade as given below, depending on the range in which the marks obtained by the student fall.

Structure of Grading of Academic Performance

Range in which the marks in the subject fall	Grade	Grade Points
		Assigned
90 & above	S (Superior)	10
80 - 89	A (Excellent)	9
70 - 79	B (Very Good)	8
60 - 69	C (Good)	7
50 - 59	D (Average)	6
40 - 49	E (Pass)	5
< 40	F (Fail)	0
Absent	Ab (Absent)	0

- i) A student obtaining Grade 'F' or Grade 'Ab' in a subject shall be considered failed and will be required to reappear for that subject when it is offered the next supplementary examination.
- ii) For non-credit audit courses, "Satisfactory" or "Unsatisfactory" shall be indicated instead of the letter grade and this will not be counted for the computation of SGPA/CGPA/Percentage.

Computation of Semester Grade Point Average (SGPA) and Cumulative Grade Point Average (CGPA):

The Semester Grade Point Average (SGPA) is the ratio of sum of the product of the number of credits with the grade point scored by a student in all the courses taken by a student and the sum of the number of credits of all the courses undergone by a student, i.e.,

$$SGPA = \frac{\sum(C_i \times G_i)}{\sum C_i}$$

Where, C_i is the number of credits of the i^{th} subject and G_i is the grade point scored by the student in the i^{th} course.

The Cumulative Grade Point Average (CGPA) will be computed in the same manner considering all the courses undergone by a student over all the Semesters of a program, i.e.,

$$CGPA = \frac{\sum(C_i \times S_i)}{\sum C_i}$$

Where " S_i " is the SGPA of the i^{th} Semester and C_i is the total number of credits up to that Semester.

Both SGPA and CGPA shall be rounded off to 2 decimal points and reported in the transcripts.

While computing the SGPA the subjects in which the student is awarded Zero grade points will also be included.

Grade Point: It is a numerical weight allotted to each letter grade on a 10-point scale. Letter Grade: It is an index of the performance of students in a said course. Grades are denoted by the letters S, A, B, C, D and F.

Award of Class:

After a student has satisfied the requirements prescribed for the completion of the program and is eligible for the award of B. Tech. Degree, he/she shall be placed in one of the following four classes:

Class Awarded	CGPA Secured
First Class with Distinction	≥ 7.5
First Class	$\geq 6.5 < 7.5$
Second Class	$\geq 5.5 < 6.5$
Pass Class	$\geq 5.0 < 5.5$

CGPA to Percentage Conversion Formula – $(CGPA - 0.5) \times 10$

20. With-holding of Results

If the candidate has any dues not paid to the college or if any case of indiscipline or malpractice is pending against him/her, the result of the candidate shall be withheld in such cases.

21. Multiple Entry / Exit Option

(a) Exit Policy:

The students can choose to exit the four-year program at the end of first/second/third year.

- i) **UG Certificate in (Field of study/discipline)** - Program duration: First year (first two Semesters) of the undergraduate program, 40 credits followed by an additional exit 10-credit bridge course(s) lasting two months, including at least 6- credit job-specific internship/ apprenticeship that would help the candidates acquire job- ready competencies required to enter the workforce.
- ii) **UG Diploma (in Field of study/discipline)** - Program duration: First two years (first four Semesters) of the undergraduate program, 80 credits followed by an additional exit 10-credit bridge course(s) lasting two months, including at least 6-credit job-specific internship/ apprenticeship that would help the candidates acquire job-ready competencies required to enter the workforce.
- iii) **Bachelor of Science (in Field of study/discipline) i.e., B.Sc. Engineering in (Field of study/discipline)**- Program duration: First three years (first six Semesters) of the undergraduate program, 120 credits.

(b) Entry Policy:

Modalities on multiple entry by the student into the B.Tech. program will be provided by the university in due course of time.

Note: The College Academic council shall resolve any issues that may arise in the implementation of Multiple Entry and Exit policies from time to time and shall review the policies in the light of periodic changes brought by UGC, AICTE and State government.

22. Gap Year Concept:

Gap year concept for Student Entrepreneur in Residence is introduced and outstanding students who wish to pursue entrepreneurship / become entrepreneur are allowed to take a break of one year at any time after II year to pursue full-time entrepreneurship program / to establish start-ups. This period may be extended to two years at the most and these two years would not be counted for the time for the maximum time for graduation. The Head of the department shall forward such proposals submitted by the students to the principal. An evaluation committee constituted by the College Academic Council shall evaluate the proposal submitted by the student and the committee shall decide whether to permit the student(s) to avail the Gap Year or not.

23. Transitory Regulations

Discontinued, detained, or failed candidates are eligible for readmission as and when the Semester is offered after fulfilment of academic regulations. Candidates who have been detained for want of attendance or not fulfilled academic requirements or who have failed after having undergone the course in earlier regulations or have discontinued and wish to continue the course are eligible for admission into the unfinished Semester from the date of commencement of class work with the same or equivalent subjects as and when subjects are offered, subject to Section 2 and they will follow the academic regulations into which they are readmitted.

Candidates who are permitted to avail Gap Year shall be eligible for re-joining into the succeeding year of their B. Tech from the date of commencement of class work, subject to Section 2 and they will follow the academic regulations into which they are readmitted.

24. Minimum Instruction Days for a Semester:

The minimum instruction days including exams for each Semester shall be 90 days.

25. Medium of Instruction:

The medium of instruction of the entire B. Tech undergraduate program in Engineering & Technology (including examinations and project reports) will be in English only.

26. Student Transfers:

Student transfers shall be as per the guidelines issued by the Government of Andhra Pradesh and the Universities from time to time.

27. General Instructions:

- i. The academic regulations should be read as a whole for purpose of any interpretation.
- ii. Malpractices rules-nature and punishments are appended.
- iii. Where the words “he”, “him”, “his”, occur in the regulations, they also include “she”, “her”, “hers”, respectively.
- iv. In the case of any doubt or ambiguity in the interpretation of the above rules, the decision of the Head of the Institution is final.
- v. The Institution may change or amend the academic regulations or syllabi at any time and the changes or amendments shall be made applicable to all the students on rolls with effect from the dates notified.
- vi. In the case of any doubt or ambiguity in the interpretation of the guidelines given, the decision of the Head of the institution is final.

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ACADEMIC REGULATIONS (R23UG)

FOR B.TECH. (LATERAL ENTRY SCHEME)

(Effective for the students admitted into II year through Lateral Entry Scheme from the Academic Year 2024-25 onwards)

1. Award of the Degree

- (a) Award of the B.Tech. Degree / B.Tech. Degree with a Minor if he/she fulfils the following:
 - (i) Pursues a course of study for not less than three academic years and not more than six academic years. However, for the students availing Gap year facility this period shall be extended by two years at the most and these two years would in addition to the maximum period permitted for graduation (Six years).
 - (ii) Registers for 120 credits and secures all 120 credits.
- (b) Award of B.Tech. degree with Honors if he/she fulfils the following:
 - (i) Student secures additional 15 credits fulfilling all the requisites of a B.Tech. program i.e., 120 credits.
 - (ii) Registering for Honors is optional.
 - (iii) Honors is to be completed simultaneously with B.Tech. program.

- 2. Students, who fail to fulfil the requirement for the award of the degree within six consecutive academic years from the year of admission, shall forfeit their seat.

3. Minimum Academic Requirements

The following academic requirements have to be satisfied in addition to the requirements mentioned in item no.2

- i. A student shall be deemed to have satisfied the minimum academic requirements and earned the credits allotted to each theory, practical, design, drawing subject or project if he secures not less than 35% of marks in the end examination and a minimum of 40% of marks in the sum total of the mid Semester evaluation and endexamination taken together.
- ii. A student shall be promoted from III year to IV year if he/she fulfils the academic requirements of securing 40% of the credits (any decimal fraction should be rounded off to lower digit) in the subjects that have been studied up to V Semester.

And in case if student is already detained for want of credits for particular academic year, the student may make up the credits through supplementary exams of the above exams before the commencement of IV year I Semester class work of next year.

Course Pattern

- i) The entire course of study is three academic years on Semester pattern.
- ii) A student eligible to appear for the end examination in a subject but absent at it or has failed in the end examination may appear for that subject at the next supplementary examination offered.
- iii) When a student is detained due to lack of credits/shortage of attendance the student may be re-admitted when the Semester is offered after fulfilment of academic regulations, the student shall be in the academic regulations into which he/she is readmitted.

All other regulations as applicable for B. Tech. Four-year degree course (Regular) will hold good for B. Tech. (Lateral Entry Scheme).

B.Tech. Course Structure (R23UG)

B.Tech. I-Semester

S.No.	Category	Course Code	Course Name	L	T	P	Credits
1	BS&H	2321101	Linear Algebra and Calculus	3	0	0	3
2	BS&H	2323102	Chemistry	3	0	0	3
3	Engg Science	23CM105	Basic Civil and Mechanical Engineering	3	0	0	3
4	Engg Science	2305107	Introduction to Programming	3	0	0	3
5	Engg Science	2305109	IT Workshop	0	0	2	1
6	BS&H	2323111	Chemistry Lab	0	0	2	1
7	Engg Science	2303112	Engineering Workshop	0	0	3	1.5
8	Engg Science	2305113	Computer Programming Lab	0	0	3	1.5
9	BS&H	2306117	Health And Wellness, Yoga and Sports	-	-	1	0.5
Total				12	0	11	17.5

B.Tech. II-Semester

S. No.	Category	CourseCode	Course Name	L	T	P	Credits
1	BS&H	2321201	Differential Equations & Vector Calculus	3	0	0	3
2	BS&H	2324201	Communicative English	2	0	0	2
3	PC	2305202	Data Structures	3	0	0	3
4	BS&H	2322204	Engineering Physics	3	0	0	3
5	Engg Science	23EE206	Basic Electrical & Electronics Engineering	3	0	0	3
6	Engg Science	2303208	Engineering Graphics	1	0	4	3
7	PC	2305203	Data Structures Lab	0	0	3	1.5
8	BS&H	2324210	Communicative EnglishLab	0	0	2	1
9	Engg Science	23EE214	Electrical & ElectronicsEngineering Workshop	0	0	3	1.5
10	BS&H	2322215	Engineering Physics Lab	0	0	2	1
11	BS&H	2306216	NSS / NCC / Scouts & Guides / Community Service	-	-	1	0.5
Total				15	0	15	22.5

B.Tech. III-Semester

S. No.	Category	Course Code	Course Name	Hours per Week			IM	EM	Credits
				L	T	P	30	70	
1	BS&H	23HS302	Discrete Mathematics and Graph Theory	3	0	0	30	70	3
2	HSMC	23HS311	Universal Human Values – Understanding Harmony and Ethical Human Conduct	2	1	0	30	70	3
3	Professional Core	2305301	Digital Logic and Computer Organization	3	0	0	30	70	3
4	Professional Core	2305302	Advanced Data Structures and Algorithms Analysis	3	0	0	30	70	3
5	Professional Core	2305303	Object Oriented Programming through Java	3	0	0	30	70	3
6	Professional Core	2305351	Advanced Data Structures and Algorithms Analysis Lab	0	0	3	30	70	1.5
7	Professional Core	2305352	Object Oriented Programming through Java Lab	0	0	3	30	70	1.5
8	SEC	2305353	Python Programming	0	1	2	30	70	2
9	AC	23HS331	Environmental Science	2	0	0	30	--	--
Total				16	2	8	270	560	20

B.Tech. IV-Semester

S. No.	Category	Course Code	Course Name	Hours per Week			IM	EM	Credits
				L	T	P	30	70	
1	Management Elective	23HS421	Managerial Economics and Financial Analysis	2	0	0	30	70	2
2	BS&H	23HS402	Probability and Statistics	3	0	0	30	70	3
3	Professional Core	2305401	Operating Systems	3	0	0	30	70	3
4	Professional Core	2339402	Database Management Systems	3	0	0	30	70	3
5	Professional Core	2305402	Software Engineering	3	0	0	30	70	3
6	Professional Core	2305451	Operating Systems Lab	0	0	3	30	70	1.5
7	Professional Core	2339452	Database Management Systems Lab	0	0	3	30	70	1.5
8	SEC	2305452	Full Stack Development-1	0	1	2	30	70	2
9	Engg Science	2304453	Design Thinking and Innovation	1	0	2	30	70	2
Total				15	1	10	270	630	21

I Semester

Course Title	LINEAR ALGEBRA & CALCULUS					B. Tech. I Sem. (Common to All Branches)		
Course Code	Category	Hours/Week			Credits	Maximum Marks		
2321101	BS&H	L	T	P	C	Continuous Internal Assessment	End Exams	Total
		3	0	0	3	30	70	100
Mid Exam Duration : 2Hrs					End Exam Duration : 3 Hrs.			
Course Objectives: To equip the students with standard concepts and tools at an intermediate to advanced level mathematics to develop the confidence and ability among the students to handle various real-world problems and their applications.								
Course Outcomes: On successful completion of this course, the students will be able to								
CO1	Develop and use of matrix algebra techniques that are needed by engineers for practical applications.							
CO2	Utilize mean value theorems to real life problems.							
CO3	Familiarize with functions of several variables which are useful in optimization.							
CO4	Learn important tools of calculus in higher dimensions.							
CO5	Familiarize with double and triple integrals of functions of several variables in two dimensions using Cartesian and polar coordinates and in three dimensions using cylindrical and spherical coordinates.							

UNIT I: Matrices

Rank of a matrix by echelon form, normal form, **normal form in PAQ**. Cauchy–Binet formula (without proof). Inverse of Non- singular matrices by Gauss-Jordan method, System of linear equations: **Consistency and inconsistency of system of equations**, solving system of Homogeneous and Non-Homogeneous equations by Gauss elimination method. **Iterative Methods:** Jacobi and Gauss Seidel Iteration Methods.

UNIT II: Eigenvalues, Eigenvectors and Orthogonal Transformation

Eigenvalues, Eigenvectors and their properties, Diagonalization of a matrix, Cayley-Hamilton Theorem (without proof), finding inverse and power of a matrix by Cayley-Hamilton Theorem, Quadratic forms, Nature, **Signature and Index** of the Quadratic Forms, Reduction of Quadratic form to canonical forms by Orthogonal Transformation.

UNIT III: Calculus

Basic concepts of Calculus. Mean Value Theorems: Rolle's Theorem, Lagrange's mean value theorem with their geometrical interpretation, Cauchy's mean value theorem, Taylor's and Maclaurin's theorems with remainders (without proof), Problems and applications on the above theorems.

UNIT IV: Partial differentiation and Applications (Multi variable calculus)

Functions of several variables: Continuity and Differentiability, Partial derivatives, total derivatives, chain rule, Taylor's and Maclaurin's series expansion of functions of two variables. Jacobians, Functional dependence, maxima and minima of functions of two variables, method of Lagrange multipliers.

UNIT V: Multiple Integrals (Multi variable Calculus)

Double integrals: **Evaluation of double integrals in cartesian and polar coordinates**, triple integrals, change of order of integration, change of variables to polar, cylindrical and spherical coordinates. Finding areas (by double integrals) and volumes (by double integrals and triple integrals).

Text Books:

1. Higher Engineering Mathematics, B. S. Grewal, Khanna Publishers, 2017, 44th Edition.
2. Advanced Engineering Mathematics, Erwin Kreyszig, John Wiley & Sons, 2018, 10th Edition.

Reference Books:

1. Thomas Calculus, George B. Thomas, Maurice D. Weir and Joel Hass, Pearson Publishers, 2018, 14th Edition.
2. Advanced Engineering Mathematics, R. K. Jain and S. R. K. Iyengar, Alpha Science International Ltd., 2021 5th Edition (9th reprint).
3. Advanced Modern Engineering Mathematics, Glyn James, Pearson publishers, 2018, 5th Edition.
4. Advanced Engineering Mathematics, Micheael Greenberg, Pearson publishers, 9th Edition.
5. Higher Engineering Mathematics, H. K Das, Er. Rajnish Verma, S. Chand Publications, 2014, Third Edition (Reprint 2021).
6. A Text Book of Engineering Mathematics, N.P. Bali and Manish Goyal, Lakshmi Publications, Reprint 2008.

Course Title	CHEMISTRY					B. Tech. I Sem. EEE, CSE & AIML		
Course Code	Category	Hours/Week			Credits	Maximum Marks		
2323102	BS&H	L	T	P	C	Continuous Internal Assessment	End Exams	Total
		3	0	0	3	30	70	100
Mid Exam Duration: 2Hrs					End Exam Duration: 3Hrs			
Course Objectives: - To familiarize engineering chemistry and its applications - To train the students on the principles and applications of electrochemistry and polymers - To introduce instrumental methods, molecular machines and switches.								
Course Outcomes: On successful completion of this course, the students will be able to								
CO1	Explain Schrodinger Wave equation & energy level diagrams of homo and heteronuclear diatomic molecules							
CO2	Apply the principle of Band diagrams in the application of Semiconductors, superconductors and explain the importance of supercapacitors & nanomaterials.							
CO3	Compare the materials of construction for battery and electrochemical sensors							
CO4	Explain the preparation, properties, and applications of thermoplastics & thermosetting & elastomers conducting polymers							
CO5	Explain the principles of spectrometry & chromatography and its applications							

UNIT I : Structure and Bonding Models:

Fundamentals of Quantum mechanics, Schrodinger Wave equation, significance of Ψ and Ψ^2 , particle in one dimensional box, **Differences between atomic & molecular orbitals**, molecular orbital theory –bonding in homo- and heteronucleardiatomic molecules – energy level diagrams of O_2 , N_2 and CO, etc. π -molecular orbitals of butadiene and benzene, calculation of bond order.

UNIT II : Modern Engineering materials

Semiconductors – Introduction, basic concept, **Types-Intrinsic & Extrinsic Semiconductors**, application.

Super conductors-Introduction basic concept, applications.

Supercapacitors: Introduction, Basic Concept-Classification – Applications.

Nano materials: Introduction, classification, properties and applications of Fullerenes, carbon nano tubes and Graphines nanoparticles.

UNIT III : Electrochemistry and Applications

Electrochemical cell, Nernst equation, **Reference Electrodes-Primary & Secondary Reference Electrodes**, cell potential calculations and numerical problems, potentiometry- potentiometric titrations (redox titrations), concept of conductivity, conductivity cell, conductometric titrations (acid-base titrations).

Electrochemical sensors – potentiometric sensors with examples, amperometric sensors with examples.

Primary cells – Zinc-air battery, Secondary cells –lithium-ion batteries, **Lead acid battery**- working of the batteries including cell reactions; Fuel cells, hydrogen-oxygen fuel cell– working of the cells. Polymer Electrolyte Membrane Fuel cells (PEMFC).

UNIT IV : Polymer Chemistry

Introduction to polymers, functionality of monomers, chain growth and step growth polymerization, coordination polymerization, with specific examples and mechanisms of polymer formation.

Plastics –Thermo and Thermosetting plastics, Preparation, properties and applications of – PVC, Teflon, Bakelite, Nylon-6,6, carbon fibres.

Elastomers–Buna-S, Buna-N–preparation, properties and applications.

Conducting polymers – polyacetylene, polyaniline, – mechanism of conduction and applications.

Bio-Degradable polymers - Poly Glycolic Acid (PGA), Polyl Lactic Acid (PLA).

UNIT V : Instrumental Methods and Applications

Electromagnetic spectrum. Absorption of radiation: Beer-Lambert's law. UV-Visible Spectroscopy, electronic transition, Instrumentation, IR spectroscopies, fundamental modes and selection rules, Instrumentation. Chromatography-Basic Principle, Classification-HPLC: Principle, Instrumentation and Applications., **Gas Chromatography- Principle, Instrumentation and Applications**

Text Books:

1. Jain and Jain, Engineering Chemistry, 16/e, Dhanpat Rai, 2013.
2. Peter Atkins, Julio de Paula and James Keeler, Atkins' Physical Chemistry, 10/e, Oxford University Press, 2010.
3. A Text book of Engineering chemistry by Shashi Chawla, Dhanpat Rai & Co publications

Reference Books:

1. Skoog and West, Principles of Instrumental Analysis, 6/e, Thomson, 2007.
2. J.D. Lee, Concise Inorganic Chemistry, 5th Edition, Wiley Publications, Feb.2008
3. Text book of Polymer Science, Fred W. Billmayer Jr, 3rd Edition
4. Principles of Instrumental Analysis, 6th edition, Douglas A. Skoog, Cengage Publications.
5. Advanced Inorganic Chemistry, Cotton F Albert, Wilkinson Geoffrey, Prism Publications
6. An Introduction to Electrochemistry, Glasstone, Arihant Publications

Course Title	BASIC CIVIL AND MECHANICAL ENGINEERING					B. Tech. I Sem. EEE, CSE & AIML		
Course Code	Category	Hours/Week			Credits	Maximum Marks		
23CM105	Engineering Science	L	T	P	C	Continuous Internal Assessment	End Exams	Total
		3	0	0	3	30	70	100
Mid Exam Duration: 2Hrs					End Exam Duration: 3Hrs			
Course Objectives: • Get familiarized with the scope and importance of Civil and Mechanical Engineering in different sectors and industries. • Introduce the preliminary concepts of Building Planning, Building Construction, Materials and the related tests. • Acquire preliminary knowledge of surveying and understand the importance of the quality of the drinking water. • Explain different engineering materials and manufacturing processes. • Provide an overview of different thermal and mechanical systems; introduce basics of robotics and its applications.								
Course Outcomes: On completion of the course, the student should be able to								
CO1	Understand various sub-divisions of Civil Engineering and to appreciate their role in ensuring better society. Know the concepts of surveying and to understand the measurement of distances, angles and levels through surveying.							
CO2	Realize the importance of Transportation in nation's economy and the engineering measures related to highways in terms of geometrics.							
CO3	Understand the importance of water resources and storage structures so that the social responsibilities of water conservation will be appreciated. Understand the different manufacturing processes and explain the basics of thermal engineering and its applications.							
CO4	Describe the working of different mechanical and power plants; learn basics of robotics.							
CO5	Analyze various power transmission systems applications.							

PART A: BASIC CIVIL ENGINEERING

UNIT I

Basics of Civil Engineering: Role of Civil Engineers in Society- Various Disciplines of Civil Engineering- Structural Engineering- Geo-technical Engineering- Transportation Engineering - Hydraulics and Water Resources Engineering - Environmental Engineering -Scope of each discipline - Building Construction and Planning- Construction Materials-Cement – Aggregate Bricks - Cement concrete- Steel-Tests on these materials.

Factors to be considered in Building Planning- Nature of Buildings- Typical Layouts of a Residential Building- Industrial Building- Commercial Building like a Supermarket / Hotel / Theatre.

UNIT II

Surveying: Objectives of Surveying- Horizontal Measurements- Vertical Measurements- Angular Measurements- Leveling instruments used for leveling- Introduction to Bearings- Simple problems on leveling and bearings-Contour mapping.

UNIT III

Transportation Engineering, Water Resources and Environmental Engineering: Importance of Transportation in Nation's economic development- Types of Highway Pavements- Flexible Pavements and Rigid Pavements - Simple Differences - Basic geometric design elements of a highway- Camber- Stopping Sight Distance- Super elevation- Introduction.

Water Resources and Environmental Engineering: Sources of water- Quality of water- Specifications and Tests- Introduction to Hydrology- Hydrograph –Rain water Harvesting- Rain water runoff- Water Storage Structures (Simple introduction to Dams and Reservoirs).

Text Books:

1. G. Shanmugam and M.S. Palanisamy, Basic Civil and the Mechanical Engineering, TataMcGraw Hill publications (India) Pvt. Ltd.
2. Basic Civil Engineering, S.S. Bhavikatti, New Age International Publishers.
3. Engineering Materials, Dr. S.C. Rangwala, Charotar Publishing House.
4. Highway Engineering, S.K. Khanna, C.E.G. Justo and Veeraraghavan, Nemchand andBrothers Publications.
5. Irrigation Engineering and Hydraulic Structures - Santosh Kumar Garg, Khanna Publishers, Delhi.
6. Building Construction, Dr. B. C. Punmia, Lakshmi Publications, Delhi.

Reference Books:

1. Surveying, Vol- I and Vol-II, S.K. Duggal, Tata McGraw Hill Publishers.
2. Hydrology and Water Resources Engineering, Santosh Kumar Garg, KhannaPublishers, Delhi.

PART B: BASIC MECHANICAL ENGINEERING

UNIT I:

Introduction to Mechanical Engineering: Role of Mechanical Engineering in Industries and Society- Technologies in different sectors such as Energy, Manufacturing, Automotive, Aerospace, and Marine sectors.

Engineering Materials - Metals-Ferrous and Non-ferrous, Ceramics, Composites, Smart materials.

UNIT II

Manufacturing Processes: Principles of Casting, Forming, joining processes, Machining, Introduction to CNC machines, 3D printing, and Smart manufacturing.

Thermal Engineering – working principle of Boilers, Otto cycle, Diesel cycle, Refrigeration and air-conditioning cycles, IC engines, 2-Stroke and 4-Stroke engines, SI/CI Engines, Components of Electric and Hybrid Vehicles.

UNIT III

Power plants – working principle of Steam, Diesel, Hydro, Nuclear power plants. **Mechanical Power Transmission** - Belt Drives, Chain, Rope drives, Gear Drives and their applications.

Introduction to Robotics - Joints & links, configurations, and applications of robotics.

(Note: The subject covers only the basic principles of Civil and Mechanical Engineering systems. The evaluation shall be intended to test only the fundamentals of the subject)

Text Books:

1. Internal Combustion Engines by V. Ganesan, By Tata McGraw Hill publications (India) Pvt.Ltd.
2. A Text book of Theory of Machines by S.S. Rattan, Tata McGraw Hill Publications,(India) Pvt. Ltd.
3. An introduction to Mechanical Engineering by Jonathan Wicker and Kemper Lewis, Cengage Learning India Pvt. Ltd.

Reference Books:

1. Appuu Kuttan KK, Robotics, I.K. International Publishing House Pvt. Ltd. Volume-I.
2. 3D printing & Additive Manufacturing Technology- L. Jyothish Kumar, Pulak MPandey, Springer publications.
3. Thermal Engineering by Mahesh M Rathore Tata McGraw Hill publications (India) Pvt.Ltd.
4. G. Shanmugam and M.S. Palanisamy, Basic Civil and the Mechanical Engineering, Tata McGraw Hill publications (India) Pvt. Ltd.

Course Title	INTRODUCTION TO PROGRAMMING					B. Tech. I Sem. EEE, CSE & AIML		
Course Code	Category	Hours/Week			Credits	Maximum Marks		
2305107	Engineering Science	L	T	P	C	Continuous Internal Assessment	End Exams	Total
		3	0	0	3	30	70	100
Mid Exam Duration: 2Hrs					End Exam Duration: 3Hrs			
Course Objectives: - To introduce students to the fundamentals of computer programming. - To provide hands-on experience with coding and debugging. - To foster logical thinking and problem-solving skills using programming. - To familiarize students with programming concepts such as data types, control structures, functions, and arrays. - To encourage collaborative learning and teamwork in coding projects.								
Course Outcomes: A student after completion of the course will be able to								
CO1	Understand basics of computers, the concept of algorithm and algorithmic thinking.							
CO2	Analyse a problem and develop an algorithm to solve it.							
CO3	Implement various algorithms using the C programming language.							
CO4	Understand more advanced features of C language.							
CO5	Develop problem-solving skills and the ability to debug and optimize the code.							

UNIT I : Introduction to Programming and Problem Solving

History of Computers, Basic organization of a computer: ALU, input-output units, memory, program counter, Introduction to Programming Languages, Program Development Life Cycle, Basics of a Computer Program Algorithms, flowcharts (Using Dia Tool), pseudo code. Introduction to Compilation and Execution, Primitive Data Types, Variables, and Constants, Basic Input and Output, Operations, Type Conversion, and Casting.

Problem solving techniques: Algorithmic approach, characteristics of algorithm.

Problem solving strategies: Top-down approach, Bottom-up approach, Time and space complexities of algorithms.

UNIT II : Control Structures

Simple sequential programs Conditional Statements (if, if-else, switch), Loops (for, while, dowhile) Break and Continue. Go to statement, Nested Loops.

UNIT III : Arrays and Strings

Arrays indexing, memory model, programs with array of integers, two dimensional arrays, Introduction to Strings. String handling functions, and Command line arguments.

UNIT IV : Pointers & User Defined Data types

Pointers, dereferencing and address operators, pointer and address arithmetic, array manipulation using pointers, Dynamic Memory Allocation, Storage classes – auto, register, static, extern.

User-defined data types-Structures and Unions.

UNIT V : Functions & File Handling

Introduction to Functions, Function Declaration and Definition, Function call Return Types and Arguments, Parameter Passing techniques, Recursion, modifying parameters inside functions using pointers, arrays as parameters. Scope and Lifetime of Variables, Basics of File Handling.

Note: The syllabus is designed with C Language as the fundamental language of implementation.

Text Books:

1. "The C Programming Language", Brian W. Kernighan and Dennis M. Ritchie, Prentice-Hall, 1988
2. Schaum's Outline of Programming with C, Byron S Gottfried, McGraw-Hill Education, 1996
3. Computer Science: A Structured Programming Approach Using C 3rd Edition by Behrouz A. Forouzan, Richard F. Gilberg)

Reference Books:

1. Computing fundamentals and C Programming, Balagurusamy, E., McGraw-Hill Education, 2008.
2. Programming in C, Rema Theraja, Oxford, 2016, 2nd edition.
3. C Programming, A Problem Solving Approach, Forouzan, Gilberg, Prasad, CENGAGE, 3rd edition.

Course Title	IT WORKSHOP					B. Tech. I Sem. EEE, CSE & AIML		
Course Code	Category	Hours/Week			Credits	Maximum Marks		
2305109	Engineering Science	L	T	P	C	Continuous Internal Assessment	End Exams	Total
		0	0	2	1	30	70	100
					End Exam Duration: 3Hrs			
Course Objectives: - To introduce the internal parts of a computer, peripherals, I/O ports, connecting cables. - To demonstrate configuring the system as Dual boot both Windows and other Operating Systems Viz. Linux, BOSS. - To teach basic command line interface commands on Linux. - To teach the usage of Internet for productivity and self-paced life-long learning. - To introduce Compression, Multimedia and Antivirus tools and Office Tools such as Word processors, Spread sheets and Presentation tools.								
Course Outcomes:								
CO1	Perform Hardware troubleshooting.							
CO2	Understand Hardware components and inter dependencies.							
CO3	Safeguard computer systems from viruses/worms.							
CO4	Document/ Presentation preparation.							
CO5	Perform calculations using spreadsheets.							

PC Hardware & Software Installation

Task 1: Identify the peripherals of a computer, components in a CPU and its functions. Draw the block diagram of the CPU along with the configuration of each peripheral and submit to your instructor.

Task 2: Student should disassemble and assemble the PC back to working condition. Lab instructors should verify the work and follow it up with a Viva. Also students need to go through the video which shows the process of assembling a PC. A video would be given as part of the course content.

Task 3: Student should individually install MS windows on the personal computer. Lab instructor should verify the installation and follow it up with a Viva.

Task 4: Student should install Linux on the computer. This computer should have windows installed. The system should be configured as dual boot (VMWare) with both Windows and Linux. Lab instructors should verify the installation and follow it up with a Viva.

Task 5: Student should install BOSS on the computer. The system should be configured as dual boot (VMWare) with both Windows and BOSS. Lab instructors should verify the installation and follow it up with a Viva.

Internet & World Wide Web

Task1: Orientation & Connectivity Boot Camp: Students should get connected to their Local Area Network and access the Internet. In the process they configure the TCP/IP setting. Finally, students should demonstrate, to the instructor, how to access the websites and email. If there are no internet connectivity preparations need to be made by the instructors to simulate the WWW on the LAN.

Task 2: Web Browsers, Surfing the Web: Students customize their web browsers with the LAN proxy settings, bookmarks, search toolbars and pop-up blockers. Also, plug-ins like Macromedia Flash and JRE for applets should be configured.

Task 3: Search Engines & Netiquette: Students should know what search engines are and how to use the search engines. A few topics would be given to the students for which they need to search on Google. This should be demonstrated to the instructors by the student.

Task 4: Cyber Hygiene: Students would be exposed to the various threats on the internet and would be asked to configure their computer to be safe on the internet. They need to customize their browsers to block pop ups, block active x downloads to avoid viruses and/or worms.

LaTeX and WORD

Task 1 – Word Orientation: The mentor needs to give an overview of La TeX and Microsoft (MS) office or equivalent (FOSS) tool word: Importance of La TeX and MS office or equivalent (FOSS) tool Word as word Processors, Details of the four tasks and features that would be covered in each, Using La TeX and word – Accessing, overview of toolbars, saving files, Using help and resources, rulers, format painter in word.

Task 2: Using La TeX and Word to create a project certificate. Features to be covered:- Formatting Fonts in word, Drop Cap in word, Applying Text effects, Using Character Spacing, Borders and Colors, Inserting Header and Footer, Using Date and Time option in both La TeXand Word.

Task 3: Creating project abstract Features to be covered:- Formatting Styles, Inserting table, Bullets and Numbering, Changing Text Direction, Cell alignment, Footnote, Hyperlink, Symbols, Spell Check, Track Changes.

Task 4: Creating a Newsletter: Features to be covered:- Table of Content, Newspaper columns, Images from files and clipart, Drawing toolbar and Word Art, Formatting Images, Textboxes, Paragraphs and Mail Merge in word.

EXCEL

Excel Orientation: The mentor needs to tell the importance of MS office or equivalent (FOSS) tool Excel as a Spreadsheet tool, give the details of the four tasks and features that would be covered in each. Using Excel – Accessing, overview of toolbars, saving excel files, Using help and resources.

Task 1: Creating a Scheduler - Features to be covered: Gridlines, Format Cells, Summation, auto fill, Formatting Text

Task 2: Calculating GPA -. Features to be covered:- Cell Referencing, Formulae in excel – average, std. deviation, Charts, Renaming and Inserting worksheets, Hyper linking, Count function,

LOOKUP/VLOOKUP

Task 3: Split cells, freeze panes, group and outline, Sorting, Boolean and logical operators, Conditional formatting

POWER POINT

Task 1: Students will be working on basic power point utilities and tools which help them create basic power point presentations. PPT Orientation, Slide Layouts, Inserting Text, Word Art, Formatting Text, Bullets and Numbering, Auto Shapes, Lines and Arrows in PowerPoint.

Task 2: Interactive presentations - Hyperlinks, Inserting –Images, Clip Art, Audio, Video, Objects, Tables and Charts.

Task 3: Master Layouts (slide, template, and notes), Types of views (basic, presentation, slide slotter, notes etc), and Inserting – Background, textures, Design Templates, Hidden slides.

ADOBE PHOTOSHOP

Task 1: Scanning images, editing their resolution and size, learning about bitmap and vector images, creating new images, and placing files are the basic concepts that are covered here.

Task 2: The students learn about the colour modes, colour management, colour mode conversion, colour picker functions, etc., in this module.

Task 3: An exposure to painting tools, brush tools, pencils, erasers, magic erasers, blending modes, etc.

AI TOOLS – ChatGPT

Task 1: Prompt Engineering: Experiment with different types of prompts to see how the model responds. Try asking questions, starting conversations, or even providing incomplete sentences to see how the model completes them.

- Ex: Prompt: "You are a knowledgeable AI. Please answer the following question: What is the capital of France?"

Task 2: Creative Writing: Use the model as a writing assistant. Provide the beginning of a story or a description of a scene, and let the model generate the rest of the content. This can be a fun way to brainstorm creative ideas

- Ex: Prompt: "In a world where gravity suddenly stopped working, people started floating upwards. Write a story about how society adapted to this new reality."

Task 3: Language Translation: Experiment with translation tasks by providing a sentence in one language and asking the model to translate it into another language. Compare the output to see how accurate and fluent the translations are.

- Ex: Prompt: "Translate the following English sentence to French: 'Hello, how are you doing today?'"

Reference Books:

1. Comdex Information Technology course tool kit, Vikas Gupta, WILEY Dream tech, 2003
2. The Complete Computer upgrade and repair book, Cheryl A Schmidt, WILEY Dream tech, 2013, 3rd edition
3. Introduction to Information Technology, ITL Education Solutions limited, Pearson Education, 2012, 2nd edition
4. PC Hardware - A Handbook, Kate J. Chase, PHI (Microsoft)
5. LaTeX Companion, Leslie Lamport, PHI/Pearson.
6. IT Essentials PC Hardware and Software Companion Guide, David Anfins on and Ken Quamme. – CISCO Press, Pearson Education, 3rd edition
7. IT Essentials PC Hardware and Software Labs and Study Guide, Patrick Regan– CISCO Press, Pearson Education, 3rd edition.

Course Title	CHEMISTRY LAB					B. Tech. I Sem. EEE, CSE & AIML		
Course Code	Category	Hours/Week			Credits	Maximum Marks		
2323111	BS & H	L	T	P	C	Continuous Internal Assessment	End lab Exams	Total
		0	0	2	1	30	70	100
					End Exam Duration: 3Hrs			
Course Objectives: - To verify the fundamental concepts with experiments. - The student will have exposure to various experimental skills and hand-on experience which is very essential for an Engineering student.								
Course Outcomes: At the end of the course, the students will be able to								
CO1	Determine the cell constant, redox potentials and conductance of solutions.							
CO2	Prepare advanced polymer Bakelite materials							
CO3	Measure the strength of an acid present in secondary batteries & copper in given sample							
CO4	Analyse the IR spectra of some organic compounds & verify Beer Lamberts law							
CO5	Calculate different parameters in water sample							

Note: In the following list, out of 13 experiments, any 10 experiments must be performed in a Semester

List of Experiments:

- Measurement of 10Dq by spectrophotometric method
- Conductometric titration of strong acid vs. strong base
- Conductometric titration of weak acid vs. strong base
- Determination of cell constant and conductance of solutions
- Potentiometry - determination of redox potentials and emfs
- Determination of Strength of an acid in Pb-Acid battery
- Preparation of a Bakelite
- Verify Lambert-Beer's law
- Wavelength measurement of sample through UV-Visible Spectroscopy
- Identification of simple organic compounds by IR
- Preparation of nanomaterials by precipitation method
- Estimation of Ferrous Iron by Dichrometry.
- Estimation of Hardness of Water by EDTA Method**
- Determination of Dissolved Oxygen present in Water sample by Winkler's method.**
- Estimation of Copper by EDTA Method**
- Determine the strength of chlorides present in water sample by AgNO₃ solution**

Text Books:

- Vogel's Text book of Quantitative Chemical Analysis, J. Mendham et.al., Pearson Education, Sixth Edition, 2012.

2. Laboratory manual on Engineering Chemistry, Anupama Rajput, Dhanpat Rai & Co Publications.
3. Essentials of Experimental Engineering Chemistry, Shashi Chawla, Dhanpat Rai & Co Publications.

Reference Books:

1. Practical Engineering Chemistry by K. Mukkanti, et al, B.S. Publications, Hyderabad.
2. Instrumental methods of chemical analysis, Chatwal, Anand, Himalaya Publications.
3. Essentials of Physical Chemistry, Bhal& Tuli. (S. Chand Publications).
4. Advanced Inorganic Analysis, Agarwal & Keemtilal (Pragati prakashan)

Course Title	Engineering Workshop				B. Tech. I Sem. EEE, CSE & AIML			
Course Code	Category	Hours/Week			Credits	Maximum Marks		
2303112	BS&H	L	T	P	C	Continuous Internal Assessment	End lab Exams	Total
		0	0	3	1.5	30	70	100
					End Exam Duration: 3Hrs			
Course Objectives: ➤ To familiarize students with wood working, sheet metal operations, fitting and electrical house wiring skills								
Course Outcomes:								
CO1	Identify workshop tools and their operational capabilities.							
CO2	Practice on manufacturing of components using workshop trades including fitting, carpentry,and foundry and welding							
CO3	Apply fitting operations in various applications.							
CO4	Apply basic electrical engineering knowledge for House Wiring Practice							
CO5								

- Demonstration:** Safety practices and precautions to be observed in workshop.
- Wood Working:** Familiarity with different types of woods and tools used in woodworking and make following joints.
 - Half – Lap joint
 - Mortise and Tenon joint
 - Corner Dovetail joint or Bridle joint
- Sheet Metal Working:** Familiarity with different types of tools used in sheet metalworking, Developments of following sheet metal job from GI sheets.
- Fitting:** Familiarity with different types of tools used in fitting and do the following fitting exercises.
 - V-fit
 - Dovetail fit
 - Semi-circular fit
 - Bicycle tire puncture and change of two-wheeler tyre
- Electrical Wiring:** Familiarity with different types of basic electrical circuits and make the following connections.
 - Parallel and series
 - Two-way switch
 - Godown lighting
 - Tube light
 - Three phase motor
 - Soldering of wires
- Foundry Trade:** Demonstration and practice on Moulding tools and processes, Preparation of Green Sand Moulds for given Patterns.
- Welding Shop:** Demonstration and practice on Arc Welding and Gas welding. Preparation of Lap joint and Butt joint.
- Plumbing:** Demonstration and practice of Plumbing tools, Preparation of Pipe joints with coupling for same diameter and with reducer for different diameters.

Text Books:

1. Basic Workshop Technology: Manufacturing Process, Felix W.; Independently Published, 2019. Workshop Processes, Practices and Materials; Bruce J. Black, Routledge publishers, 5th Edn. 2015.
2. A Course in Workshop Technology Vol I. & II, B.S. Raghuwanshi, Dhanpath Rai & Co., 2015 & 2017.

Reference Books:

1. Elements of Workshop Technology, Vol. I by S. K. Hajra Choudhury & Others, Media Promoters and Publishers, Mumbai. 2007, 14th edition
2. Workshop Practice by H. S. Bawa, Tata-McGraw Hill, 2004.
3. Wiring Estimating, Costing and Contracting; Soni P.M. & Upadhyay P.A.; Atul Prakashan, 2021-22.

Course Title	Computer Programming Lab					B. Tech. I Sem. EEE, CSE & AIML		
Course Code	Category	Hours/Week			Credits	Maximum Marks		
2305113	Engineering Sciences	L	T	P	C	Continuous Internal Assessment	End lab Exams	Total
		0	0	3	1.5	30	70	100
					End Exam Duration: 3Hrs			
Course Objectives: ➤ The course aims to give students hands – on experience and train them on the concepts of the C-programming language								
Course Outcomes:								
CO1	Read, understand, and trace the execution of programs written in C language.							
CO2	Select the right control structure for solving the problem.							
CO3	Develop C programs which utilize memory efficiently using programming constructs.							
CO4	Develop, Debug and Execute programs to demonstrate the applications of arrays and functions.							
CO5	Develop, Debug and Execute programs to demonstrate the applications of basic concepts of pointers and Structures in C							

UNIT I

WEEK 1

Objective: Getting familiar with the programming environment on the computer and writing the first program.

Suggested Experiments/Activities:

Tutorial 1: Problem-solving using Computers.

Lab1: Familiarization with programming environment

i) Basic Linux environment and its editors like Vi, Vim & Emacs etc., **MS-DOS**

Environment and its commands.

ii) Exposure to Turbo C, gcc

iii) Writing simple programs using printf(), scanf()

WEEK 2

Objective: Getting familiar with how to formally describe a solution to a problem in a series of finite steps both using textual notation and graphic notation.

Suggested Experiments /Activities:

Tutorial 2: Problem-solving using Algorithms and Flow charts.

Lab 1: Converting algorithms/flow charts into C Source code.

Developing the algorithms/flowcharts for the following sample programs

i) Sum and average of 3 numbers

ii) Conversion of Fahrenheit to Celsius and vice versa

iii) Simple interest calculation

iv) Finding area of circle

WEEK 3

Objective: Learn how to define variables with the desired data-type, initialize them with

appropriate values and how arithmetic operators can be used with variables and constants.

Suggested Experiments/Activities:

Tutorial 3: Variable types and type conversions:

Lab 3: Simple computational problems using arithmetic expressions.

- i) Finding the square root of a given number
- ii) Finding compound interest
- iii) Area of a triangle using heron's formulae
- iv) Distance travelled by an object
- v) Find out $e=mc^2$

UNIT II

WEEK 4

Objective: Explore the full scope of expressions, type-compatibility of variables & constants and operators used in the expression and how operator precedence works.

Suggested Experiments/Activities:

Tutorial4: Operators and the precedence and as associativity:

Lab4: Simple computational problems using the operator' precedence and associativity

i) Evaluate the following expressions.

- a. $A+B*C+(D*E) + F*G$
- b. $A/B*C-B+A*D/3$
- c. $A+++B---A$
- d. $J= (i++) + (++i)$

ii) Find the maximum of three numbers using conditional operator

iii) Take marks of 5 subjects in integers, and find the total, average in float

WEEK 5

Objective: Explore the full scope of different variants of “if construct” namely if-else, null else, if-else if*-else, switch and nested-if including in what scenario each one of them can be used and how to use them. Explore all relational and logical operators while writing conditionals for “if construct”.

Suggested Experiments/Activities:

Tutorial 5: Branching and logical expressions:

Lab 5: Problems involving if-then-else structures.

- i) Write a C program to find the max and min of four numbers using if-else.
- ii) Write a C program to generate electricity bill.
- iii) Find the roots of the quadratic equation.
- iv) Write a C program to simulate a calculator using switch case.
- v) Write a C program to find the given year is a leap year or not.
- vi) Write a C program to find out the given number is even or odd.
- vii) Write a C program to find out the given phone number is valid or not.
- viii) Write a C program to find out the given number is positive or negative.

WEEK 6

Objective: Explore the full scope of iterative constructs namely while loop, do-while loop and for loop in addition to structured jump constructs like break and continue including when each of these statements is more appropriate to use.

Suggested Experiments/Activities:

Tutorial 6: Loops, while and for loops

Lab 6: Iterative problems e.g., the sum of series

- i) Find the factorial of given number using any loop.

- ii) Find the given number is a prime or not.
- iii) Compute sine and cos series
- iv) Checking a number palindrome
- v) Construct a pyramid of numbers.
- vi) Find out sum of individual digits of a given positive integer
- vii) Find out the given number is strong number or not.

UNIT III

WEEK 7:

Objective: Explore the full scope of Arrays construct namely defining and initializing 1-D and 2-D and more generically n-D arrays and referencing individual array elements from the defined array. Using integer 1-D arrays, explore search solution linear search.

Suggested Experiments/Activities:

Tutorial 7: 1 D Arrays: searching.

Lab 7: 1D Array manipulation, linear search

- i) Find the min and max of a 1-D integer array.
- ii) Perform linear search on 1D array.
- iii) The reverse of a 1D integer array
- iv) Find 2's complement of the given binary number.
- v) Eliminate duplicate elements in an array.
- vi) Find out smallest and biggest element in an 1D Float Array.
- vii) Count frequency of each element.

WEEK 8:

Objective: Explore the difference between other arrays and character arrays that can be used as Strings by using null character and get comfortable with string by doing experiments that will reverse a string and concatenate two strings. Explore sorting solution bubble sort using integer arrays.

Suggested Experiments/Activities:

Tutorial 8: 2 D arrays, sorting and Strings.

Lab 8: Matrix problems, String operations, Bubble sort

- i) Addition of two matrices
- ii) Multiplication two matrices
- iii) Sort array elements using bubble sort
- iv) Concatenate two strings without built-in functions
- v) Reverse a string using built-in and without built-in string functions
- vi) String palindrome or not

UNIT IV

WEEK 9:

Objective: Explore pointers to manage a dynamic array of integers, including memory allocation & value initialization, resizing changing and reordering the contents of an array and memory de-allocation using malloc (), calloc (), realloc () and free () functions. Gain experience processing command-line arguments received by C

Suggested Experiments/Activities:

Tutorial 9: Pointers, structures and dynamic memory allocation

Lab 9: Pointers and structures, memory dereference.

- i) Write a C program to find the sum of a 1D array using malloc()
- ii) Write a C program to find the total, average of n students using structures
- iii) Enter n students data using calloc() and display failed students list

- iv) Read student name and marks from the command line and display the student details along with the total.
- v) Write a C program to implement realloc()
- vi) C program to calculate employee gross salary using structures

WEEK 10:

Objective: Experiment with C Structures, Unions, bit fields and self-referential structures (Singly linked lists) and nested structures

Suggested Experiments/Activities:

Tutorial 10: Bitfields, Self-Referential Structures, Linked lists

Lab10 : Bitfields, linked lists

Read and print a date using dd/mm/yyyy format using bit-fields and differentiate the same without using bit- fields

- i) Create and display a singly linked list using self-referential structure.
- ii) Demonstrate the differences between structures and unions using a C program.
- iii) Write a C program to shift/rotate using bitfields.
- iv) Write a C program to copy one structure variable to another structure of the same type.
- v) Demonstrate nested structures using a C program.

UNIT V

WEEK 11:

Objective: Explore the Functions, sub-routines, scope and extent of variables, doing some experiments by parameter passing using call by value. Basic methods of numerical integration

Suggested Experiments/Activities:

Tutorial 11: Functions, call by value, scope and extent,

Lab 11: Simple functions using call by value, solving differential equations using Eulers theorem.

- i) Write a C function to calculate NCR value.
- ii) Write a C function to find the length of a string.
- iii) Write a C function to transpose of a matrix.
- iv) Write a C function to demonstrate numerical integration of differential equations using Euler's method

WEEK 12:

Objective: Explore how recursive solutions can be programmed by writing recursive functions that can be invoked from the main by programming at-least five distinct problems that have naturally recursive solutions.

Suggested Experiments/Activities:

Tutorial 12: Recursion, the structure of recursive calls

Lab 12: Recursive functions

- i) Write a recursive function to generate Fibonacci series.
- ii) Write a recursive function to find the lcm of two numbers.
- iii) Write a recursive function to find the factorial of a number.
- iv) Write a C Program to implement Ackermann function using recursion.
- v) Write a recursive function to find the sum of series.
- vi) Write a program in C to calculate the sum of numbers from 1 to n using recursion

WEEK 13:

Objective: Explore the basic difference between normal and pointer variables, Arithmetic operations using pointers and passing variables to functions using pointers

Suggested Experiments/Activities:

Tutorial 13: Call by reference, dangling pointers

Lab 13: Simple functions using Call by reference, Dangling pointers.

- i) Write a C program to swap two numbers using call by reference.
- ii) Demonstrate Dangling pointer problem using a C program.
- iii) Write a C program to copy one string into another using pointer.
- iv) Write a C program to find no of lowercase, uppercase, digits and other characters using pointers.

WEEK14:

Objective: To understand data files and file handling with various file I/O functions. Explore the differences between text and binary files.

Suggested Experiments/Activities:

Tutorial 14: File handling

Lab 14: File operations

- i) Write a C program to write and read text into a file.
- ii) Write a C program to write and read text into a binary file using fread() and fwrite()
- iii) Copy the contents of one file to another file.
- iv) Write a C program to merge two files into the third file using command-line arguments.
- v) Find no. of lines, words and characters in a file
- vi) Write a C program to print last n characters of a given file.

Text Books:

1. Ajay Mittal, Programming in C: A practical approach, Pearson.
2. Byron Gottfried, Schaum's Outline of Programming with C, McGraw Hill.

Reference Books:

1. Brian W. Kernighan and Dennis M. Ritchie, The C Programming Language, Prentice Hall of India
2. C Programming, A Problem-Solving Approach, Forouzan, Gilberg, Prasad, CENGAGE

Course Title	HEALTH AND WELLNESS, YOGA AND SPORTS					B. Tech. I Sem. EEE, CSE & AIML		
Course Code	Category	Hours/Week			Credits	Maximum Marks		
2306117	BS&H	L	T	P	C	Continuous Internal Assessment	End Exams	Total
		0	0	1	0.5			
					6 X 15 = 90 + 10 Viva voce			
Course Objectives: ➤ The main objective of introducing this course is to make the students maintain their mental and physical wellness by balancing emotions in their life. It mainly enhances the essential traits required for the development of the personality.								
Course Outcomes: After completion of the course the student will be able to								
CO1	Understand the importance of yoga and sports for Physical fitness and sound health.							
CO2	Demonstrate an understanding of health-related fitness components.							
CO3	Compare and contrast various activities that help enhance their health.							
CO4	Assess current personal fitness levels.							
CO5	Develop Positive Personality.							

UNIT I

Concept of health and fitness, Nutrition and Balanced diet, basic concept of immunity Relationship between diet and fitness, Globalization and its impact on health, Body Mass Index(BMI) of all age groups.

Activities:

- Organizing health awareness programmes in community
- Preparation of health profile
- Preparation of chart for balance diet for all age groups

UNIT II

Concept of yoga, need for and importance of yoga, origin and history of yoga in Indian context, classification of yoga, Physiological effects of Asanas- Pranayama and meditation, stress management and yoga, Mental health and yoga practice.

Activities:

Yoga practices – Asana, Kriya, Mudra, Bandha, Dhyana, Surya Namaskar

UNIT III

Concept of Sports and fitness, importance, fitness components, history of sports, Ancient and Modern Olympics, Asian games and Commonwealth games.

Activities:

- Participation in one major game and one individual sport viz., Athletics, Volleyball, Basketball, Handball, Football, Badminton, Kabaddi, Kho-kho, Table tennis, Cricket etc. Practicing general and specific warm up, aerobics
- Practicing cardiorespiratory fitness, treadmill, run test, 9 min walk, skipping and running.

Reference Books:

1. Gordon Edlin, Eric Golanty. Health and Wellness, 14th Edn. Jones & Bartlett Learning, 2022
2. T.K.V.Desikachar. The Heart of Yoga: Developing a Personal Practice
3. Archie J.Bahm. Yoga Sutras of Patanjali, Jain Publishing Company, 1993
4. Wiseman, John Lofty, SAS Survival Handbook: The Ultimate Guide to Surviving Anywhere Third Edition, William Morrow Paperbacks, 2014
5. The Sports Rules Book/ Human Kinetics with Thomas Hanlon. -- 3rd ed. Human Kinetics, Inc. 2014

General Guidelines:

1. Institutes must assign slots in the Timetable for the activities of Health/Sports/Yoga.
2. Institutes must provide field/facility and offer the minimum of five choices of as many as Games/Sports.
3. Institutes are required to provide sports instructor / yoga teacher to mentor the students.

Evaluation Guidelines:

- Evaluated for a total of 100 marks.
- A student can select 6 activities of his/her choice with a minimum of 01 activity per unit. Each activity shall be evaluated by the concerned teacher for 15 marks, totalling to 90 marks.
- A student shall be evaluated by the concerned teacher for 10 marks by conducting viva-voce on the subject.

II Semester

Course Title	Differential Equations and Vector Calculus					B. Tech. II Sem. (Common to All Branches)		
Course Code	Category	Hours/Week			Credits	Maximum Marks		
2321201	BS&H	L	T	P	C	Continuous Internal Assessment	End Exams	Total
		3	0	0	3	30	70	100
Mid Exam Duration: 2 Hrs					End Exam Duration: 3 Hours			
Course Objectives: - To enlighten the learners in the concept of differential equations and Multi Variable Calculus. - To furnish the learners with basic concepts and techniques at plus two level to lead them into advanced level by handling various real-world applications								
Course Outcomes: On successful completion of this course, the students will be able to								
CO1	Solve the first order differential equations related to various engineering fields.							
CO2	Solve the second and higher differential equations related to various engineering fields.							
CO3	Identify solution methods for partial differential equations that model physical processes.							
CO4	Interpret the physical meaning of different operators such as gradient, curl and divergence.							
CO5	Estimate the work done against a field, circulation and flux using vector calculus.							

UNIT I: Differential equations of first order and first degree

Linear differential equations – Bernoulli's equations- Exact equations and equations reducible to exact form. Applications: Newton's Law of cooling – Law of natural growth and decay- Electrical circuits.

UNIT II: Linear differential equations of higher order (Constant Coefficients)

Definitions, homogenous and non-homogenous, complementary function, general solution, particular integral of the type e^{ax} , $\sin ax$, $\cos ax$, polynomials in x , $e^{ax} V(x)$, $x^n V(x)$, Wronskian, Method of variation of parameters. Simultaneous linear equations, Applications to **L-R Circuit** and L-C-R Circuit problems and Simple Harmonic motion.

UNIT III: Partial Differential Equations

Introduction and formation of Partial Differential Equations by elimination of arbitrary constants and arbitrary functions, solutions of first order linear equations using Lagrange's method. Homogeneous Linear Partial differential equations with constant coefficients.

UNIT IV: Vector differentiation

Scalar and vector point functions, vector operator Del, Del applies to scalar point functions-Gradient, Directional derivative, **Solenoidal and Irrotational vectors**, del applied to vector point functions-Divergence and Curl, vector identities.

UNIT V: Vector integration

Line integral-circulation-work done - **Scalar Potential**, surface integral-flux, Green's theorem in

the plane (without proof), Stoke's theorem (without proof), volume integral, Divergence theorem (without proof) and related problems.

Text Books:

1. Higher Engineering Mathematics, B. S. Grewal, Khanna Publishers, 2017, 44th Edition
2. Advanced Engineering Mathematics, Erwin Kreyszig, John Wiley & Sons, 2018, 10th Edition.

Reference Books:

1. Thomas Calculus, George B. Thomas, Maurice D. Weir and Joel Hass, Pearson Publishers, 2018, 14th Edition.
2. Advanced Engineering Mathematics, Dennis G. Zill and Warren S. Wright, Jones and Bartlett, 2018.
3. Advanced Modern Engineering Mathematics, Glyn James, Pearson publishers, 2018, 5th Edition.
4. Advanced Engineering Mathematics, R. K. Jain and S. R. K. Iyengar, Alpha Science International Ltd., 2021 5th Edition (9th reprint).
5. Higher Engineering Mathematics, B. V. Ramana, McGraw Hill Education, 2017
6. A Text Book of Engineering Mathematics, N.P. Bali and Manish Goyal, Lakshmi Publications, Reprint 2008.

Course Title	COMMUNICATIVE ENGLISH					B. Tech. II Sem. EEE, CSE & AIML		
Course Code	Category	Hours/Week			Credits	Maximum Marks		
2324201	BS&H	L	T	P	C	Continuous Internal Assessment	End Exams	Total
		2	0	0	2	30	70	100
Mid Exam Duration: 2Hrs					End Exam Duration: 3Hrs			
COURSE OBJECTIVES								
1	Facilitate effective listening, Reading, Speaking and Writing skills among the students							
2	Enhances the same in their comprehending abilities, oral presentations, reporting useful information and providing knowledge of grammatical structures and vocabulary							
3	Provide knowledge of grammatical structures and vocabulary and encourage their appropriate use in speech and writing							
4	Helps the students to make them effective in speaking and writing skills and to make them industry ready.							
5	Impart effective strategies for good writing and demonstrate the same in summarizing, writing well organized essays, record and report useful information							
COURSE OUTCOMES								
CO1	Understand the context, topic, and pieces of specific information from social or transactional dialogues.							
CO2	Apply grammatical structures to formulate sentences and correct word forms.							
CO3	Analyze discourse markers to speak clearly on a specific topic in informal discussions							
CO4	Evaluate reading / listening texts and to write summaries based on global comprehension of these texts.							
CO5	Create a coherent paragraph, essay, and resume.							

UNIT I

Lesson : **HUMAN VALUES:** Gift of Magi (Short Story)

Listening : Identifying the topic, the context and specific pieces of information by listening to short audio texts and answering a series of questions.

Speaking : Asking and answering general questions on familiar topics such as home, family, work, studies and interests; introducing oneself and others.

Reading : Skimming to get the main idea of a text; scanning to look for specific pieces of information.

Writing : Mechanics of Writing-Capitalization, Spellings, Punctuation-Parts of Sentences.

Grammar : Parts of Speech, Basic Sentence Structures-forming questions

Vocabulary: Synonyms, Antonyms, Affixes (Prefixes/Suffixes), Root words, **One word Substitutes.**

UNIT II

Lesson : **NATURE:** The Brook by Alfred Tennyson (Poem)

Listening : Answering a series of questions about main ideas and supporting ideas after listening to audio texts.

Speaking : Discussion in pairs/small groups on specific topics followed by short structure talks.

Reading : Identifying sequence of ideas; recognizing verbal techniques that help to link the ideas in a paragraph together.

Writing : Structure of a paragraph - Paragraph writing (specific topics) Grammar: Cohesive devices - linkers, use of articles and zero article; prepositions.

Vocabulary: Homonyms, Homophones, Homographs, **Idioms and Phrases.**

UNIT III

Lesson : **BIOGRAPHY:** Elon Musk

Listening : Listening for global comprehension and summarizing what is listened to.

Speaking : Discussing specific topics in pairs or small groups and reporting what is discussed

Reading : Reading a text in detail by making basic inferences -recognizing and interpreting specific context clues; strategies to use text clues for comprehension.

Writing : Summarizing, Note-making, paraphrasing

Grammar : Verbs - tenses; subject-verb agreement; Compound words, Collocations, **Question Tags**

Vocabulary: Compound words, Collocations

UNIT IV

Lesson : **INSPIRATION:** The Toys of Peace by Saki

Listening : Making predictions while listening to conversations/ transactional dialogues without video; listening with video.

Speaking : Role plays for practice of conversational English in academic contexts (formal and informal) - asking for and giving information/directions.

Reading : Studying the use of graphic elements in texts to convey information, reveal trends / patterns / relationships, communicate processes or display complicated data.

Writing : Letter Writing: Official Letters, Resumes and **Technical Report Writing**

Grammar : Reporting verbs, Direct & Indirect speech, Active & Passive Voice

Vocabulary: Words often confused, Jargons

UNIT V

Lesson : **MOTIVATION:** The Power of Intrapersonal Communication (An Essay)

Listening : Identifying key terms, understanding concepts and answering a series of relevant questions that test comprehension.

Speaking : Formal oral presentations on topics from academic contexts

Reading : Reading comprehension.

Writing : Writing structured essays on specific topics.

Grammar : Editing short texts –identifying and correcting common errors in grammar and usage (articles, prepositions, tenses, subject verb agreement) **Graphic Presentation**

Vocabulary: Technical Jargons

Text Books:

1. Pathfinder: Communicative English for Undergraduate Students, 1st Edition, Orient Black Swan, 2023 (Units 1,2 & 3)
2. Empowering with Language by Cengage Publications, 2023 (Units 4 & 5)

Reference Books:

1. Raman, Meenakshi and Sangeeta Sharma, 2011. Technical Communication: Principles and Practice. Second Edition. New Delhi: Oxford University Press.
2. Dubey, Sham Ji & Co. English for Engineers, Vikas Publishers, 2020

3. Bailey, Stephen. Academic writing: A Handbook for International Students. Routledge, 2014.
4. Murphy, Raymond. English Grammar in Use, Fourth Edition, Cambridge University Press, 2019.
5. Lewis, Norman. Word Power Made Easy- The Complete Handbook for Building a Superior Vocabulary. Anchor, 2014.

Web Resources:

GRAMMAR:

1. www.bbc.co.uk/learningenglish
2. <https://dictionary.cambridge.org/grammar/british-grammar/>
3. www.eslpod.com/index.html
4. <https://www.learngrammar.net/>
5. <https://english4today.com/english-grammar-online-with-quizzes/>
6. <https://www.talkenglish.com/grammar/grammar.aspx>

VOCABULARY

1. <https://www.youtube.com/c/DailyVideoVocabulary/videos>
2. https://www.youtube.com/channel/UC4cmBAit8i_NJZE8qK8sfpA

Course Title	Data Structures					B. Tech. CSE & AIML (II Sem.)		
Course Code	Category	Hours/Week			Credits	Maximum Marks		
2305202	Professional Core	L	T	P	C	Continuous Internal Assessment	End Exams	Total
		3	0	0	3	30	70	100
Mid Exam Duration: 2Hrs					End Exam Duration: 3Hrs			
Course Objectives: ➤ To provide the knowledge of basic data structures and their implementations. ➤ To understand importance of data structures in context of writing efficient programs. ➤ To develop skills to apply appropriate data structures in problem solving.								
Course Outcomes: At the end of the course, Student will be able to								
CO1	Explain the role of linear data structures in organizing and accessing data efficiently in algorithms.							
CO2	Design, implement, and apply linked lists for dynamic data storage, demonstrating understanding of memory allocation.							
CO3	Develop programs using stacks to handle recursive algorithms, manage program states, and solve related problems.							
CO4	Apply queue-based algorithms for efficient task scheduling and breadth-first traversal in graphs and distinguish between dequeues and priority queues, and apply them appropriately to solve data management challenges.							
CO5	Devise novel solutions to small scale programming challenges involving data structures such as stacks, queues, Trees.							
CO6	Recognize scenarios where hashing is advantageous, and design hash-based solutions for specific problems.							

UNIT I

Introduction to Linear Data Structures: Definition and importance of linear data structures, Abstract data types (ADTs) and their implementation, Overview of time and space complexity analysis for linear data structures. Searching Techniques: Linear & Binary Search, Sorting Techniques: Bubble sort, Selection sort, Insertion Sort

UNIT II

Linked Lists: Singly linked lists: representation and operations, doubly linked lists and circular linked lists, comparing arrays and linked lists, Applications of linked lists.

UNIT III

Stacks: Introduction to stacks: properties and operations, implementing stacks using arrays and linked lists, Applications of stacks in expression evaluation, backtracking, reversing list etc.

UNIT IV

Queues: Introduction to queues: properties and operations, implementing queues using arrays and linked lists, Applications of queues in breadth-first search, scheduling, etc.

Dequeues: Introduction to dequeues (double-ended queues), Operations on dequeues and their applications.

UNIT V

Trees: Introduction to Trees, Binary Search Tree – Insertion, Deletion & Traversal

Hashing: Brief introduction to hashing and hash functions, Collision resolution techniques: chaining and open addressing, Hash tables: basic implementation and operations, Applications of hashing in unique identifier generation, caching, etc.

Text Books:

1. Data Structures and algorithm analysis in C, Mark Allen Weiss, Pearson, 2nd Edition.
2. Fundamentals of data structures in C, Ellis Horowitz, Sartaj Sahni, Susan Anderson-Freed, Silicon Press, 2008

Reference Books:

1. Algorithms and Data Structures: The Basic Toolbox by Kurt Mehlhorn and Peter Sanders
2. C Data Structures and Algorithms by Alfred V. Aho, Jeffrey D. Ullman, and John E. Hopcroft
3. Problem Solving with Algorithms and Data Structures" by Brad Miller and David Ranum
4. Introduction to Algorithms by Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein
5. Algorithms in C, Parts 1-5 (Bundle): Fundamentals, Data Structures, Sorting, Searching, and Graph Algorithms" by Robert Sedgewick

Course Title	ENGINEERING PHYSICS					B. Tech. II Sem. EEE, CSE & AIML		
Course Code	Category	Hours/Week			Credits	Maximum Marks		
2322204	BS&H	L	T	P	C	Continuous Internal Assessment	End Exams	Total
		3	0	0	3	30	70	100
Mid Exam Duration: 2Hrs					End Exam Duration: 3Hrs			
Course Objectives: • To bridge the gap between the Physics in school at 10+2 level and UG level engineering courses by identifying the importance of the optical phenomenon like interference, diffraction etc, • Enlightening the periodic arrangement of atoms in crystalline solids and concepts of quantum mechanics. • Introduce novel concepts of dielectric and magnetic materials, physics of semiconductors.								
Course Outcomes:								
CO1	Analyze the intensity variation of light due to interference, diffraction and polarization							
CO2	Familiarize with the basics of crystals and their structures							
CO3	Summarize various types of polarization of dielectrics and classify the magnetic materials.							
CO4	Explain fundamentals of quantum mechanics and apply to one dimensional motion of particles.							
CO5	Identify the type of semiconductor using Hall Effect.							

UNIT I : Wave Optics

Interference: Introduction- Principle of superposition –Interference of light –**Conditions for sustained Interference**-Interference in thin films (Reflection Geometry) & applications - Colors in thin films- Newton’s Rings- Determination of wavelength and refractive index. **Diffraction:** Introduction-Fresnel and Fraunhofer diffractions- Fraunhofer diffraction due to single slit, double slit & N-slits (Qualitative) – Diffraction Grating - Dispersive power and resolving power of Grating (Qualitative)- **Engineering applications of diffraction.**

Polarization: Introduction-Types of polarization- Polarization by reflection, refraction and Double refraction - Nicol’s Prism -Half wave and Quarter wave plates- **Engineering applications of polarization.**

UNIT II : Crystallography and X-ray diffraction

Crystallography: Introduction - Space lattice, Basis, Unit Cell and lattice parameters – Bravais Lattices – crystal systems (3D) – coordination number - packing fraction of SC, BCC & FCC - Miller indices – separation between successive (hkl) planes.

X-ray diffraction: Introduction - Bragg’s law - X-ray Diffractometer – crystal structure determination by Laue’s and powder methods- **applications.**

UNIT III : Dielectric and Magnetic Materials

Dielectric Materials: Introduction - Dielectric polarization - Dielectric polarizability, Susceptibility, Dielectric constant and Displacement Vector – Relation between the electric vectors-Types of polarizations-Electronic (Quantitative), Ionic (Quantitative) and Orientation polarizations (Qualitative) - Lorentz internal field - Clausius-Mossotti equation - complex dielectric constant – Frequency dependence of polarization – dielectric loss- **applications.**

Magnetic Materials: Introduction - Magnetic dipole moment - Magnetization-Magnetic susceptibility and permeability – Atomic origin of magnetism - Classification of magnetic materials: Dia, Para, Ferro, Anti-Ferro & Ferri magnetic materials - Domain concept for Ferro magnetism & Domain walls (Qualitative)-Hysteresis-soft and hard magnetic materials- **applications.**

UNIT IV : Quantum Mechanics and Free electron theory

Quantum Mechanics: Introduction – Dual nature of matter–Heisenberg’s Uncertainty Principle – Significance and properties of wave function – Schrodinger’s time independent and dependent wave equations– Particle in a one-dimensional infinite potential well- **applications.**

Free Electron Theory: Introduction - Classical free electron theory (Qualitative with discussion of merits and demerits) – Quantum free electron theory– electrical conductivity based on quantum free electron theory - Fermi-Dirac distribution - Density of states - Fermi energy.

UNIT V : Semiconductors

Semiconductors: Introduction - Formation of energy bands – classification of crystalline solids - Intrinsic semiconductors: Density of charge carriers – Electrical conductivity– Fermi level– Extrinsic semiconductors: density of charge carriers – dependence of Fermi energy on carrier concentration and temperature -Drift and diffusion currents–Einstein’s equation–**Direct and indirect band gap of semiconductors**-Hall effect and its applications-**Formation of p-n junction.**

Text Books:

1. A Text book of Engineering Physics - M.N. Avadhanulu, P.G.Kshirsagar & TVS Arun Murthy, S. Chand Publications, 11th Edition 2019.
2. Engineering Physics-D.K. Bhattacharya and Poonam Tandon, Oxford press (2015).

Reference Books:

1. Engineering Physics - B.K. Pandey and S.Chaturvedi, Cengage Learning
2. Engineering Physics - Shatendra Sharma, Jyotsna Sharma, Pearson Education, 2018.
3. Engineering Physics - Sanjay D. Jain, D. Sahasrabudhe and Girish, University Press.
4. Engineering Physics - M.R. Srinivasan, New Age international publishers (2009).

Course Title	Basic Electrical & Electronics Engineering					B. Tech. II Sem. EEE, CSE & AIML		
Course Code	Category	Hours/Week			Credits	Maximum Marks		
23EE206	Engineering Science	L	T	P	C	Continuous Internal Assessment	End Exams	Total
		3	0	0	3	30	70	100
Mid Exam Duration: 2Hrs					End Exam Duration: 3Hrs			
Course Objectives: ➤ To expose to the field of electrical & electronics engineering, laws and principles of electrical/ electronic engineering and to acquire fundamental knowledge in the relevant field.								
Course Outcomes: O\After the completion of this course students will be able to								
CO1	Remember the fundamental laws, operating principles of motors, generators, MC and MI instruments.							
CO2	Understand the problem solving concepts associated to AC and DC circuits, construction and operation of AC and DC machines, measuring instruments; different power generation mechanisms, Electricity billing concept and important safety measures related to electrical operations.							
CO3	Apply mathematical tools and fundamental concepts to derive various equations related to machines, circuits and measuring instruments; electricity bill calculations and layout representation of electrical power systems.							
CO4	Analyze different electrical circuits, performance of machines and measuring instruments.							
CO5	Evaluate different circuit configurations, Machine performance and Power systems operation.							

PART A: BASIC ELECTRICAL ENGINEERING

UNIT I : DC & AC Circuits

DC Circuits: Electrical circuit elements (R, L and C), Ohm's Law and its limitations, KCL & KVL, series, parallel, series-parallel circuits, Super Position theorem, Simple numerical problems.

AC Circuits: A.C. Fundamentals: Equation of AC Voltage and current, waveform, time period, frequency, amplitude, phase, phase difference, average value, RMS value, form factor, peakfactor, Voltage and current relationship with phasor diagrams in R, L, and C circuits, Concept of Impedance, Active power, reactive power and apparent power, Concept of power factor (Simple Numerical problems).

UNIT II : Machines and Measuring Instruments

Machines: Construction, principle and operation of (i) DC Motor, (ii) DC Generator, (iii) Single Phase Transformer, (iv) Three Phase Induction Motor and (v) Alternator, Applications of electrical machines.

Measuring Instruments: Construction and working principle of Permanent Magnet Moving Coil (PMMC), Moving Iron (MI) Instruments and Wheat Stone bridge.

UNIT III : Energy Resources, Electricity Bill & Safety Measures

Energy Resources: Conventional and non-conventional energy resources; Layout and operation of various Power Generation systems: Hydel, Nuclear, Solar & Wind power generation.

Electricity Bill: Power rating of household appliances including air conditioners, PCs, Laptops, Printers, etc. Definition of “unit” used for consumption of electrical energy, two-part electricity tariff, calculation of electricity bill for domestic consumers.

Equipment Safety Measures: Working principle of Fuse and Miniature circuit breaker (MCB), merits and demerits. Personal safety measures: Electric Shock, Earthing and its types, Safety Precautions to avoid shock.

Text Books:

1. Basic Electrical Engineering, D. C. Kulshreshtha, Tata McGraw Hill, 2019, First Edition
2. Power System Engineering, P.V. Gupta, M.L. Soni, U.S. Bhatnagar and A. Chakrabarti, Dhanpat Rai & Co, 2013
3. Fundamentals of Electrical Engineering, Rajendra Prasad, PHI publishers, 2014, Third Edition

Reference Books:

1. Basic Electrical Engineering, D. P. Kothari and I. J. Nagrath, Mc Graw Hill, 2019, Fourth Edition
2. Principles of Power Systems, V.K. Mehtha, S.Chand Technical Publishers, 2020
3. Basic Electrical Engineering, T. K. Nagsarkar and M. S. Sukhija, Oxford University Press, 2017
4. Basic Electrical and Electronics Engineering, S. K. Bhattacharya, Person Publications, 2018, Second Edition.

Web Resources:

1. <https://nptel.ac.in/courses/108105053>
2. <https://nptel.ac.in/courses/108108076>

PART B: BASIC ELECTRONICS ENGINEERING

Course Objectives: ➤ To understand the principles of digital electronics, basics of semiconductor devices like diodes & transistors, characteristics and its applications.	
Course Outcomes: On successful completion of this course, the students will be able to	
CO1	Understand the working mechanism of diodes, transistors, logic gates, different combinational, sequential circuits and their role in the digital systems.
CO2	Apply diodes, transistors in the electronic circuits and number systems, logic gates, Boolean algebra in logic circuits.
CO3	Analyze the circuits with diodes, transistors and logic circuits with logic gates.

UNIT I : SEMI CONDUCTOR DEVICES

Introduction - Evolution of electronics – Vacuum tubes to nano electronics Characteristics of PN Junction Diode — Zener Effect — Zener Diode and its Characteristics. Bipolar Junction

Transistor — CB, CE, CC Configurations and Characteristics — Elementary Treatment of Small Signal CE Amplifier.

UNIT II : BASIC ELECTRONIC CIRCUITS AND INSTRUMENTATION

Rectifiers and power supplies: Block diagram description of a dc power supply, working of a full wave bridge rectifier, capacitor filter (no analysis), working of simple Zener voltage regulator. Amplifiers: Block diagram of Public Address system, Circuit diagram and working of common emitter (RC coupled) amplifier with its frequency response. Electronic Instrumentation: Block diagram of an electronic instrumentation system.

UNIT III : DIGITAL ELECTRONICS

Overview of Number Systems, Logic gates including Universal Gates, BCD codes, Excess-3 code, Gray code, Hamming code. Boolean Algebra, Basic Theorems and properties of Boolean Algebra, Truth Tables and Functionality of Logic Gates—NOT, OR, AND, NOR, NAND, XOR and XNOR. Simple combinational circuits—Half and Full Adder, Introduction to sequential circuits, Flip flops, Registers and counters (Elementary Treatment only)

Text Books:

1. R.L. Boylestad & Louis Nashlesky, Electronic Devices & Circuit Theory, Pearson Education, 2021.
2. R. P. Jain, Modern Digital Electronics, 4th Edition, Tata McGraw Hill, 2009

Reference Books:

1. R.S. Sedha, A Text book of Electronic Devices and Circuits, S. Chand & Co, 2010.
2. Santiram Kal, Basic Electronics-Devices, Circuits and IT Fundamentals, Prentice Hall, India, 2002.
3. R.T. Paynter, Introductory Electronic

Course Title	Engineering Graphics					B. Tech. II Sem. EEE, CSE & AIML		
Course Code	Category	Hours/Week			Credits	Maximum Marks		
2303208	Engineering Science	L	T	P	C	Continuous Internal Assessment	End Exams	Total
		1	0	4	3	30	70	100
Mid Exam Duration: 2Hrs					End Exam Duration: 3Hrs			
Course Objectives: The students completing the course are expected to: ○ Understand the basic principles and conventions of engineering drawing use engineering instruments and draw engineering curves. ○ Use orthographic projections and make the students draw the projections of lines and planes inclined to both the planes. ○ Draw the projections of the solids in different positions with respect to the reference planes. ○ Understand the importance of sectioning and concept of development of surfaces. ○ Represent and convert isometric views to orthographic views and vice versa.								
Course Outcomes: On completion of the course, the student should be able to								
CO1	Understand the principles of engineering drawing, including engineering curves, scales, orthographic and isometric projections.							
CO2	Draw and interpret orthographic projections of points, lines, planes and solids in front, top and side views.							
CO3	Apply concepts of sectional views to represent details of solids in simple positions.							
CO4	Gain a clear understanding of the principles behind development of surfaces and to understand how to unfold basic geometric shapes into flat patterns.							
CO5	Develop the ability to draw isometric views and orthographic views and should be able to convert isometric views to orthographic views and vice versa.							

UNIT I

Introduction: Lines, Lettering and Dimensioning, Geometrical Constructions and Constructing regular polygons by general methods.

Curves: construction of ellipse, parabola and hyperbola by general, Cycloids, Involute, Normal and tangent to Curves.

Scales: Plain scales, diagonal scales and vernier scales.

UNIT II

Orthographic Projections: Reference plane, importance of reference lines or Plane, Projections of a point situated in any one of the four quadrants.

Projections of Straight Lines: Projections of straight lines parallel to both reference planes, perpendicular to one reference plane and parallel to other reference plane, inclined to one reference plane and parallel to the other reference plane. Projections of Straight Line Inclined to both the reference planes

Projections of Planes: regular planes Perpendicular to both reference planes, parallel to one

reference plane and inclined to the other reference plane; plane inclined to both the reference planes.

UNIT III

Projections of Solids: Types of solids: Polyhedra and Solids of revolution. Projections of solids in simple positions: Axis perpendicular to horizontal plane, Axis perpendicular to vertical plane and Axis parallel to both the reference planes, Projection of Solids with axis inclined to one reference plane and parallel to other and axes inclined to both the reference planes.

UNIT IV

Sections of Solids: Perpendicular and inclined section planes, Sectional views and True shape of section, Sections of solids in simple position only.

Development of Surfaces: Methods of Development: Parallel line development and radial line development. Development of a cube, prism, cylinder, pyramid and cone.

UNIT V

Conversion of Views: Conversion of isometric views to orthographic views; Conversion of orthographic views to isometric views.

Computer graphics: Creating 2D&3D drawings of objects including PCB and Transformations using Auto CAD (*Not for end Examination*).

Text Books:

1. N. D. Bhatt, Engineering Drawing, Charotar Publishing House, 2016.
2. K. Vengopal Engineering Drawing & Graphics.2018
3. Harwinder Singh Engineering Drawing & Computer Graphics.2016

Reference Books:

1. Engineering Drawing, K.L. Narayana and P. Kannaiah, Tata McGraw Hill, 2013.
2. Engineering Drawing, M.B.Shah and B.C. Rana, Pearson Education Inc,2009.
3. Engineering Drawing with an Introduction to AutoCAD, Dhananjay Jolhe,Tata McGraw Hill, 2017.

Course Title	Data Structures Lab					B. Tech. CSE & AIML (II Sem.)		
Course Code	Category	Hours/Week			Credits	Maximum Marks		
2305203	PC	L	T	P	C	Continuous Internal Assessment	End Exams	Total
		0	0	3	1.5	30	70	100
					End Exam Duration: 3Hrs			
Course Objectives: ➤ The course aims to strengthen the ability of the students to identify and apply the suitable data structure for the given real-world problem. It enables them to gain knowledge in practical applications of data structures.								
Course Outcomes: At the end of the course, Student will be able to								
CO1	Explain the role of linear data structures in organizing and accessing data efficiently in algorithms.							
CO2	Design, implement, and apply linked lists for dynamic data storage, demonstrating understanding of memory allocation.							
CO3	Develop programs using stacks to handle recursive algorithms, manage program states, and solve related problems.							
CO4	Apply queue-based algorithms for efficient task scheduling and breadth-first traversal in graphs and distinguish between dequeues and priority queues and apply them appropriately to solve data management challenges.							
CO5	Recognize scenarios where hashing is advantageous, and design hash-based solutions for specific problems.							

List of Experiments:

Exercise 1: Array Manipulation

- Write a program to reverse an array.
- C Programs to implement the Searching Techniques – Linear & Binary Search
- C Programs to implement Sorting Techniques – Bubble, Selection and Insertion Sort

Exercise 2: Linked List Implementation

- Implement a singly linked list and perform insertion and deletion operations.
- Develop a program to reverse a linked list iteratively and recursively.
- Solve problems involving linked list traversal and manipulation.

Exercise 3: Linked List Applications

- Create a program to detect and remove duplicates from a linked list.
- Implement a linked list to represent polynomials and perform addition.
- Implement a double-ended queue (deque) with essential operations.

Exercise 4: Double Linked List Implementation

- Implement a doubly linked list and perform various operations to understand its properties and applications.
- Implement a circular linked list and perform insertion, deletion, and traversal.

Exercise 5: Stack Operations

- Implement a stack using arrays and linked lists.
- Write a program to evaluate a postfix expression using a stack.
- Implement a program to check for balanced parentheses using a stack.

Exercise 6: Queue Operations

- i) Implement a queue using arrays and linked lists.
- ii) Develop a program to simulate a simple printer queue system.
- iii) Solve problems involving circular queues.

Exercise 7: Stack and Queue Applications

- i) Use a stack to evaluate an infix expression and convert it to postfix.
- ii) Create a program to determine whether a given string is a palindrome or not.
- iii) Implement a stack or queue to perform comparison and check for symmetry.

Exercise 8: Binary Search Tree

- i) Implementing a BST using Linked List.
- ii) Traversing of BST.

Exercise 9: Hashing

- i) Implement a hash table with collision resolution techniques.
- ii) Write a program to implement a simple cache using hashing.

Text Books:

1. Data Structures and algorithm analysis in C, Mark Allen Weiss, Pearson, 2nd Edition.
2. Fundamentals of data structures in C, Ellis Horowitz, Sartaj Sahni, Susan Anderson-Freed, Silicon Press, 2008

Reference Books:

1. Algorithms and Data Structures: The Basic Toolbox by Kurt Mehlhorn and Peter Sanders
2. C Data Structures and Algorithms by Alfred V. Aho, Jeffrey D. Ullman, and John E. Hopcroft
3. Problem Solving with Algorithms and Data Structures" by Brad Miller and David Ranum
4. Introduction to Algorithms by Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein
5. Algorithms in C, Parts 1-5 (Bundle): Fundamentals, Data Structures, Sorting, Searching, and Graph Algorithms by Robert Sedgewick.

Course Title	COMMUNICATIVE ENGLISH LAB					B. Tech. II Sem. EEE, CSE & AIML		
Course Code	Category	Hours/Week			Credits	Maximum Marks		
2324210	BS&H	L	T	P	C	Continuous Internal Assessment	End Exams	Total
		0	0	2	1	30	70	100
					End Exam Duration: 3Hrs			
Course Objectives - Students will be exposed to a variety of self-instructional, learner friendly modes of language learning. - The students will get trained in basic communication skills and also make them ready to face job interviews. - Students will learn better pronunciation through stress, intonation and rhythm. - Students will be initiated into greater use of the computer in resume preparation, report writing, format making etc.								
Course Outcomes								
CO1	Understand the different aspects of the English language proficiency with emphasis on LSRW skills.							
CO2	Apply communication skills through various language learning activities.							
CO3	Analyze the English speech sounds, stress, rhythm, intonation and syllable division for better listening and speaking comprehension.							
CO4	Evaluate and exhibit professionalism in participating in debates and group discussions.							
CO5	Create effective Course Objectives:							

List of Topics:

1. Vowels & Consonants
2. Neutralization/Accent Rules
3. Communication Skills & JAM
4. Role Play or Conversational Practice
5. E-mail Writing
6. Resume Writing, Cover letter, SOP
7. Group Discussions-methods & practice
8. Debates - Methods & Practice
9. PPT Presentations/ Poster Presentation
10. Interviews Skills
11. Listening Skills
12. Describing Objects

Suggested Software:

- Walden Infotech
- Young India Films

Reference Books:

1. Raman Meenakshi, Sangeeta-Sharma. *Technical Communication*. Oxford Press.2018.
2. Taylor Grant: *English Conversation Practice*, Tata McGraw-Hill Education India, 2016
3. Hewing's, Martin. *Cambridge Academic English* (B2). CUP, 2012.
4. J. Sethi & P.V. Dhamija. *A Course in Phonetics and Spoken English*, (2nd Ed) , Kindle, 2013

Web Resources:

Spoken English:

1. www.esl-lab.com
2. www.englishmedialab.com
3. www.englishinteractive.net
4. <https://www.britishcouncil.in/english/online>
5. <http://www.letstalkpodcast.com/>
6. https://www.youtube.com/c/mmmEnglish_Emma/featured
7. <https://www.youtube.com/c/ArnelsEverydayEnglish/featured>
8. <https://www.youtube.com/c/engvidAdam/featured>
9. <https://www.youtube.com/c/EnglishClass101/featured>
10. <https://www.youtube.com/c/SpeakEnglishWithTiffani/playlists>
11. https://www.youtube.com/channel/UCV1h_cBE0Drdx19qkTM0WNw

Voice & Accent:

1. <https://www.youtube.com/user/letstalkaccent/videos>
2. <https://www.youtube.com/c/EngLanguageClub/featured>
3. https://www.youtube.com/channel/UC_OskgZBoS4dAnVUgJVexc
4. https://www.youtube.com/channel/UCNfm92h83W2i2ijc5Xwp_IA

Course Title	Electrical & Electronics Engineering Workshop					B. Tech. II Sem. EEE, CSE & AIML		
Course Code	Category	Hours/Week			Credits	Maximum Marks		
23EE214	Engineering Science	L	T	P	C	Continuous Internal Assessment	End Exams	Total
		0	0	3	1.5	30	70	100
					End Exam Duration: 3Hrs			
Course Objectives: To impart knowledge on the principles of digital electronics and fundamentals of electron devices & its applications.								
Course Outcomes: On successful completion of this course, the students will be able to								
CO1	Identify &testing of various electronic components.							
CO2	Understand the usage of electronic measuring instruments.							
CO3	Plot and discuss the characteristics of various electronic devices.							
CO4	Explain the operation of a digital circuit.							

Activities:

1. Familiarization of commonly used Electrical & Electronic Workshop Tools: Breadboard, Solder, cables, relays, switches, connectors, fuses, Cutter, plier, screwdriver set, wire stripper, flux, knife / blade, soldering iron, de-soldering pump etc.
 - Provide some exercises so that hand ware tools and instruments are learned to be used by the students.
2. Familiarization of Measuring Instruments like Voltmeters, Ammeters, multimeter, LCR-Q meter, Power Supplies, CRO, DSO, Function Generator, Frequency counter.
 - Provide some exercises so that measuring instruments are learned to be used by the students.
3. Components:
 - Familiarization / Identification of components (Resistors, Capacitors, Inductors, Diodes, transistors, IC's etc.) – Functionality, type, size, colour coding package, symbol, cost etc.
 - Testing of components like Resistor, Capacitor, Diode, Transistor, IC set, Compare values of components like resistors, inductors, capacitors etc with the measured values by using instruments.

PART A: ELECTRICAL ENGINEERING LAB

List of experiments:

1. Verification of KCL and KVL
2. Verification of Superposition theorem
3. Measurement of Resistance using Wheat stone bridge
4. Magnetization Characteristics of DC shunt Generator
5. Measurement of Power and Power factor using Single-phase wattmeter
6. Measurement of Earth Resistance using Megger
7. Calculation of Electrical Energy for Domestic Premises
8. OC and SC tests on single phase transformer
9. Brake test on DC Shunt Motor

PARTB: ELECTRONICS ENGINEERING LAB

List of Experiments:

1. Plot V-I characteristics of PN Junction diode A) Forward bias B) Reverse bias.
2. Plot V-I characteristics of Zener Diode and its application as voltage Regulator.
3. Implementation of half wave and full wave rectifiers.
4. Plot Input & Output characteristics of BJT in CE and CB configurations
5. Frequency response of CE amplifier.
6. Simulation of RC coupled amplifier with the design supplied
7. Verification of Truth Table of AND, OR, NOT, NAND, NOR, Ex-OR, Ex-NOR gates using ICs.
8. Verification of Truth Tables of S-R, J-K & D flip flops using respective ICs.

Tools / Equipment Required: DC Power supplies, Multimeters, DC Ammeters, DC Voltmeters, AC Voltmeters, CROs, all the required active devices.

Reference Books:

1. R.L. Boylestad & Louis Nashlesky, Electronic Devices & Circuit Theory, Pearson education, 2021.
2. R. P. Jain, Modern Digital Electronics, 4th Edition, Tata Mc Graw Hill, 2009.
3. R.T. Paynter, Introductory Electronic Devices & Circuits–Conventional Flow Version, Pearson Education, 2009.

Note: Minimum Six Experiments to be performed. All the experiments shall be implemented using both Hardware and Software.

Course Title	ENGINEERING PHYSICS LAB					B. Tech. II Sem. EEE, CSE & AIML		
Course Code	Category	Hours/Week			Credits	Maximum Marks		
2322215	BSC	L	T	P	C	Continuous Internal Assessment	End Exams	Total
		0	0	2	1	30	70	100
					End Exam Duration: 3Hrs			
Course Objectives: ➤ To study the concepts of optical phenomenon like interference, diffraction etc., recognize the importance of energy gap in the study of conductivity and Hall effect in semiconductors and study the parameters and applications of dielectric and magnetic materials by conducting experiments.								
Course Outcomes:								
CO1	Operate optical instruments like travelling microscope and spectrometer.							
CO2	Estimate the wavelengths of different colors using diffraction grating.							
CO3	Plot the intensity of the magnetic field of circular coil carrying current with distance.							
CO4	Evaluate dielectric constant and magnetic susceptibility for dielectric and magnetic materials respectively.							
CO5	Calculate the band gap of a given semiconductor. Identify the type of semiconductor using Hall Effect.							

List of Experiments:

1. Determination of radius of curvature of a given Plano convex lens by Newton's rings.
2. Determination of wavelengths of different spectral lines in mercury spectrum using Diffraction grating in normal incidence on figuration.
3. Verification of Brewster's law
4. Determination of dielectric constant using charging and discharging method.
5. Study the variation of B versus H by magnetizing the magnetic material (B-H curve).
6. Determination of wavelength of Laser light using diffraction grating.
7. Estimation of Planck's constant using photo electric effect.
8. Determination of the resistivity of semiconductors by four probe methods.
9. Determination of energy gap of a semiconductor using p-n junction diode.
10. Magnetic field along the axis of a current carrying circular coil by Stewart Gee's Method.
11. Determination of Hall voltage and Hall coefficient of a given semiconductor using Hall Effect.
12. Determination of temperature coefficients of a thermistor.
13. Determination of acceleration due to gravity and radius of Gyration by using a compound pendulum.
14. Determination of magnetic susceptibility by Kundt's tube method.
15. Determination of rigidity modulus of the material of the given wire using Torsional pendulum.
16. Sonometer: Verification of laws of stretched string.
17. Determination of young's modulus for the given material of wooden scale by non- uniform bending (or double cantilever) method.
18. Determination of Frequency of electrically maintained tuning fork by Melde's experiment.
19. Fraunhofer diffraction due to single slit.

Note: Any **TEN** of the listed experiments are to be conducted. Out of which any **TWO** Experiments may be conducted in virtual mode.

Text Books:

1. A Text book of Practical Physics-S.Balasubramanian, M.N.Srinivasan, S.Chand Publishers, 2017.
2. Practical Physics by [K.Venugopalan](#) (Author), [Vimal Saraswat](#) (Author), Himanshu Publications (1 January 2018)

Reference Books:

1. Lab manual Physics, R Rangarajan, R P Manchanda, R K Gupta, Rajesh Kumar Neena Sinha-New Saraswati House.
2. Practical Physics by Kumar P. R. Sasi, Prentice-Hall of India Pvt. Ltd

URL:www.vlab.co.in

Course Title	NSS / NCC / SCOUTS & GUIDES / COMMUNITY SERVICE					B. Tech. II Sem. EEE, CSE & AIML		
Course Code	Category	Hours/Week			Credits	Maximum Marks		
2306216	BS&H	L	T	P	C	Continuous Internal Assessment	End Exams	Total
		0	0	1	0.5			
					6 X 15 = 90 + 10 Viva voce			
Course Objectives: ➤ The objective of introducing this course is to impart discipline, character, fraternity, teamwork, social consciousness among the students and engaging them in selfless service.								
Course Outcomes: After completion of the course the students will be able to								
CO1	Understand the importance of discipline, character and service motto.							
CO2	Solve some societal issues by applying acquired knowledge, facts, and techniques.							
CO3	Explore human relationships by analyzing social problems.							
CO4	Determine to extend their help for the fellow beings and downtrodden people.							
CO5	Develop leadership skills and civic responsibilities.							

UNIT I : Orientation

General Orientation on NSS/NCC/ Scouts & Guides/Community Service activities, career guidance.

Activities:

- Conducting –ice breaking sessions-expectations from the course-knowing personal talents and skills
- Conducting orientations programs for the students –future plans-activities-releasing roadmap etc.
- Displaying success stories-motivational biopics- award winning movies on societal issues etc.
- Conducting talent show in singing patriotic songs-paintings- any other contribution.

UNIT II : Nature & Care Activities:

- Best out of waste competition.
- Poster and signs making competition to spread environmental awareness.
- Recycling and environmental pollution article writing competition.
- Organizing Zero-waste day.
- Digital Environmental awareness activity via various social media platforms.
- Virtual demonstration of different eco-friendly approaches for sustainable living.
- Write a summary on any book related to environmental issues.

UNIT III : Community Service Activities:

- Conducting One Day Special Camp in a village contacting village-area leaders- Survey in the village, identification of problems- helping them to solve via media- authorities-experts-etc.
- Conducting awareness programs on Health-related issues such as General Health, Mental health, Spiritual Health, HIV/AIDS,
- Conducting consumer Awareness. Explaining various legal provisions etc.
- Women Empowerment Programmes- Sexual Abuse, Adolescent Health and Population Education.

- v) Any other programmes in collaboration with local charities, NGOs etc.

Reference Books:

1. Nirmalya Kumar Sinha & Surajit Majumder, *A Text Book of National Service Scheme* Vol.;I, Vidya Kutir Publication, 2021 (ISBN 978-81-952368-8-6)
2. *Red Book - National Cadet Corps* – Standing Instructions Vol I & II, DirectorateGeneral of NCC, Ministry of Defence, New Delhi
3. Davis M. L. and Cornwell D. A., “Introduction to Environmental Engineering”, McGrawHill, New York 4/e 2008
4. Masters G. M., Joseph K. and Nagendran R. “Introduction to Environmental Engineering and Science”, Pearson Education, New Delhi. 2/e 2007
5. Ram Ahuja. *Social Problems in India*, Rawat Publications, New Delhi.

General Guidelines:

1. Institutes must assign slots in the Timetable for the activities.
2. Institutes are required to provide instructor to mentor the students.

Evaluation Guidelines:

- Evaluated for a total of 100 marks.
- A student can select 6 activities of his/her choice with a minimum of 01 activity per unit. Each activity shall be evaluated by the concerned teacher for 15 marks, totalling to 90 marks.
- A student shall be evaluated by the concerned teacher for 10 marks by conducting vivavoce on the subject.

III Semester

DISCRETE MATHEMATICS AND GRAPH THEORY								
(Common to CSE&AIML (III SEMESTER)								
(Open Elective / Inter Disciplinary Elective / Professional Elective)								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
23HS302	Basic Sciences (BS)	L	T	P	C	Continuous Internal Assessment	Sem.-End Exam	Total
		3	0	--	3	30	70	100
Mid Exam Duration: 2 Hrs						Sem.-End Exam Duration: 3 Hrs		
Pre-Requisites:								
Course Objectives:								
CEO1. To introduce the concepts of mathematical logic and the concepts sets, relations, functions.								
CEO2. To introduce generating functions and recurrence relations and to use Graph Theory for solving problems.								
Course Outcomes: On successful completion of this course, the students will be able to								
CO1. Apply mathematical logic to solve problems.								
CO2. Understand the concepts and perform the operations related to sets, relations and functions								
CO3. Apply basic counting techniques to solve combinatorial problems.								
CO4.								
CO5. Formulate problems and solve recurrence relations.								
CO6. Apply Graph Theory in solving computer science problems.								

UNIT-I: Mathematical Logic

Introduction, Statements and Notation, Connectives, Well-formed formulas, Tautology, Duality law, Equivalence, Implication, Normal Forms, Functionally complete set of connectives.

UNIT-II: Set theory

The Principle of Inclusion- Exclusion (Statements only), Pigeon hole principle and its application, Functions composition of functions, Inverse Functions, Recursive Functions, Lattices and its properties.

UNIT-III: Elementary Combinatorics

Combinations and Permutations, Enumeration of Combinations and Permutations, Enumerating Combinations and Permutations with Repetitions, Enumerating Permutations with Constrained Repetitions, Binomial Coefficients, The Binomial and Multinomial Theorems.

UNIT-IV: Recurrence Relations

Generating functions of Sequences, Calculating Coefficients of Generating Functions, Recurrence relations, Solving Recurrence Relations by Substitution and Generating functions, The Method of Characteristic roots, Solutions of Inhomogeneous, Recurrence Relations.

UNIT-V: Graphs

Basic Concepts, Isomorphism and Sub graphs, Trees and their Properties, Spanning Trees, Directed Trees, Binary Trees, Planar Graphs, Euler's Formula, Multigraphs and Euler Circuits, Hamiltonian Graphs.

Total Periods: 45

TEXT BOOKS:

1. J.P. Tremblay and R. Manohar, Discrete Mathematical Structures with Applications to Computer Science, Tata McGraw Hill, 2002.
2. Kenneth H. Rosen, Discrete Mathematics and its Applications with Combinatorics and Graph Theory, 7th Edition, McGraw Hill Education (India) Private Limited.

REFERENCE BOOKS:

1. Joe L. Mott, Abraham Kandel and Theodore P. Baker, Discrete Mathematics for Computer Scientists & Mathematicians, 2nd Edition, Pearson Education.
2. Narsingh Deo, Graph Theory with Applications to Engineering and Computer Science.

UNIVERSAL HUMAN VALUES – UNDERSTANDING HARMONY AND ETHICAL HUMAN CONDUCT								
(Common to AIML, CE, CSE, ECE, EEE, ME)								
(Open Elective / Inter Disciplinary Elective / Professional Elective)								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
23HS311	Humanities, Social Science and Management Course (HSMC)	L	T	P	C	Continuous Internal Assessment	Sem.-End Exam	Total
		2	1	0	3	30	70	100
Mid Exam Duration: 2 Hrs						Sem.-End Exam Duration: 3 Hrs		
Pre-Requisites:								
Course Objectives:								
CEO1. To help the students appreciate the essential complementary between 'VALUES' and 'SKILLS' to ensure sustained happiness and prosperity which are the core aspirations of all human beings.								
CEO2. To facilitate the development of a Holistic perspective among students towards life and profession as well as towards happiness and prosperity based on a correct understanding of the Human reality and the rest of existence. Such holistic perspective forms the basis of Universal Human Values and movement towards value-based living in a natural way.								
CEO3. To highlight plausible implications of such a Holistic understanding in terms of ethical human conduct, trustful and mutually fulfilling human behaviour and mutually enriching interaction with Nature.								
Course Outcomes: On successful completion of this course, the students will be able to								
CO1. Define the terms like Natural Acceptance, Happiness and Prosperity.								
CO2. Identify one’s self, and one’s surroundings (family, society nature).								
CO3. Apply what they have learnt to their own self in different day-to-day settings in real Life.								
CO4. Relate human values with human relationship and human society.								
CO5. Justify the need for universal human values and harmonious existence.								
CO6. Develop as socially and ecologically responsible engineers.								

UNIT-I: INTRODUCTION TO VALUE EDUCATION

General Introduction to Values - Right Understanding, Relationship and Physical Facility (Holistic Development and the Role of Education) - Understanding Value Education - Practice Session PS1 Sharing about Oneself - self-exploration as the Process for Value Education – Continuous Happiness and Prosperity – the Basic Human Aspirations - Practice Session PS2 Exploring Human Consciousness - Happiness and Prosperity – Current Scenario - Method to Fulfill the Basic Human Aspirations - Practice Session PS3 Exploring Natural Acceptance.

UNIT-II: HARMONY IN THE HUMAN BEING

Human Nature Relationships - Understanding Human being as the Co-existence of the self and the body - Distinguishing between the Needs of the self and the body - Practice Session PS4 Exploring the difference of Needs of self and body - The body as an Instrument of the self - Understanding Harmony in the self - Practice Session PS5 Exploring Sources of Imagination in the self - Harmony of the self with the body - Programme to ensure self-regulation and Health - Practice Session PS6 Exploring Harmony of self with the body.

UNIT-III: HARMONY IN THE FAMILY AND SOCIETY

(9 periods)

Harmony in the Family – the Basic Unit of Human Interaction - 'Trust' – the Foundational Value in Relationship - Practice Session PS7 Exploring the Feeling of Trust - 'Respect' – as the Right Evaluation - Practice Session PS8 Exploring the Feeling of Respect - Other Feelings, Justice in Human-to-Human Relationship - Understanding Harmony in the Society - Vision for the Universal Human Order.

UNIT-IV: HARMONY IN THE NATURE/EXISTENCE

(8 periods)

Understanding Harmony in the Nature - Interconnectedness, self-regulation and Mutual Fulfilment among the Four Orders of Nature - Practice Session PS10 Exploring the Four Orders of Nature - Realizing Existence as Co-existence at All Levels - The Holistic Perception of Harmony in Existence - Practice Session PS11 Exploring Co-existence in Existence.

UNIT-V: HOLISTIC PERCEPTION OF HARMONY

(9 periods)

Understanding the Harmony in the society - Universal Order - Practice Session PS12 Understanding the Harmony in the society - Critical appreciation of Human values - Justice, Trust.

Total Periods: 45

TEXT BOOKS:

1. R R Gaur, R Asthana, G P Bagaria, A Foundation Course in Human Values and Professional Ethics, 2nd Revised Edition, Excel Books, New Delhi, 2019. ISBN 978-93-87034-47-1 b. The Teacher's Manual.
2. R R Gaur, R Asthana, G P Bagaria, Teachers' Manual for A Foundation Course in Human Values and Professional Ethics, 2nd Revised Edition, Excel Books, New Delhi, 2019. ISBN 978-93-87034-53-2.

REFERENCE BOOKS:

1. Jeevan Vidya: Ek Parichaya, A Nagaraj, Jeevan Vidya Prakashan, Amarkantak, 1999.
2. Human Values, A.N. Tripathi, New Age Intl. Publishers, New Delhi, 2004.
3. The Story of Stuff (Book).
4. The Story of My Experiments with Truth - by Mohandas Karamchand Gandhi

5. Small is Beautiful - E. F Schumacher.
6. Slow is Beautiful - Cecile Andrews
7. Economy of Permanence - J C Kumarappa
8. Bharat Mein Angreji Raj – PanditSunderlal
9. Rediscovering India - by Dharampal
10. Hind Swaraj or Indian Home Rule - by Mohandas K. Gandhi
11. India Wins Freedom - Maulana Abdul Kalam Azad

DIGITAL LOGIC AND COMPUTER ORGANIZATION ((Professional Core)								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
2305301	Professional Core	L	T	P	C	Continuous Internal Assessment	Sem.-End Exam	Total
		3	0	0	3	30	70	100
Mid Exam Duration: 1.5 Hrs					Sem.-End Exam Duration: 3 Hrs			
Pre-Requisites:								
Course Objectives: CEO1. Provide students with a comprehensive understanding of digital logic design principles and computer organization fundamentals CEO2. Describe memory hierarchy concepts CEO3. Explain input/output (I/O) systems and their interaction with the CPU, memory, and peripheral devices								
Course Outcomes: On successful completion of this course, the students will be able to CO1. Demonstrate combinational circuits, sequential circuits and functional units of computers. (L2) CO2. Analyze the design and operation of processors, including instruction execution, pipelining, and control unit mechanisms, to comprehend their role in computer systems. (L4) CO3. Describe memory hierarchy concepts, including cache memory, virtual memory, and secondary storage, and evaluate their impact on system performance and scalability. (L3) CO4. Explain input/output (I/O) systems and their interaction with the CPU, memory, and peripheral devices, including interrupts, DMA, and I/O mapping techniques. (L3) CO5. Design Sequential and Combinational Circuits. (L6)								

UNIT-I:

Data Representation: Binary Numbers, Fixed Point Representation. Floating Point Representation. Number base conversions, Octal and Hexadecimal Numbers, Signed binary numbers, Binary codes
Digital Logic Circuits-I: Basic Logic Functions, Logic gates, universal logic gates, Minimization of Logic expressions. K-Map Simplification, Combinational Circuits, Decoders, Multiplexers

UNIT-II:

Digital Logic Circuits-II: Sequential Circuits, Flip-Flops, Binary counters, Registers, Shift Registers,
Basic Structure of Computers: Computer Types, Functional units, Basic operational concepts, Bus structures, Software, Performance, multiprocessors and multi computers, Computer Generations.

UNIT-III:

Computer Arithmetic: Addition and Subtraction of Signed Numbers, Multiplication of Positive Numbers, Signed-operand Multiplication, Integer Division, Floating-Point Numbers and Operations

Processor Organization: Fundamental Concepts, Execution of a Complete Instruction, Multiple-Bus Organization, Hardwired Control and Multi programmed Control

UNIT-IV:

The Memory Organization: Basic Concepts, Semiconductor RAM Memories, Read-Only Memories, Speed, Size and Cost, Cache Memory, Performance Considerations, Virtual Memory, Memory Management Requirements, Secondary Storage

UNIT-V:

Input /Output Organization: Accessing I/O Devices, Interrupts, Direct Memory Access, Buses, Interface Circuits, Standard I/O Interfaces

TEXT BOOKS:

1. Computer Organization, Carl Hamacher, Zvonko Vranesic, Safwat Zaky, 6th edition, McGraw Hill, 2023.
2. Digital Design, 6th Edition, M. Morris Mano, Pearson Education, 2018.
3. Computer Organization and Architecture, William Stallings, 11thEdition, Pearson, 2022.

REFERENCE BOOKS:

1. Computer Systems Architecture, M. Moris Mano, 3rdEdition, Pearson, 2017.
2. Computer Organization and Design, David A. Paterson, John L. Hennessy, Elsevier, 2004.
Fundamentals of Logic Design, Roth, 5thEdition, Thomson, 2003.

ADVANCED DATA STRUCTURES AND ALGORITHM ANALYSIS (Common To CSE, CSE(DS), AI&ML, CSE(AI&ML)) (Professional Core)								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
2305302	(Professional Core)	L	T	P	C	Continuous Internal Assessment	Sem.-End Exam	Total
		3	0	0	3	30	70	100
Mid Exam Duration: 1.5 Hrs					Sem.-End Exam Duration: 3 Hrs			
Pre-Requisites: Data Structures								
Course Objectives: CEO1. provide knowledge on advance data structures frequently used in Computer Science domain CEO2. Develop skills in algorithm design techniques popularly used CEO3. Understand the use of various data structures in the algorithm design								
Course Outcomes: On successful completion of this course, the students will be able to CO1. Demonstrate the advanced tree data structures, and their applications (L2) CO2. Analyze graph traversals methods, connected and biconnected components (L4) CO3. Apply greedy and dynamic methods to find out feasible and optimal solutions (L3) CO4. Evaluate and verify the possible solutions using backtracking method (L5) CO5. Construct nondeterministic algorithms, branch and bound solutions for complex problems (L6)								

UNIT-I:

Introduction to Algorithm Analysis, Space and Time Complexity analysis, Asymptotic Notations.
AVL Trees – Creation, Insertion, Deletion operations and Applications

UNIT-II:

Heap Trees (Priority Queues) – Min and Max Heaps, Operations and Applications
Graphs – Terminology, Representations, Basic Search and Traversals, Connected Components and Biconnected Components, applications.

UNIT-III:

Divide and Conquer: The General Method, Quick Sort, Merge Sort, Strassen's matrix multiplication.
Greedy Method: General Method, Job Sequencing with deadlines, Knapsack Problem, Minimum cost spanning trees, Single Source Shortest Paths.

UNIT-IV:

Dynamic Programming: General Method, All pairs shortest paths , 0/1 Knapsack, Travelling Salesperson problem, Optimal Binary Search Tree.

Backtracking: General Method, 8-Queens Problem, Sum of Subsets problem, Graph Coloring.

UNIT-V:

Branch and Bound: The General Method, 0/1 Knapsack Problem, Travelling Salesperson problem

NP Hard and NP Complete Problems: Basic Concepts, P, NP, NP Hard and NP complete classes.

TEXT BOOKS:

1. Fundamentals of Data Structures, Horowitz, Ellis; Sahni, Sartaj; Mehta, Dinesh 2nd Edition Universities Press
2. Computer Algorithms, Ellis Horowitz, Sartaj Sahni, Sanguthevar Rajasekaran 2nd Edition University Press

REFERENCE BOOKS:

1. Data Structures and program design in C, Robert Kruse, Pearson Education Asia
2. Introduction to the Design and Analysis of Algorithms, Levitin, Anany, Pearson Education.
3. An introduction to Data Structures with applications, Trembley & Sorenson, McGraw Hill
4. The Art of Computer Programming, Vol.1: Fundamental Algorithms, Donald E Knuth, Addison-Wesley, 1997.
5. Data Structures using C & C++: Langsam, Augenstein & Tanenbaum, Pearson, 1995
6. Algorithms + Data Structures & Programs:, N. Wirth, PHI
7. Fundamentals of Data Structures in C++: Horowitz Sahni& Mehta, Galgottia Pub.
8. Data structures in Java:, Thomas Standish, Pearson Education Asia
9. Classic Data Structures, Debasis Samanta, PHI.

Online Learning Resources:

1. https://www.tutorialspoint.com/advanced_data_structures/index.asp
2. <http://peterindia.net/Algorithms.html>
3. Abdul Bari, 1. Introduction to Algorithms (youtube.com)

OBJECT ORIENTED PROGRAMMING THROUGH JAVA
(Common to CSE, CSE(DS), AI&ML, CSE(AI&ML))
(Professional Core)

Course Code	Category	Hours/Week			Credits	Maximum Marks		
2305303	Professional Core	L	T	P	C	Continuous Internal Assessment	Sem.-End Exam	Total
		3	0	0	3	30	70	100

Mid Exam Duration: 1.5 Hrs

Sem.-End Exam Duration: 3 Hrs

Pre-Requisites: C programming

Course Objectives:

CEO1. Identify Java language components and how they work together in applications

CEO2. Learn the fundamentals of object-oriented programming in Java, including defining classes, invoking methods, using class libraries.

CEO3. Learn how to extend Java classes with inheritance and dynamic binding and how to use exception handling in Java applications

CEO4. Understand how to design applications with threads in Java

CEO5. Understand how to use Java APIs for program development

Course Outcomes: On successful completion of this course, the students will be able to

CO1. Analyze problems, design solutions using OOP principles, and implement them efficiently in Java. (L4)

CO2. Design and implement classes to model real-world entities, with a focus on attributes, behaviors, and relationships between objects (L3)

CO3. Demonstrate an understanding of inheritance hierarchies and polymorphic behavior, including method overriding and dynamic method dispatch. (L3)

CO4. Apply Competence in handling exceptions and errors to write robust and fault-tolerant code. (L3)

CO5. Perform file input/output operations, including reading from and writing to files using Java I/O classes, graphical user interface (GUI) programming using JavaFX. (L4)

UNIT-I: Object Oriented Programming

Basic concepts, Principles, Program Structure in Java: Introduction, Writing Simple Java Programs, Elements or Tokens in Java Programs, Java Statements ,Command Line Arguments, User Input to Programs, Escape Sequences Comments, Programming Style.

Data Types, Variables, and Operators : Introduction, Data Types in Java, Declaration of Variables, Data Types, Type Casting, Scope of Variable Identifier ,Literal Constants ,Symbolic Constants ,Formatted Output with printf () Method, Static Variables and Methods, Attribute Final,

Introduction to Operators, Precedence and Associativity of Operators, Assignment Operator(=), Basic Arithmetic Operators, Increment (++) and Decrement (- -) Operators, Ternary Operator, RelationalOperators,BooleanLogicalOperators,BitwiseLogicalOperators.

Control Statements: Introduction, if Expression, Nested if Expressions, if–else Expressions, Ternary Operator?:, Switch Statement, Iteration Statements, while Expression, do–while Loop,forLoop,NestedforLoop,For–EachforLoop,BreakStatement,ContinueStatement

UNIT-II: Arrays

Introduction, Declaration and Initialization of Arrays, Storage of Array in Computer Memory, Accessing Elements of Arrays, Operations on Array Elements, Assigning Array to Another Array, Dynamic Change of Array Size, Sorting of Arrays, Search for Values in Arrays, Class Arrays, Two-dimensional Arrays, Arrays of Varying Lengths, Three-dimensional Arrays.

Classes and Objects

Introduction, Class Declaration and Modifiers, Class Members, Declaration of Class Objects, Assigning One Object to Another, Access Control for Class Members, Accessing Private Members of Class, Constructor Methods for Class, Overloaded Constructor Methods, Nested Classes, Final Class and Methods, Passing Arguments by Value and by Reference, Keyword this.

Methods: Introduction, Defining Methods, Overloaded Methods, Overloaded Constructor Methods, Class Objects as Parameters in Methods, Access Control, Recursive Methods, Overriding Methods, Attributes Final and Static.

UNIT-III: Inheritance and Interfaces

Inheritance: Introduction, Process of Inheritance, Types of Inheritances, Universal Super Class-Object Class, Inhibiting Inheritance of Class Using Final, Access Control and Inheritance, Multilevel Inheritance, Application of Keyword Super, Constructor Method and Inheritance, Method Overriding, Abstract Classes, Interfaces and Inheritance.

Interfaces: Introduction, Declaration of Interface, Implementation of Interface, Multiple Interfaces, Nested Interfaces, Inheritance of Interfaces.

UNIT-IV: Packages and Java Library

Introduction, Defining Package, Importing Packages and Classes into Programs, Path and Class Path, Access Control, Packages in Java SE, Java.lang Package and its Classes, Class Object, Enumeration, class Math, Wrapper Classes, Java util Classes and Interfaces, Time Package, Formatting for Date/Time in Java.

Exception Handling: Introduction, Hierarchy of Standard Exception Classes, Keywords throws and throw, try, catch, and finally Blocks, Multiple Catch Clauses, Class Throwable, Unchecked Exceptions, Checked Exceptions.

Java I/O and File: Java I/O API, standard I/O streams, types, Byte streams, Character streams, Scanner class, Files in Java

UNIT-V: String Handling in Java

Introduction, Class String, Methods for Extracting Characters from Strings, Comparison, Modifying, Searching; Class String Buffer.

Multithreaded Programming: Introduction, Need for Multiple Threads Multithreaded Programming for Multi-core Processor, Thread Class, Main Thread-Creation of New Threads, Thread States, Thread Priority-Synchronization.

Java FX GUI: Java FX Scene Builder, Java FX App Window Structure, displaying text and image, event handling, mouse events

Text Books:

1. JAVA one step ahead, Anitha Seth, B.L. Juneja, Oxford.
2. Joy with JAVA, Fundamentals of Object Oriented Programming, Debasis Samanta, Monalisa Sarma, Cambridge, 2023.
3. JAVA9forProgrammers, Paul Deitel, Harvey Deitel, 4th Edition, Pearson.

References Books:

1. The complete Reference Java, 11th edition, Herbert Schildt, TMH
2. Introduction to Java programming, 7th Edition, Y Daniel Liang, Pearson

ADVANCED DATA STRUCTURES AND ALGORITHMS ANALYSIS LAB (Common to CSE, CSE(DS), AI&ML, CSE(AI&ML)) (Professional Core)								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
2305351	Professional Core	L	T	P	C	Continuous Internal Assessment	Sem.-End Exam	Total
		0	0	3	1.5	30	70	100
Pre-Requisites: Data Structures								
Course Objectives: CEO1. Acquire practical skills in constructing and managing Data structures CEO2. Apply the popular algorithm design methods in problem-solving								
Course Outcomes: On successful completion of this course, the students will be able to CO1. Demonstrate the Non-Linear data structures by developing and implementing the operations on AVL Trees, Heaps and Graphs. (L2) CO2. Implement Divide and Conquer methods for sorting elements (L5) CO3. Utilize appropriate data structures and algorithms to optimize solutions for specific computational problems. (L3) CO4. Design and develop programs to solve real world problems using Backtracking methods. (L5) CO5. Design algorithms to solve real-world problems (L6)								

Experiments covering the Topics:

- Operations on AVL trees, B-Trees, Heap Trees
- Graph Traversals
- Sorting techniques
- Minimum cost spanning trees
- Shortest path algorithms
- 0/1 Knapsack Problem
- Travelling Salesperson problem
- Optimal Binary Search Trees
- N-Queens Problem
- Job Sequencing

Minimum Ten experiments are to be conducted.

List of Exercises/List of Experiments:

Sample Programs:

1. Construct an AVL tree for a given set of elements which are stored in a file. And implement insert and delete operation on the constructed tree.
2. Construct Min and Max Heap using arrays, delete any element and display the content of the Heap.
3. Implement Breadth First Traversal and Depth First Traversal for a given graph.
4. Implement Quick sort and Merge sort and observe the execution time for various input sizes (Average, Worst and Best cases).
5. Implement the Single Source Shortest Paths using Greedy method when the graph is represented by adjacency matrix.
6. Implement Job sequencing with deadlines using Greedy strategy.
7. Implement a all pair shortest path for a given path.
8. Write a program to solve 0/1 Knapsack problem Using Dynamic Programming.
9. Implement Travelling Sales Person problem using Dynamic Programming.
10. Implement N-Queens Problem Using Backtracking.

REFERENCE BOOKS/LABORATORY MANUALS:

3. Fundamentals of Data Structures in C++, Horowitz Ellis, Sahni Sartaj, Mehta, Dinesh, 2ndEdition, Universities Press
4. Computer Algorithms/C++ Ellis Horowitz, SartajSahni, Sanguthevar Rajasekaran, 2ndEdition, University Press
5. Data Structures and program design in C, Robert Kruse, Pearson Education Asia
6. An introduction to Data Structures with applications, Trembley& Sorenson, McGraw Hill

ADDITIONAL LEARNING RESOURCES:

1. <http://cse01-iiith.vlabs.ac.in/>
2. <http://peterindia.net/Algorithms.html>

OBJECT ORIENTED PROGRAMMING THROUGH JAVA LAB (Common to CSE, CSE(DS), AI&ML, CSE(AI&ML)) (Professional Core)								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
2305352	Professional Core	L	T	P	C	Continuous Internal Assessment	Sem.-End Exam	Total
		0	0	3	1.5	30	70	100
Pre-Requisites: C programming								
Course Objectives: CEO1. Practice object-oriented programming in the Java programming language CEO2. Implement Classes, Objects, Methods, Inheritance, Exception, Runtime Polymorphism, User defined Exception handling mechanism CEO3. Illustrate inheritance, Exception handling mechanism CEO4. Construct Threads, Event Handling, implement packages, Java FXGUI								
Course Outcomes: On successful completion of this course, the students will be able to CO1. Demonstrate a solid understanding of Java syntax, including data types, control structures, methods, classes, objects, inheritance, polymorphism, and exception handling. (L2) CO2. Apply fundamental OOP principles such as encapsulation, inheritance, polymorphism, and abstraction to solve programming problems effectively. (L3) CO3. Familiar with commonly used Java libraries and APIs, including the Collections Framework, Java I/O and other utility classes. (L2) CO4. Develop problem-solving skills and algorithmic thinking, applying OOP concepts to design efficient solutions to various programming challenges. (L3) CO5. Proficiently construct graphical user interface (GUI) applications using JavaFX (L4)								

Sample Experiments:

Exercise-1:

- Write a JAVA program to display default value of all primitive data types of JAVA
- Write a java program that display the roots of a quadratic equation $ax^2+bx=0$. Calculate the discriminate D and basing on value of D, describe the nature of root.

Exercise-2

- Write a JAVA program to search for an element in a given list of elements using binary search mechanism.
- Write a JAVA program to sort for an element in a given list of elements using bubble sort

c) Write a JAVA program using StringBuffer to delete ,remove character.

Exercise-3

- a) Write a JAVA program to implement class mechanism. Create a class, methods and invoke them inside main method.
- b) Write a JAVA program implement method overloading.
- c) Write a JAVA program to implement constructor.
- d) Write a JAVA program to implement constructor overloading.

Exercise-4

- a) Write a JAVA program to implement Single Inheritance
- b) Write a JAVA program to implement multilevel Inheritance
- c) Write a JAVA program for abstract class to find areas of different shapes

Exercise-5

- a) Write a JAVA program give example for “super” keyword.
- b) Write a JAVA program to implement Interface. What kind of Inheritance can be achieved?
- c) Write a JAVA program that implements Runtime polymorphism

Exercise-6

- a) Write a JAVA program that describes exception handling mechanism
- b) Write a JAVA program Illustrating Multiple catch clauses
- c) Write a JAVA program for creation of Java Built-in Exceptions
- d) Write a JAVA program for creation of User Defined Exception

Exercise-7

- a) Write a JAVA program that creates threads by extending Thread class. First thread display “Good Morning “every 1 sec, the second thread displays “Hello “every 2 seconds and the third display “Welcome” every 3 seconds, (Repeat the same by implementing Runnable)

Exercise-8

- a) Write a JAVA program that import and use the user defined packages
- b) Without writing any code, build a GUI that display text in label and image in an Image View (use JavaFX)
- c) Build a Tip Calculator app using several JavaFX components and learn how to respond to user interactions with the GUI

Textbooks:

1. JAVA one step ahead, Anitha Seth, B.L. Juneja, Oxford.
2. Joy with JAVA, Fundamentals of Object Oriented Programming, Debasis Samanta, Monalisa Sarma, Cambridge, 2023.
3. JAVA9 for Programmers, Paul Deitel, Harvey Deitel, 4th Edition, Pearson.

References Books:

1. ThecompleteReferenceJava, 11th edition, Herbert Schildt, TMH
2. IntroductiontoJavaprogramming, 7th Edition, Y Daniel Liang, Pearson

S/W and Tools Used:

- a) JavaSE (JDK)
- b) MYSQL
- c) Editor

Online Resources:

- 1. <https://nptel.ac.in/courses/106/105/106105191/>
- 2. https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_012880464547618816347_shared/overview

PYTHON PROGRAMMING (Common to CSE, CSE(DS), AI&ML, CSE(AI&ML), ECE, EEE, ME) (Skill Enhancement Course)								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
2305353	Skill Enhancement Course	L	T	P	C	Continuous Internal Assessment	Sem.-End Exam	Total
		0	1	2	2	30	70	100
Pre-Requisites: Programming in C Language.								
Course Objectives: CEO1. Introduce core programming concepts of Python programming language. CEO2. Introduce core programming concepts of Python programming language. CEO3. Demonstrate about Python data structures like Lists, Tuples, Sets and dictionaries. CEO4. Implement Functions, Modules and Regular Expressions in Python Programming and to create practical and contemporary applications using these								
Course Outcomes: On successful completion of this course, the students will be able to CO1. Classify data structures of Python (L4) CO2. Apply Python programming concepts to solve a variety of computational problems (L3) CO3. Understand the principles of object-oriented programming (OOP) in Python, including classes, objects, inheritance, polymorphism, and encapsulation, and apply them to design and implement Python programs (L3) CO4. Become proficient in using commonly used Python libraries and frameworks such as JSON, XML, NumPy, pandas (L2) CO5. Exhibit competence in implementing and manipulating fundamental data structures such as lists, tuples, sets, dictionaries (L3)								

List of Exercises/List of Experiments:

Minimum Ten experiments are to be conducted.

- Write a program to find the largest element among three Numbers.
- Write a Program to display all prime numbers within an interval
- Write a program to swap two numbers without using a temporary variable.
- Demonstrate the following Operators in Python with suitable examples.
 - Arithmetic Operators
 - Relational Operators
 - Assignment Operators
 - Logical Operators
 - Bit wise Operators
 - Ternary Operator
 - Membership Operators
 - Identity Operators
- Write a program to add and multiply complex numbers
- Write a program to print multiplication table of a given number
- Write a program to define a function with multiple return values.
- Write a program to define a function using default arguments.

9. Write a program to find the length of the string without using any library functions.
10. Write a program to check if the substring is present in a given string or not.
11. Write a program to perform the given operations on a list:
 - i) Addition
 - ii. Insertion
 - iii. Slicing
12. Write a program to perform any 5 built-in-functions by taking any list
13. Write a program to create tuples (name, age, address, college) for at least two members and concatenate the tuples and print the concatenated tuples.
14. Write a program to count the number of vowels in a string (No control flow allowed).
15. Write a program to check if a given key exists in a dictionary or not.
16. Write a program to add a new key-value pair to an existing dictionary.
17. Write a program to sum all the items in a given dictionary.
18. Write a program to sort words in a file and put them in another file. The output file should have only lower-case words, so any upper-case words from source must be lowered.
19. Python program to print each line of a file in reverse order.
20. Python program to compute the number of characters, words and lines in a file.
21. Write a program to create, display, append, insert and reverse the order of the items in the array.
22. Write a program to add, transpose and multiply two matrices.
23. Write a Python program to create a class that represents a shape. Include methods to calculate its area and perimeter. Implement subclasses for different shapes like circle, triangle, and square.
24. Python program to check whether a JSON string contains complex object or not.
25. Python Program to demonstrate NumPy arrays creation using array () function.
26. Python program to demonstrate basic slicing, integer and Boolean indexing.
27. Python program to find min, max, sum, cumulative sum of array
28. Create a dictionary with at least five keys and each key represent value as a list where this list contains at least ten values and convert this dictionary as a pandas data frame and explore the data through the data frame as follows:
 - a) Apply head () function to the pandas data frame
 - b) Perform various data selection operations on Data Frame
29. Select any two columns from the above data frame, and observe the change in one attribute with respect to other attribute with scatter and plot operations in matplotlib

Text Books:

1. Gowrishankar S, Veena A., Introduction to Python Programming, CRC Press.
2. Python Programming, S Sridhar, J Indumathi, V M Hariharan, 2nd Edition, Pearson, 2024
3. Introduction to Programming Using Python, Y. Daniel Liang, Pearson.

SOFTWARE/Tools used:

- . Anaconda
- . Jupyter Notebook.

ADDITIONAL LEARNING RESOURCES:

Online Learning Resources/Virtual Labs:

- <https://www.coursera.org/learn/python-for-applied-data-science-ai>
- <https://www.coursera.org/learn/python?specialization=python#syllabus>

ENVIRONMENTAL SCIENCE (Common to AIML, CE, CSE, ECE, EEE, ME) (Open Elective / Inter Disciplinary Elective / Professional Elective)								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
23HS331	Basic Sciences (BS)	L	T	P	C	Continuous Internal Assessment	Sem.-End Exam	Total
		2	0	0	0	30	00	30
Mid Exam Duration: 2 Hrs					Sem.-End Exam Duration: 0 Hrs			
Pre-Requisites:								
Course Objectives: CEO1. To make the students to get awareness on environment. CEO2. To stress the importance of protecting natural resources, ecosystems for future generations and pollution cause due to the day-to-day activities of human life. CEO3. To save earth from the inventions by the engineers.								
Course Outcomes: On successful completion of this course, the students will be able to CO1. Influence the society in proper utilization of Natural resources. CO2. Quantify the interconnection of human dependence on this ecosystem. CO3. Recall the concepts of biodiversity & gain knowledge on distribution at different levels. CO4. Analyze the impact of environmental pollution on environment & solving environmental problems CO5. Discuss environmental laws & analyze the environmental concerns and follow sustainable developmental activities.								

UNIT-I:

Multidisciplinary Nature of Environmental Studies: – Definition, Scope and Importance – Need for Public Awareness.

Natural Resources: Renewable and non-renewable resources – Natural resources and associated problems – Forest resources – Use and over – exploitation, deforestation, case studies – Timber extraction – Mining, dams and other effects on forest and tribal people – Water resources – Use and over utilization of surface and ground water – Floods, drought, conflicts over water, dams – benefits and problems – Mineral resources: Use and exploitation, environmental effects of extracting and using mineral resources, case studies – Food resources: World food problems, changes caused by agriculture and overgrazing, effects of modern agriculture, fertilizer-pesticide problems, waterlogging, salinity, case studies. – Energy resources:

UNIT-II:

Ecosystems: Concept of an ecosystem. – Structure and function of an ecosystem – Producers, consumers and decomposers – Energy flow in the ecosystem – Ecological succession – Food chains, food webs and ecological pyramids – Introduction, types, characteristic features, structure and function of the following ecosystem:

- a. Forest ecosystem.
- b. Grassland ecosystem
- c. Desert ecosystem.
- d. Aquatic ecosystems (ponds, streams, lakes, rivers, oceans, estuaries).

UNIT-III:

Biodiversity and its Conservation : Introduction 0 Definition: genetic, species and ecosystem diversity – Bio-geographical classification of India – Value of biodiversity: consumptive use, Productive use, social, ethical, aesthetic and option values – Biodiversity at global, National and local levels – India as a mega-diversity nation – Hot-spots of biodiversity – Threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts – Endangered and endemic species of India – Conservation of biodiversity: In-situ and Ex-situ conservation of biodiversity.

UNIT-IV:

Environmental Pollution: Definition, Cause, effects and control measures of :

- a. Air Pollution.
- b. Water pollution
- c. Soil pollution
- d. Marine pollution
- e. Noise pollution
- f. Thermal pollution
- g. Nuclear hazards

Solid Waste Management: Causes, effects and control measures of urban and industrial wastes – Role of an individual in prevention of pollution – Pollution case studies – Disaster management: floods, earthquake, cyclone and landslides.

UNIT-V:

Social Issues and the Environment: From Unsustainable to Sustainable development. Issues and possible solutions – Climate change, global warming, acid rain, ozone layer depletion, nuclear accidents. Environment Protection Act. – Air (Prevention and Control of Pollution) Act. -Water (Prevention and control of Pollution) Act – Wildlife Protection Act – Forest Conservation Act.

Human Population and the Environment: Population growth, variation among nations. Population explosion – Family Welfare Programmes. – Environment and human health – Human Rights – Value Education – HIV/AIDS – Women and Child Welfare – Role of information Technology in Environment and human health – Case studies.

Field Work: Visit to a local area to document environmental assets River/forest grassland/ hill/

mountain – Visit to a local polluted site-Urban/Rural/Industrial/Agricultural Study of common plants, insects, and birds – river, hill slopes, etc.

TEXT BOOKS:

1. Text book of Environmental Studies for Undergraduate Courses by Erach Bharucha for University Grants Commission, Universities Press.
2. Environmental studies by Benny Joseph, Mc, Graw Hill Publications.
3. Principles and a basic course of Environmental science for under graduate course by Kousic, KouShic.
4. Text book of Environmental science and Technology by M. Anji Reddy, BS Publication.

REFERENCE BOOKS:

1. Environmental sciences and engineering – J. Glynn Henry and Gary W. Heinke – Printice hall of India Private limited.
2. Environmental Studies by Anindita Basak – Pearson education.
3. Introduction to Environmental engineering and science by Gilbert M. Masters and Wendell P. Ela - Printice hall of India Private limited.
4. Environmental Science, A Global Concerns, William P. Cunningham, Mary Ann Cunningham, Mc Graw Hill publications.

IV Semester

MANAGERIAL ECONOMICS AND FINANCIAL ANALYSIS								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
23HS421	Humanities (Management)	L	T	P	C	Continuous Internal Assessment	Sem.- End Exam	Total
		2	0	0	2	30	70	100
Mid Exam Duration: 2 Hrs						Sem.-End Exam Duration: 3 Hrs		
Pre-Requisites:								
Title of the Pre-requisite Course(s)								
Course Objectives:								
CEO1. To inculcate the basic knowledge of microeconomics and financial accounting								
CEO2. To make the students learn how demand is estimated for different products, input-output relationship for optimizing production and cost minimization.								
CEO3. To Know the Various types of market structure and pricing methods and strategy								
CEO4. To give an overview on investment appraisal methods to promote the students to learn how to plan long-term investment decisions.								
CEO5. To provide fundamental knowledge on accounting and to explain the process of preparing financial statemen								
Course Outcomes: On successful completion of this course, the students will be able to								
CO1. Define the concepts related to Managerial Economics, financial accounting and management(L2)								
CO2. Understand the fundamentals of Economics viz., Demand, Production, cost, revenue and markets (L2)								
CO3. Apply the Concept of Production cost and revenues for effective Business decision (L3)								
CO4. Analyzing and evaluating investment decisions using Capital budgeting techniques to maximize the returns. (L5)								
CO5. Able to prepare financial statements and evaluate the financial performance of business firms (L5)								

UNIT-I: Managerial Economics

Introduction –Meaning, Nature and scope, significance. Demand-Concept, Function, Law of Demand - Demand Elasticity- Types – Measurement. Demand Forecasting- Methods. Managerial Economics and Financial Accounting and Management.

UNIT - II Production and Cost Analysis

Introduction – Nature, meaning, significance. Production Function with one variable and two variables– Least- cost combination– - Isoquants and Iso costs, Break-Even Analysis - Cost concepts - Determination of Break-Even Point (Simple Problems).

UNIT - III Capital Budgeting

Capital Budgeting - Nature, meaning, significance, Features, Proposals, Methods and

Evaluation of Projects – Pay Back period (PBP) Accounting Rate of Return (ARR), Net Present Value (NPV), Internal Rate of Return (IRR) Method and Profitability Index (P/I) (simple problems)

UNIT - IV Business Organizations and Markets

Introduction – Forms of Business Organizations- Sole Proprietary - Partnership - Joint Stock Companies - Public Sector Enterprises. Types of Markets - Perfect and Imperfect Competition - Features of Perfect Competition Monopoly- Monopolistic Competition– Oligopoly-Price-Output Determination.

UNIT - V Financial Accounting and Analysis

(9 periods)

Introduction – Meaning, Concepts and Conventions- Double-Entry Bookkeeping, Journal, Ledger, Trial Balance- Final Accounts (Trading Account, Profit and Loss Account and Balance Sheet with simple adjustments). Financial Ratios – Types of Ratios.

Total Periods: 38

Textbooks:

1. Varshney & Maheswari: Managerial Economics, Sultan Chand.
2. Aryasri: Business Economics and Financial Analysis, 4/e, MGH.

Reference Books:

1. Ahuja Hl Managerial economics, S. Chand.
2. S.A. Siddiqui and A.S. Siddiqui: Managerial Economics and Financial Analysis, New Age International.
3. Joseph G. Nellis and David Parker: Principles of Business Economics, Pearson, 2/e, New Delhi.
4. Domnick Salvatore: Managerial Economics in a Global Economy, Cengage.

Online Learning Resources: <https://www.slideshare.net/123ps/managerial-economics-ppt> <https://www.slideshare.net/rossanz/production-and-cost-45827016> <https://www.slideshare.net/darkyla/business-organizations-19917607> <https://www.slideshare.net/balarajbl/market-and-classification-of-market>

PROBABILITY AND STATISTICS (Common to CSE&AIML) (Open Elective / Inter Disciplinary Elective / Professional Elective)								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
23HS402	Basic Sciences (BS)	L	T	P	C	Continuous Internal Assessment	Sem.-End Exam	Total
		3	0	--	3	30	70	100
Mid Exam Duration: 2 Hrs						Sem.-End Exam Duration: 3 Hrs		
Pre-Requisites:								
Course Objectives: CEO1. To help the students in getting a thorough understanding of statistical elementary tools and the fundamentals of Probability. CEO2. To help the students in getting a thorough understanding and usage of statistical techniques like testing of hypothesis.								
Course Outcomes: On successful completion of this course, the students will be able to CO1. Acquire knowledge in finding the analysis of the data quantitatively or categorically and various statistical elementary tools. CO2. Develop skills in designing mathematical models involving probability, random variables and the critical thinking in the theory of probability and its applications in real life problems. CO3. Apply the theoretical probability distributions like binomial, Poisson, and Normal in the relevant application areas. CO4. Analyze to test various hypotheses included in theory and types of errors for large samples. CO5. Apply the different testing tools like t-test, F-test, chi-square test to analyze the relevant real-life problems.								

UNIT-I: Descriptive statistics

Statistics: Introduction, Population vs Sample, Collection of data, primary and secondary data, Measures of Central tendency, Measures of Variability (spread or variance) Skewness, Kurtosis, correlation, correlation coefficient, rank correlation.

UNIT-II: Probability

Probability, probability axioms, addition law and multiplicative law of probability, conditional probability, Baye's theorem, random variables (discrete and continuous), probability density functions, properties, mathematical expectation.

UNIT-III: Probability distributions

Probability distributions: Binomial, Poisson and Normal-their properties (Chebyshevs inequality). Approximation of the binomial distribution to normal distribution.

UNIT-IV: Testing of hypothesis, large sample tests

Formulation of null hypothesis, alternative hypothesis, the critical and acceptance regions, level of significance, two types of errors and power of the test. Large Sample Tests: Test for single proportion, difference of proportions, test for single mean and difference of means. Confidence interval for parameters in one sample and two sample problems

UNIT-V: Small sample tests

Student's t-distribution (test for single mean, two means and paired t-test), testing of equality of variances (F-test), χ^2 - test for goodness of fit, χ^2 - test for independence of attributes.

TEXT BOOKS:

1. Miller and Freunds, Probability and Statistics for Engineers, 7/e, Pearson, 2008.
2. S.C. Gupta and V.K. Kapoor, Fundamentals of Mathematical Statistics, 11/e, Sultan Chand & Sons Publications, 2012.

REFERENCE BOOKS:

1. S. Ross, a First Course in Probability, Pearson Education India, 2002.
2. W. Feller, an Introduction to Probability Theory and its Applications, 1/e, Wiley, 1968.
3. B. V. Ramana, Higher Engineering Mathematics, Mc Graw Hill Education.
4. B. S. Grewal, Higher Engineering Mathematics, Khanna Publishers, 2017, 44th Edition.

ADDITIONAL LEARNING RESOURCES:

1. https://onlinecourses.nptel.ac.in/noc21_ma74/preview
2. https://onlinecourses.nptel.ac.in/noc22_mg31/preview

OPERATING SYSTEMS (Professional Core)								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
2305401	Engineering Science (ESC)	L	T	P	C	Continuous Internal Assessment	Sem.-End Exam	Total
		3	0	0	3	30	70	100
Mid Exam Duration: 1.5 Hrs					Sem.-End Exam Duration: 3 Hrs			
Pre-Requisites: Nill								
Course Objectives: CEO1. Understand the basic concepts and principles of operating systems, including process management, memory management, file systems, and Protection CEO2. Make use of process scheduling algorithms and synchronization techniques to achieve better performance of a computer system. CEO3. Illustrate different conditions for deadlock and their possible solutions.								
Course Outcomes: On successful completion of this course, the students will be able to								
CO1	Understand the basic concepts related to the operating systems.							
CO2	Analyze the various process scheduling algorithms and Understand the Thread concepts							
CO3	Analyze Process synchronization mechanisms & Understand the way to deal the deadlocks							
CO4	Analyze the various memory management schemes							
CO5	Understand the basic concepts related to files in the system & Analyze the protection and security mechanisms.							

UNIT-I:

Operating Systems Overview: Introduction, operating system functions, operating systems operations, Computing environments. **System Structures:** Operating System Services, User and Operating-System Interface, system calls, Types of System Calls, system programs, operating system structure.

UNIT-II:

Processes: Process Concept, Process scheduling, Operations on processes, Inter-process communication. **Threads and Concurrency:** Multithreading models, Thread libraries, Threading issues. **CPU Scheduling:** Basic concepts, Scheduling criteria, Scheduling algorithms, Multiple processor scheduling.

UNIT-III:

Synchronization Tools: The Critical Section Problem, Peterson's Solution, Mutex Locks, Semaphores, Monitors, Classic problems of Synchronization. **Deadlocks:** system Model, Deadlock characterization, Methods for handling Deadlocks, Deadlock prevention, Deadlock avoidance, Deadlock detection, Recovery from Deadlock.

UNIT-IV:

Memory-Management Strategies: Introduction, Contiguous memory allocation, Paging, Structure of the Page Table, Swapping. **Virtual Memory Management:** Introduction, Demand paging, Page replacement, Allocation of frames,

UNIT-V:

File System: File System Interface: File concept, Access methods, Directory Structure; File system Implementation: File-system structure, File-system Operations, Allocation method. **Protection:** Goals of protection, Principles of protection, Protection Rings, Domain of protection, Access matrix. **Security:** The security problem, Program threats, User authentication.

TEXT BOOKS:

1. Operating System Concepts, Silberschatz A, Galvin P B, Gagne G, 10th Edition, Wiley, 2018.
2. Modern Operating Systems, Tanenbaum A S, 4th Edition, Pearson , 2016

REFERENCE BOOKS:

1. Operating Systems -Internals and Design Principles, Stallings W, 9th edition, Pearson, 2018
2. Operating Systems: A Concept Based Approach, D.M Dhamdhare, 3rd Edition, McGraw- Hill, 2013

ADDITIONAL LEARNING RESOURCES:

1. <https://nptel.ac.in/courses/106/106/106106144/>
2. <http://peterindia.net/OperatingSystems.html>

DATABASE MANAGEMENT SYSTEMS (Common to CSE, CSE(DS), AI&ML, CSE(AI&ML)) (Professional Core)								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
2339402	Professional Core	L	T	P	C	Continuous Internal Assessment	Sem.-End Exam	Total
		3	0	0	3	30	70	100
Mid Exam Duration: 1.5 Hrs						Sem.-End Exam Duration: 3 Hrs		
Pre-Requisites: Data structures, File organization and management								
Course Objectives: CEO1. Introduce database management systems and to give a good formal foundation on the relational model of data and usage of Relational Algebra CEO2. Introduce the concepts of basic SQL as a universal Database language CEO3. Demonstrate the principles behind systematic database design approaches by covering conceptual design, logical design through normalization CEO4. Provide an overview of physical design of a database system, by discussing Database indexing techniques and storage techniques								
Course Outcomes: On successful completion of this course, the students will be able to CO1. Understand the basic concepts of database management systems CO2. Analyze a given database application scenario to use ER model for conceptual design of the database CO3. Utilize SQL proficiently to address diverse query challenges CO4. Employ normalization methods to enhance database structure CO5. Assess and implement transaction processing, concurrency control and database recovery protocols in databases								

UNIT-I:

Introduction: Database system, Characteristics (Database Vs File System), Database Users, Advantages of Database systems, Database applications. Brief introduction of different Data Models; Concepts of Schema, Instance and data independence; Three tier schema architecture for data independence; Database system structure, environment, Centralized and Client Server architecture for the database.

Entity Relationship Model: Introduction, Representation of entities, attributes, entity set, relationship, relationship set, constraints, sub classes, super class, inheritance, specialization, generalization using ER Diagrams.

UNIT-II:

Relational Model: Introduction to relational model, concepts of domain, attribute, tuple, relation, importance of null values, constraints (Domain, Key constraints, integrity constraints) and their importance, Relational Algebra, Relational Calculus. BASIC SQL: Simple Database schema, data types, table definitions (create, alter), different DML operations (insert, delete,

update).

UNIT-III: TITLE OF THE UNIT

SQL: Basic SQL querying (select and project) using where clause, arithmetic & logical operations, SQL functions (Date and Time, Numeric, String conversion). Creating tables with relationship, implementation of key and integrity constraints, nested queries, sub queries, grouping, aggregation, ordering, implementation of different types of joins, view(updatable and non-updatable), relational set operations.

UNIT-IV: TITLE OF THE UNIT

Schema Refinement (Normalization): Purpose of Normalization or schema refinement, concept of functional dependency, normal forms based on functional dependency Lossless join and dependency preserving decomposition, (1NF, 2NF and 3 NF), concept of surrogate key, Boyce-Codd normal form(BCNF), MVD, Fourth normal form(4NF), Fifth Normal Form (5NF).

UNIT-V: TITLE OF THE UNIT

Transaction Concept: Transaction State, ACID properties, Concurrent Executions, Serializability, Recoverability, Implementation of Isolation, Testing for Serializability, lock based, time stamp based, optimistic, concurrency protocols, Deadlocks, Failure Classification, Storage, Recovery and Atomicity, Recovery algorithm.

Introduction to Indexing Techniques: B+ Trees, operations on B+Trees, Hash Based Indexing:

Textbooks:

1. Database Management Systems, 3rd edition, Raghurama Krishnan, Johannes Gehrke, TMH (For Chapters 2, 3, 4)
2. Database System Concepts, 5th edition, Silberschatz, Korth, Sudarsan, TMH (For Chapter 1 and Chapter 5)

Reference Books:

1. Introduction to Database Systems, 8th edition, C J Date, Pearson.
2. Database Management System, 6th edition, Ramez Elmasri, Shamkant B. Navathe, Pearson
3. Database Principles Fundamentals of Design Implementation and Management, Corlos Coronel, Steven Morris, Peter Robb, Cengage Learning.

Web-Resources:

1. <https://nptel.ac.in/courses/106/105/106105175/>
2. https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_01275806667282022456_shared/overview

SOFTWARE ENGINEERING (Professional Core)								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
2305402	Engineering Science (ESC)	L	T	P	C	Continuous Internal Assessment	Sem.-End Exam	Total
		3	0	0	3	30	70	100
Mid Exam Duration: 1.5 Hrs					Sem.-End Exam Duration:3Hrs			
Pre-Requisites: Introduction to Programming								
Course Objectives: CEO1. Software life cycle models, Software requirements and SRS document CEO2. Project Planning, quality control and ensuring good quality software CEO3. Software Testing strategies, use of CASE tools, Implementation issues, validation & verification procedures								
Course Outcomes: On successful completion of this course, the students will be able to CO1. Perform various life cycle activities like Analysis, Design, Implementation, Testing and Maintenance (L3) CO2. Analyse various software engineering models and apply methods for design and development of software projects. (L4) CO3. Develop system designs using appropriate techniques. (L3) CO4. Understand various testing techniques for a software project. (L2) CO5. Apply standards, CASE tools and techniques for engineering software projects (L3)								

UNIT-I: INTRODUCTION

Introduction: Evolution, Software development projects, Computer system engineering.

Software Life Cycle Models: Basic concepts, Waterfall model and its extensions, Rapid application development, Agile development model, Spiral model.

UNIT-II: SOFTWARE PROJECT MANAGEMENT

Software Project Management: Software project management complexities, Responsibilities of a software project manager, Metrics for project size estimation, Project estimation techniques, Empirical Estimation techniques, COCOMO, Halstead's software science, risk management.

Requirements Analysis And Specification: Requirements gathering and analysis, Software Requirements Specification (SRS).

UNIT-III: SOFTWARE DESIGN

Software Design: Overview of the design process, How to characterize a good software design? Layered arrangement of modules, Cohesion and Coupling. approaches to software design.
Agility: Agility and the Cost of Change, Agile Process, Extreme Programming (XP)
User Interface Design: Characteristics of a good user interface, Basic concepts, Types of user interfaces.

UNIT-IV: CODING AND TESTING

Coding And Testing: Coding, Code review, Software documentation, Testing, Black-box testing, White-Box testing, Debugging, Integration testing, Smoke testing, and Some general issues associated with testing.

Software Reliability And Quality Management: Software reliability. Statistical testing, Software quality, Software quality management system, SEI Capability maturity model.

UNIT-V: COMPUTER AIDED SOFTWARE DESIGN

Computer-Aided Software Engineering (Case): CASE and its scope, CASE environment, CASE support in the software life cycle, other characteristics of CASE tools, Towards second generation CASE Tool, and Architecture of a CASE Environment.

Software Maintenance: Characteristics of software maintenance, Software reverse engineering, Software maintenance process models.

Software Reuse: reuse- definition, introduction, reason behind no reuse so far, Basic issues in any reuse program.

TEXT BOOKS:

1. Fundamentals of Software Engineering, Rajib Mall, 5th Edition, PHI.
2. Software Engineering A practitioner's Approach, Roger S. Pressman, 9th Edition, Mc- Graw Hill International Edition.

Reference Books:

1. Software Engineering, Ian Sommerville, 10th Edition, Pearson.
2. Software Engineering, Principles and Practices, Deepak Jain, Oxford University Press.

ADDITIONAL LEARNING RESOURCES:

- 1) <https://nptel.ac.in/courses/106/105/106105182/>
- 2) https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_01260589506387148827_shared/overview
- 3) https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_013382690411003904735_shared/overview

OPERATING SYSTEMS LAB (Professional Core)								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
2305451	Engineering	L	T	P	C	Continuous Internal Assessment	Sem.-End Exam	Total
		0	0	3	1.5	30	70	100
Pre-Requisites: Nill								
Course Objectives: CEO3. Provide insights into system calls, file systems, semaphores, CEO1. Develop and debug CPU Scheduling algorithms, page replacement algorithms. CEO2. Implement Bankers Algorithms to Avoid the Dead Lock								
Course Outcomes: On successful completion of this course, the students will be able to CO1. Trace different CPU Scheduling algorithms (L2). CO2. Implement Bankers Algorithms to Avoid the Dead Lock (L3). CO3. Evaluate Page replacement algorithms (L5). CO4. Illustrate the file organization techniques (L4). CO5. Illustrate Inter process Communication (L4)								

List of Exercises/List of Experiments:

1. Practicing of Basic UNIX Commands.
2. Simulate UNIX commands like cp, ls, grep, etc.,
3. Simulate the following CPU scheduling algorithms
a) FCFS b) SJF c) Priority d) Round Robin
4. Write a program to illustrate concurrent execution of threads using pthreads library.
5. Write a program to solve producer-consumer problem using Semaphores.
6. Implement the following memory allocation methods for fixed partition
a) First fit b) Worst fit c) Best fit
7. Simulate the following page replacement algorithms
a) FIFO b) LRU c) LFU
8. Simulate Paging Technique of memory management.
9. Implement Bankers Algorithm for Dead Lock avoidance and prevention
10. Simulate the following file allocation strategies
a) Sequential b) Indexed c) Linked

REFERENCE BOOKS/LABORATORY MANUALS:

1. Operating System Concepts, Silberschatz A, Galvin P B, Gagne G, 10th Edition, Wiley, 2018.
2. Modern Operating Systems, Tanenbaum A S, 4th Edition, Pearson, 2016
3. Operating Systems -Internals and Design Principles, Stallings W, 9th edition, Pearson, 2018
4. Operating Systems: A Concept Based Approach, D.M Dhamdhere, 3rd Edition, McGraw- Hill, 2013

DATABASE MANAGEMENT SYSTEMS LAB (Common to CSE, CSE(DS), AI&ML, CSE(AI&ML)) (Professional Core)								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
2339452	Professional Core	L	T	P	C	Continuous Internal Assessment	Sem.-End Exam	Total
		0	0	3	1.5	30	70	100
Pre-Requisites: Data Structures, File Organization.								
Course Objectives: CEO1. Populate and query a database using SQL DDL/DML Commands CEO2. Declare and enforce integrity constraints on a database CEO3. Writing Queries using advanced concepts of SQL CEO4. Programming PL/SQL including procedures, functions, cursors and triggers.								
Course Outcomes: On successful completion of this course, the students will be able to CO1. Utilizing Data Definition Language (DDL), Data Manipulation Language (DML), and Data Control Language (DCL) commands effectively within a database environment CO2. Constructing and execute queries to manipulate and retrieve data from databases CO3. Develop application programs using PL/SQL. CO4. Analyze requirements and design custom Procedures, Functions, Cursors, and Triggers, leveraging their capabilities to automate tasks and optimize database functionality								

List of Exercises/List of Experiments:

1. Creation, altering and dropping of tables and inserting rows into a table (use constraints while creating tables) examples using SELECT command.
2. Queries (along with sub Queries) using ANY, ALL, IN, EXISTS, NOTEXISTS, UNION, INTERSET, Constraints. Example:- Select the roll number and name of the student who secured fourth rank in the class.
3. Queries using Aggregate functions (COUNT, SUM, AVG, MAX and MIN), GROUP BY, HAVING and Creation and dropping of Views.
4. Queries using Conversion functions (to_char, to_number and to_date), string functions (Concatenation, lpad, rpad, ltrim, rtrim, lower, upper, initcap, length, substr and instr), date functions (Sysdate, next_day, add_months, last_day, months_between, least, greatest, trunc, round, to_char, to_date)
5.
 - i. Create a simple PL/SQL program which includes declaration section, executable section and exception –Handling section (Ex. Student marks can be selected from the table and printed for those who secured first class and an exception can be raised if no records were found)

- ii. Insert data into student table and use COMMIT, ROLLBACK and SAVEPOINT in PL/SQL block.
- 6. Develop a program that includes the features NESTED IF, CASE and CASE expression. The program can be extended using the NULLIF and COALESCE functions.
- 7. Program development using WHILE LOOPS, numeric FOR LOOPS, nested loops using ERROR Handling, BUILT –IN Exceptions, USE defined Exceptions, RAISE-APPLICATION ERROR.
- 8. Programs development using creation of procedures, passing parameters IN and OUT of PROCEDURES.
- 9. Program development using creation of stored functions, invoke functions in SQL Statements and write complex functions.
- 10. Develop programs using features parameters in a CURSOR, FOR UPDATE CURSOR, WHERE CURRENT of clause and CURSOR variables.
- 11. Develop Programs using BEFORE and AFTER Triggers, Row and Statement Triggers and INSTEAD OF Triggers
- 12. Create a table and perform the search operation on table using indexing and non-indexing techniques.
- 13. Write a Java program that connects to a database using JDBC
- 14. Write a Java program to connect to a database using JDBC and insert values into it
- 15. Write a Java program to connect to a database using JDBC and delete values from it.

REFERENCE BOOKS/LABORATORY MANUALS:

- 1. Oracle: The Complete Reference by Oracle Press
- 2. Nilesh Shah, "Database Systems Using Oracle", PHI, 2007
- 3. Rick F Vander Lans, "Introduction to SQL", Fourth Edition, Pearson Education, 2007

FULL STACK DEVELOPMENT – 1 (Common To CSE, CSE(DS), AI&ML, CSE(AI&ML)) (Skill Enhancement Course)								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
2305452	Skill Enhancement Course	L	T	P	C	Continuous Internal Assessment	Sem.-End Exam	Total
		0	1	2	2	30	70	100
Pre-Requisites: C Programming & Java Basics								
Course Objectives: CEO4. Make use of HTML elements and their attributes for designing static web pages. CEO1. Build a web page by applying appropriate CSS styles to HTML elements CEO2. Experiment with JavaScript to develop dynamic web pages and validate forms. CEO3. Developing JavaScript programs using Looping statements.								
Course Outcomes: On successful completion of this course, the students will be able to CO1.Design Websites. (L6) CO2. Apply Styling to web pages. (L4) CO3. Design Forms for applications. (L6) CO4. Choose Control Structure based on the logic to be implemented. (L3) CO5. Understand HTML tags, Attributes and CSS properties (L2)								

List of Exercises/List of Experiments:

1. Lists, Links and Images

a. Write a HTML program, to explain the working of lists.

Note: It should have an ordered list, unordered list, nested lists and ordered list in an unordered list and definition lists.

b. Write a HTML program, to explain the working of hyperlinks using <a> tag and href, target Attributes.

c. Create a HTML document that has your image and your friend's image with a specific height and width. Also, when clicked on the images it should navigate to their respective profiles.

d. Write a HTML program, in such a way that, rather than placing large images on a page, the preferred technique is to use thumbnails by setting the height and width parameters to something like to 100*100 pixels. Each thumbnail image is also a link to a full sized

version of the image. Create an image gallery using this technique.

2. HTML Tables, Forms and Frames

- Write a HTML program, to explain the working of tables. (use tags: <table>, <tr>, <th>, <td> and attributes: border, rowspan, colspan)
- Write a HTML program, to explain the working of tables by preparing a timetable. (Note: Use <caption> tag to set the caption to the table & also use cell spacing, cell padding, border, rowspan, colspan etc.).
- Write a HTML program, to explain the working of forms by designing Registration form. (Note: Include text field, password field, number field, date of birth field, checkboxes, radio buttons, list boxes using <select>&<option> tags, <text area> and two buttons ie: submit and reset. Use tables to provide a better view).
- Write a HTML program, to explain the working of frames, such that page is to be divided into 3 parts on either direction. (Note: first frame image, second frame paragraph, third frame hyperlink. And also make sure of using “no frame” attribute such that frames to be fixed)

3. HTML 5 and Cascading Style Sheets, Types of CSS

- a. Write a HTML program, that makes use of <article>, <aside>, <figure>, <figcaption>, <footer>, <header>, <main>, <nav>, <section>, <div>, tags.
- b. Write a HTML program, to embed audio and video into HTML web page.
- c. Write a program to apply different types (or levels of styles or style specification formats) - inline, internal, external styles to HTML elements. (identify selector, property and value)

4. Selector forms

- a. Write a program to apply different types of selector forms
 - Simple selector (element, id, class, group, universal)
 - Combinator selector (descendant, child, adjacent sibling, general sibling)
 - Pseudo-class selector
 - Pseudo-element selector
 - Attribute selector

5. CSS with Color, Background, Font, Text and CSS Box Model

- a. Write a program to demonstrate the various ways you can reference a color in CSS.
- b. Write a CSS rule that places a background image halfway down the page, tilting it horizontally.

The image should remain in place when the user scrolls up or down.

c. Write a program using the following terms related to CSS font and text:

i. font-size ii. font-weight iii. font-style iv. text-decoration v. text-transformation vi. text-alignment

d. Write a program, to explain the importance of CSS Box model using

i. Content ii. Border iii. Margin iv. Padding

6. Applying JavaScript - internal and external, I/O, Type Conversion

a. Write a program to embed internal and external JavaScript in a web page.

b. Write a program to explain the different ways for displaying output.

c. Write a program to explain the different ways for taking input.

d. Create a webpage which uses prompt dialogue box to ask a voter for his name and age.

Display the information in table format along with either the voter can vote or no

7. JavaScript Pre-defined and User-defined Objects

a. Write a program using document object properties and methods.

b. Write a program using window object properties and methods.

c. Write a program using array object properties and methods.

d. Write a program using math object properties and methods

e. Write a program using string object properties and methods.

f. Write a program using date object properties and methods.

g. Write a program to explain user-defined object by using properties, methods, accessors, constructors and display.

8. JavaScript Conditional Statements and Loops

a. Write a program which asks the user to enter three integers, obtains the numbers from the user and outputs HTML text that displays the larger number followed by the words

“LARGER NUMBER” in an information message dialog. If the numbers are equal, output HTML text as “EQUAL NUMBERS”.

b. Write a program to display week days using switch case.

c. Write a program to print 1 to 10 numbers using for, while and do-while loops.

d. Write a program to print data in object using for-in, for-each and for-of loops

9. JavaScript Functions and Events

a. Design a appropriate function should be called to display

- Factorial of that number
- Fibonacci series up to that number
- Prime numbers up to that number
- Is it palindrome or not

b. Design a HTML having a text box and four buttons named Factorial, Fibonacci, Prime, and Palindrome. When a button is pressed an appropriate function should be called to display

- Factorial of that number
- Fibonacci series up to that number
- Prime numbers up to that number
- Is it palindrome or not

c. Write a program to validate the following fields in a registration page

- i. Name (start with alphabet and followed by alphanumeric and the length should not be less than 6 characters)
- ii. Mobile (only numbers and length 10 digits)
- iii. E-mail (should contain format like xxxxxxx@xxxxxx.xxx)

Textbooks:

1. Programming the World Wide Web, 7th Edition, Robert W Sebesta, Pearson, 2013.
2. Web Programming with HTML5, CSS and JavaScript, John Dean, Jones & Bartlett Learning, 2019 (Chapters 1-11).
3. Pro MERN Stack: Full Stack Web App Development with Mongo, Express, React, and Node, Vasan Subramanian, 2nd edition, APress, O'Reilly.

DESIGN THINKING AND INNOVATION								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
2304453	Engineering Science (ESC)	L	T	P	C	Continuous Internal Assessment	Sem.-End Exam	Total
		1	0	2	2	30	70	100
Pre-Requisites: Management science								
Course Objectives: CEO1: To familiarize students with design thinking process as a tool for breakthrough innovation. CEO2: To equip students with design thinking skills and ignite CEO3: To create innovative ideas, develop solutions for real-time problems.								
Course Outcomes: On successful completion of this course, the students will be able to CO1: Define the concepts related to design thinking (L1, L2) CO2: Explain the fundamentals of Design Thinking and innovation (L1, L2) CO3: Apply the design thinking techniques for solving problems in various sectors. (L3) CO4: Analyze to work in a multidisciplinary environment(L4) CO5: Evaluate the value of creativity(L5)								

Unit I :

Introduction to Design Thinking: Introduction to elements and principles of Design, basics of design-dot, line, shape, form as fundamental design components. Principles of design. Introduction to design thinking, history of Design Thinking, New materials in Industry.

Unit II :

Design Thinking Process: Design thinking process (empathize, analyze, idea & prototype), implementing the process in driving inventions, design thinking in social innovations. Tools of design thinking - person, costumer, journey map, brainstorming, product development.

Activity: Every student presents their idea in three minutes, every student can present design process in the form of flow diagram or flow chart etc. Every student should explain about product development.

Unit III :

Innovation: Artofinnovation, Difference between innovation and creativity, role of creativity and innovation in organizations- Creativity to Innovation- Teams for innovation- Measuring the impact and value of creativity.

Activity: Debate on innovation and creativity, Flow and planning from idea to innovation, Debate on value-based innovation.

Unit IV :

Product Design: Problem formation, introduction to product design, Product strategies, Product value, Product planning, product Specifications-Innovation towards product design-Case studies

Activity: Importance of modeling, how to set specifications, Explaining their own product design.

Unit V :

Design thinking in Business Processes: Design Thinking applied in Business & Strategic Innovation, Design Thinking principles that redefine business – Business challenges: Growth, Predictability, Change, Maintaining Relevance, Extreme competition, Standardization. Design thinking to meet corporate needs-Design thinking for Startups- Defining and testing Business Models and Business Cases- Developing & testing prototypes.

Activity: How to market our own product, about maintenance, Reliability and plan for startup.

TEXTBOOKS:

1. Tim Brown, “*Change by design*”, Harper Bollins (2009)
2. Idris Mootee, “*Design Thinking in Business Processes*”, 2013, John Wiley & Sons.

REFERENCE BOOKS:

1. David Lee, “*Design Thinking in the Classroom*”, Ulysses press
2. Shruti N Shetty, “*Design the Future*”, Norton Press
3. William Lidwell, “*Universal Principles of Design*”- Kritinaholden, Jill Butter.
4. Chesbrough.H, “*The Era of Open Innovation*”– 2013

Online Learning Resources:

- <https://nptel.ac.in/courses/110/106/110106124/>
- <https://nptel.ac.in/courses/109/104/109104109/>
- https://swayam.gov.in/nd1_noc19_mg60/preview

COMPUTER SCIENCE AND ENGINEERING**B.Tech. (R23UG) Course Structure****V-SEMESTER**

S. No.	COURSE CODE	COURSE TITLE	SC	L	T	P	IM	EM	CR
1	2305501	Artificial Intelligence	PCC	3	0	0	30	70	3
2	2305502	Computer Networks	PCC	3	0	0	30	70	3
3	2305503	Automata Theory and Compiler Design	PCC	3	0	0	30	70	3
4	2305504	Object Oriented Analysis and Design	PEC-1	3	0	0	30	70	3
	2305505	Soft Computing							
	2304503	Microprocessors and Microcontrollers							
	2305506	Augmented Reality and Virtual Reality							
5		Open Elective Course-I	OEC-1	3	0	0	30	70	3
6	2305507	Introduction to Quantum Technologies and Applications	ESC	3	0	0	30	70	3
7	2305551	Artificial Intelligence Lab	PCC	0	0	3	30	70	1.5
8	2305552	Computer Networks Lab	PCC	0	0	3	30	70	1.5
9	2305553	Tinkering Lab	ESC	0	0	2	30	70	1
10	2305554	Community Service Project	PROJ	-	-	-	-	50	2
11	23HS451	Soft Skills	SC	0	1	2	30	70	2
Total				18	01	10	300	750	26

B.Tech. V Sem CSE (R23UG) SYLL

2305501	B.Tech. CSE, V-SEMESTER ARTIFICIAL INTELLIGENCE	L	T	P	C
		3	0	0	3

Pre-Requisites: Knowledge in Computer Programming, Mathematical Foundations of Computer Science, Background in linear algebra, data structures and algorithms, and probability.

Course Outcomes:

After completion of the course, students will be able to:

- CO1.** Explain AI foundations, history, intelligent agents, and problem formulation
- CO2.** Apply search strategies and adversarial game-playing algorithms to AI problems
- CO3.** Represent knowledge and reason under uncertainty using logic and probabilistic methods.
- CO4.** Implement inference and learning techniques, including decision trees and reinforcement learning.
- CO5.** Evaluate expert system architectures, knowledge acquisition, and typical expert system shells.

SYLLABUS:

UNIT-I: (09 PERIODS)

Introduction: AI problems, foundation of AI and history of AI intelligent agents: Agents and Environments, the concept of rationality, the nature of environments, structure of agents, problem solving agents, problem formulation.

UNIT-II: (10 PERIODS)

Searching- Searching for solutions, uniformed search strategies – Breadth first search, depth first Search. Search with partial information (Heuristic search) Hill climbing, A* ,AO* Algorithms, Problem reduction, Game Playing-Adversial search, Games, mini-max algorithm, optimal decisions in multiplayer games, Problem in Game playing, Alpha-Beta pruning, Evaluation functions.

UNIT-III: (08 PERIODS)

Representation of Knowledge: Knowledge representation issues, predicate logic- logic programming, semantic nets- frames and inheritance, constraint propagation, representing knowledge using rules, rules based deduction systems. Reasoning under uncertainty, review of probability, Bayes' probabilistic interferences and Dempstershafer theory.

UNIT-IV: (08 PERIODS)

Logic concepts: First order logic. Inference in first order logic, propositional vs. first order inference, unification & lifts forward chaining, Backward chaining, Resolution, Learning from observation Inductive learning, Decision trees, Explanation based learning, Statistical Learning methods, Reinforcement Learning.

UNIT-V: (10 PERIODS)

Expert Systems: Architecture of expert systems, Roles of expert systems – Knowledge Acquisition Meta knowledge Heuristics. Typical expert systems – MYCIN, DART, XCON: Expert systems shells.

Text Books:

- T1. S. Russel and P. Norvig, —Artificial Intelligence—A Modern Approach, Second Edition, Pearson Education.
- T2. Kevin Night and Elaine Rich, Nair B., —Artificial Intelligence (SIE), Mc Graw Hill

Reference Books:

- R1. David Poole, Alan Mack worth, Randy Goebel, Computational Intelligence: a logical approach, Oxford University Press.
- R2. G. Luger, —Artificial Intelligence: Structures and Strategies for complex problemsolving, Fourth Edition, Pearson Education.
- R3. J. Nilsson, —Artificial Intelligence: A new Synthesis, Elsevier Publishers.
- R4. Artificial Intelligence, Saroj Kaushik, CENGAGE Learning.
- R5. An Intelligent Tutoring System for the Accounting Cycle: Enhancing Textbook Homework with Artificial Intelligence by Benny G. Johnson, Fred Phillips, and Linda G. Chase, Journal of Accounting Education, Vol. 27, No. 1, 2009, pp. 30–39

Web Resources:

1. <https://ai.google/>
2. https://swayam.gov.in/nd1_noc19_me71/preview

2305502	B.Tech. CSE, V-SEMESTER COMPUTER NETWORKS AND INTERNET PROTOCOLS	L	T	P	C
		3	0	0	3

Pre-Requisites: NIL

Course Outcomes:

On successful completion of the course, student will be able to

CO1. Identify the software and hardware components of a computer network.

CO2. Design software for a computer network.

CO3. Develop new routing, and congestion control algorithms.

CO4. Assess critically the existing routing protocols.

CO5. Explain the functionality of each layer of a computer network.

SYLLABUS:

UNIT-I: COMPUTER NETWORKS AND THE INTERNET (08 Periods)

What Is the Internet? The Network Edge, The Network Core, Delay, Loss, and Throughput in Packet Switched Networks (Textbook 2), Reference Models, Example Networks, Guided Transmission Media, Wireless Transmission.

UNIT-II: THE DATA LINK LAYER, ACCESS NETWORKS, AND LAN (10 Periods)

Data Link Layer Design Issues, Elementary Data Link Protocols, Sliding Window Protocols, Introduction to the Link Layer, Error-Detection and -Correction Techniques, Multiple Access Links and Protocols.

UNIT-III: THE NETWORK LAYER (08 Periods)

Routing Algorithms, Internetworking, The Network Layer in The Internet

UNIT -IV: THE TRANSPORT LAYER (09 Periods)

Connectionless Transport: UDP (Textbook 2), The Internet Transport Protocols: TCP, Congestion Control.

UNIT-V: PRINCIPLES OF NETWORK APPLICATIONS (08 Periods)

Principles of Network Applications, The Web and HTTP, Electronic Mail in the Internet, DNS—The Internet's Directory Service.

Total Periods: 43

Text Books:

- T1. Andrew S.Tanenbaum, David j.wetherall, Computer Networks, 6th Edition, PEARSON.
- T2. James F. Kurose, Keith W. Ross, —Computer Networking: A Top-Down Approach, 6th edition, Pearson, 2019.

Reference Books:

- R1. Forouzan, Data communications and Networking, 5th Edition, McGraw Hill Publication.
R2. Youlu Zheng, Shakil Akthar, —Networks for Computer Scientists and Engineers, Oxford Publishers, 2016.

Web Resources:

1. <https://nptel.ac.in/courses/106105183/25>
2. <http://www.nptelvideos.in/2012/11/computer-networks.html>
3. <https://nptel.ac.in/courses/106105183/3>

2305503	B.Tech. CSE, V-SEMESTER AUTOMATA THEORY AND COMPILER DESIGN	L	T	P	C
		3	0	0	3

Pre-Requisites: Data Structures, Discrete Mathematics and Graph Theory, Introduction to Programming (C or Python or JAVA), Digital Logic Design and Computer Organization.

Course Outcomes:

On successful completion of the course, student will be able to

- CO1.** Design and construct finite automata and regular expressions to represent and analyze regular languages, including minimization and closure properties.
- CO2.** Develop context-free grammars and design equivalent pushdown automata to model and analyze context-free languages.
- CO3.** Design Turing machines to simulate computation and analyze the role of lexical analysis within the compiler architecture.
- CO4.** Implement top-down and bottom-up parsing techniques and generate intermediate code using three-address code representations.
- CO5.** Apply code optimization techniques and design effective code generation strategies using control flow analysis and DAGs.

SYLLABUS:

UNIT- I: INTRODUCTION TO AUTOMATA AND REGULAR EXPRESSIONS (10 Periods)

Introduction, Alphabets, Strings and Languages, Chomsky Hierarchy, Automata and Grammars, Regular Grammar and Language, Finite Automata, Deterministic finite Automata (DFA), Nondeterministic finite Automata (NFA), Equivalence of NFA and DFA, Minimization of Finite Automata, Regular Expressions, Converting Regular Grammar and Expression into Finite Automata, Pumping lemma for regular sets, Closure properties of regular sets (Without proof).

UNIT- II: CONTEXT FREE GRAMMARS AND PUSHDOWN AUTOMATA (10 Periods)

Context Free Language, Context Free Grammar, Derivation and Parse tree, Ambiguity, Simplification of CFG's, Chomsky Normal Form, Greibach Normal Form, Push Down Automata (PDA), Design of PDA, Equivalence of PDA and CFL/CFG.

UNIT- III: TURING MACHINES AND INTRODUCTION TO COMPILERS (09 Periods)

Turing Machine, TM Model, Language acceptance, Design of Turing Machine, Compilers, Phases of Compiler, The role of Lexical Analyzer, Input Buffering.

UNIT- IV: PARSERS AND INTERMEDIATE CODE GENERATION (08 Periods)

Parser, Top-Down parsers: Recursive Descent Parsers, Predictive Parsers

Bottom-up Parsers: Shift-Reduce Parsing, LR parsers, Intermediate Code Generation: Three address codes.

UNIT- V: CODE OPTIMIZATION AND CODE GENERATION (08 Periods)

Code Optimization: Peephole optimization, Basic blocks and flow graphs, DAG, Principles of Source Code Optimization, Code Generation: Issues in Design of Code Generation, Simple Code Generator.

Total Periods: 45

Text Books:

- T1. Introduction to Automata theory languages and Computation, Hopcroft H.E. and Ullman Jeffrey. D, 3/e, 2006, Pearson Education, New Delhi, India.
- T2. Theory of Computer Science - Automata, Languages and Computation, 2. Mishra K L P and Chandrasekaran N, 2/e, 2007, PHI, New Delhi, India.
- T3. Compilers: Principles, Techniques, and Tools, Alfred V. Aho , Monica S. Lam, Ravi Sethi , Jeffrey D. Ullman , Sorav Bansal, Updated 2e July 2023.

Reference Books:

- R1. Introduction to Languages and Theory of Computation, John C Martin, 1/e, 2009, Tata McGraw Hill Education, Hyderabad, India.
- R2. Introduction to Theory of Computation, Thomson, Australia, Sipser, 2/e, 2005.
- R3. Compiler Construction: Principles And Practice, Kenneth C. Louden, Thomson/ Delmar Cengage Learning, 2006.
- R4. Lex & yacc, Doug Brown, John Levine and Tony Mason, 2 nd Edition, O'reilly Media.
- R5. Engineering a compiler, Keith Cooper and Linda Torczon, 2 nd Edition, Morgan Kaufmann, 2011.

Web Resources:

1. <https://nptel.ac.in/courses/106/106/106106049/>
2. <https://nptel.ac.in/courses/106105190>
3. <https://nptel.ac.in/courses/106108113>
4. <https://nptel.ac.in/courses/106104123>
5. <https://nptel.ac.in/courses/106105196>
6. <https://nptel.ac.in/courses/106104028>

2305504	B. Tech CSE, V SEMESTER OBJECT ORIENTED ANALYSIS AND DESIGN (Professional Elective –I)	L	T	P	C
		3	0	0	3

Pre-Requisites: NIL

Course Outcomes:

On successful completion of the course, student will be able to

- CO1.** The importance of modelling in UML .
- CO2.** Compare and contrast the object-oriented model with the E-R and EER models)
- CO3.** Design Behavioural diagrams and Activity diagrams
- CO4.** Design an application using deployment diagram
- CO5.** Apply UML diagrams to build library application.

SYLLABUS:

UNIT-I:

Introduction to UML: Importance of modelling, principles of modelling, object-oriented modelling, conceptual model of the UML, Architecture, Software Development Life Cycle.

UNIT-II:

Basic Structural Modelling: Classes, Relationships, common Mechanisms, and diagrams. Advanced Structural Modelling: Advanced classes, advanced relationships, Interfaces, Types and Roles, Packages. Class & Object Diagrams: Terms, concepts, modelling techniques for Class & Object Diagrams.

UNIT-III:

Basic Behavioural Modelling-I: Interactions, Interaction diagrams. Basic Behavioural Modelling-II: Use cases, Use case Diagrams, Activity Diagrams

UNIT-IV:

Advanced Behavioral Modelling: Events and signals, state machines, processes and Threads, time and space, state chart diagrams. Architectural Modelling: Component, Deployment, Component diagrams and Deployment diagrams.

UNIT-V:

Patterns and Frameworks, Artifact Diagrams. Case Study: The Unified Library application

Total Periods: --45

Text Books:

- T1. Grady Booch, James Rumbaugh, Ivar Jacobson: The Unified Modelling Language User Guide, Pearson Education 2nd Edition.
- T2. Object-Oriented Analysis and Design with the Unified Process By John W. Satzinger, Robert B Jackson and Stephen D Burd, Cengage Learning.

Reference Books:

- R1. Meilir Page-Jones: Fundamentals of Object-Oriented Design in UML, Pearson Education.
- R2. Pascal Roques: Modelling Software Systems Using UML2, WILEY-Dreamtech India Pvt. Ltd.
- R3. Atul Kahate: Object Oriented Analysis & Design, The McGraw-Hill Companies.
- R4. Mark Priestley: Practical Object-Oriented Design with UML, TMH.
- R5. Applying UML and Patterns: An introduction to Object – Oriented Analysis and Design and Unified Prpocess, Craig Larman, Pearson Education.

2305505	B. Tech CSE, V SEMESTER SOFT COMPUTING (Professional Elective –I)	L	T	P	C
		3	0	0	3

Pre-Requisites: NIL

Course Outcomes:

On successful completion of the course, student will be able to

- CO1.** Identify the difference between Conventional Artificial Intelligence to Computational Intelligence.
- CO2.** Understand fuzzy logic and reasoning to handle and solve engineering problems
- CO3.** understand the fuzzy decision making and optimization.
- CO4.** Perform various operations of genetic algorithms.
- CO5.** understand rough sets and Integration of Soft Computing Techniques.

SYLLABUS:

UNIT - I

(09 Periods)

Introduction to Soft Computing: Evolutionary Computing, "Soft" computing versus "Hard" computing, Soft Computing Methods, Recent Trends in Soft Computing, Characteristics of Soft computing, Applications of Soft Computing Techniques.

UNIT- II

(09 Periods)

Fuzzy Systems: Fuzzy Sets, Fuzzy Relations, Fuzzy Logic, Fuzzy Rule-Based Systems

UNIT- III**(09 Periods)**

Fuzzy Decision Making, Particle Swarm Optimization.

UNIT- IV**(09 Periods)**

Genetic Algorithms: Basic Concepts, Basic Operators for Genetic Algorithms, Crossover and Mutation Properties, Genetic Algorithm Cycle, Fitness Function, Applications of Genetic Algorithm.

UNIT- V**(09 Periods)**

Rough Sets, Rough Sets, Rule Induction, and Discernibility Matrix, Integration of Soft Computing Techniques.

Total Periods: - 45**Text Book:**

T1. Soft Computing – Advances and Applications - Jan 2015 by B.K. Tripathy and J. Anuradha – Cengage Learning

Reference Books:

- R1. S. N. Sivanandam & S. N. Deepa, —Principles of Soft Computing, 2nd edition, Wiley India, 2008.
- R2. David E. Goldberg, —Genetic Algorithms-In Search, optimization and Machine learning, Pearson Education.
- R3. J. S. R. Jang, C.T. Sun and E.Mizutani, -Neuro-Fuzzy and Soft Computing, Pearson Education, 2004.
- R4. G.J. Klir & B. Yuan, —Fuzzy Sets & Fuzzy Logic, PHI, 1995.
- R5. Melanie Mitchell, —An Introduction to Genetic Algorithm, PHI, 1998.
- R6. Timothy J. Ross, —Fuzzy Logic with Engineering Applications, McGraw- Hill International editions, 1995.

2305506	B. Tech CSE, V SEMESTER AUGMENTED REALITY AND VIRTUAL REALITY (Professional Elective –I)	L	T	P	C
		3	0	0	3

Pre-Requisites: NIL

Course Outcomes:

On successful completion of the course, student will be able to

- CO1. Understand the core concepts of Virtual Reality and Augmented Reality, and their differences.
- CO2. Identify and describe input/output devices used in VR and AR systems.
- CO3. Apply geometric, kinematic, physical, and behavioral modeling to virtual environments.
- CO4. Analyze AR system architecture, challenges, and evaluate AR development tools and techniques.
- CO5. Design and assess interaction techniques and audio rendering methods in VR/AR environments.

SYLLABUS:

UNIT – I

(09 Periods)

INTRODUCTION TO VIRTUAL REALITY (VR): Defining Virtual Reality, Key elements of virtual reality experience, Virtual Reality, Telepresence, Augmented Reality and Cyberspace. Bird's-Eye View: Hardware, Software, Human Physiology and Perception.

UNIT-II

(09 Periods)

Input Devices: (Trackers, Navigation, and Gesture Interfaces): Three-dimensional position trackers, navigation and manipulation, interfaces and gesture interfaces.

Output Devices: Graphics displays, sound displays & haptic feedback.

UNIT-III

(08 Periods)

Modeling: Geometric modeling, Kinematics modeling, Physical modeling, Behaviour modeling, Model management.

UNIT-IV

(10 Periods)

Augmented Reality (AR): Taxonomy, Technology and Features of Augmented Reality, AR Vs

VR, Challenges with AR, AR systems and functionality, Augmented Reality Methods, Visualization Techniques for Augmented Reality, Enhancing interactivity in AR Environments, Evaluating AR systems.

AR software development: AR software, Camera parameters and camera calibration, Marker-based augmented reality, AR Toolkit.

UNIT-V

(10 Periods)

Interaction & Audio: Interaction - Motor Programs and Remapping, Locomotion, Manipulation, Social Interaction. Audio -The Physics of Sound, the Physiology of Human Hearing, Auditory Perception, Auditory Rendering.

Interaction - Motor Programs and Remapping, Locomotion, Manipulation, Social Interaction. Audio -The Physics of Sound, the Physiology of Human Hearing, Auditory Perception, Auditory Rendering. (From Text Book2)

Total Periods: 46

Text books:

T1. Virtual Reality Technology, Second Edition, Gregory C. Burdea & Philippe Coiffet, John Wiley & Sons, Inc, 2017.

T2. Virtual Reality, Steven M. LaValle, Cambridge University Press, 2016.

Reference Books:

R1. Rajesh K. Maurya, Computer Graphics with Virtual Reality System, 3rd Edition, Wiley Publication, 2018.

R2. William R. Sherman and Alan B. Craig, Understanding Virtual Reality Interface, Application, and Design, 2nd Edition, Morgan Kaufmann Publishers, Elsevier, 2019.

R3. Grigore C. Burdea, Philippe Coiffet, Virtual Reality Technology, 2nd Edition, Wiley, 2017.
K.S. Hale and K. M. Stanney, Handbook on Virtual Environments, 2nd Edition, CRC Press, 2015.

Web Resources:

1. <http://vr.cs.uiuc.edu/vrbook.pdf>
2. <https://nptel.ac.in/courses/106/106/106106138/>

2305507	B.Tech. CSE, V-SEMESTER INTRODUCTION TO QUANTUM TECHNOLOGIES AND APPLICATIONS (Common to CE, EEE, ME, ECE, CSE, CSE(DS), CSE(AIML), AIML)	L	T	P	C
		3	0	0	3

Pre-Requisites: Engineering Physics

Course Outcomes:

On successful completion of the course, student will be able to

CO1. explain core quantum principles in a non-mathematical manner. CO2. compare classical and quantum information systems.

CO3. identify theoretical issues in building quantum computers. CO4.

discuss quantum communication and computing concepts.

CO5. recognize applications, industry trends, and career paths in quantum technology.

SYLLABUS:

UNIT-I: INTRODUCTION TO QUANTUM THEORY AND TECHNOLOGIES (08 Periods)

The transition from classical to quantum physics, Fundamental principles explained conceptually: Superposition, Entanglement, Uncertainty Principle, Wave-particle duality, Classical vs Quantum mechanics–theoretical comparison, Quantum states and measurement: nature of observation, Overview of quantum systems: electrons, photons, atoms, The concept of quantization: discrete energy levels, Why quantum? Strategic, scientific, and technological significance, A snapshot of quantum technologies: Computing, Communication, and Sensing, National and global quantum missions: India's Quantum Mission, EU, USA, China

UNIT-II: THEORETICAL STRUCTURE OF QUANTUM INFORMATION SYSTEMS

(08 Periods)

What is a qubit? Conceptual understanding using spin and polarization, Comparison: classical bits vs quantum bits, Quantum systems: trapped ions, superconducting circuits, photons (non-engineering view), Quantum coherence and decoherence – intuitive explanation, Theoretical concepts: Hilbert spaces, quantum states, operators – only interpreted in abstract, The role of entanglement and non-locality in systems, Quantum information vs classical information: principles and differences, Philosophical implications: randomness, determinism, and observer role

UNIT-III: BUILDING A QUANTUM COMPUTER – THEORETICAL CHALLENGES AND REQUIREMENTS

(10 Periods)

What is required to build a quantum computer (conceptual overview)?, Fragility of quantum systems: decoherence, noise, and control, Conditions for a functional quantum system: Isolation, Error management, Scalability, Stability,

Theoretical barriers: Why maintaining entanglement is difficult, Error correction as a theoretical necessity, Quantum hardware platforms (brief conceptual comparison), Superconducting circuits, Trapped ions, Photonics, Vision vs reality: what's working and what remains elusive, The role of quantum software in managing theoretical complexities

UNIT-IV: QUANTUM COMMUNICATION AND COMPUTING – THEORETICAL PERSPECTIVE

(09 Periods)

Quantum vs Classical Information, Basics of Quantum Communication, Quantum Key Distribution (QKD), Role of Entanglement in Communication, The Idea of the Quantum Internet – Secure Global Networking, Introduction to Quantum Computing, Quantum Parallelism (Many States at Once), Classical vs Quantum Gates, Challenges: Decoherence and Error Correction, Real-World Importance and Future Potential

UNIT-V: APPLICATIONS, USE CASES, AND THE QUANTUM FUTURE

(10 Periods)

Real-world application domains: Healthcare (drug discovery), Material science, Logistics and

optimization, Quantum sensing and precision timing, Industrial case studies: IBM, Google, Microsoft, PsiQuantum, Ethical, societal, and policy considerations, Challenges to adoption: cost, skills, standardization, Emerging careers in quantum: roles, skillsets, and preparation pathways, Educational and research landscape – India's opportunity in the global quantum race.

Total Periods: 45

Text Books:

- T1. Quantum Computation and Quantum Information, Michael A. Nielsen and Isaac L. Chuang, Cambridge University Press, 10th Anniversary, 2010.
- T2. Quantum Computing: A Gentle Introduction, Eleanor Rieffel and Wolfgang Polak, MIT Press, 2011.
- T3. Quantum Computing for Everyone, Chris Bernhardt, MIT Press, 2019.

Reference Books:

- R1. Quantum Computing Explained, David McMahon, Wiley, 2008.
- R2. An Introduction to Quantum Computing, Phillip Kaye, Raymond Laflamme, Michele Mosca, Oxford University Press, 2007.
- R3. Quantum Computing Since Democritus, Scott Aaronson, Cambridge University Press, 2013
- R4. Quantum Physics: A Beginner's Guide, Alastair I.M. Rae, Oneworld Publications, Revised Edition, 2005.
- R5. Quantum Computing: A Gentle Introduction, Eleanor G. Rieffel, Wolfgang H. Polak, MIT Press, 2011.
- R6. Quantum Mechanics: The Theoretical Minimum, Leonard Susskind, Art Friedman, Basic Books, 2014.
- R7. Quantum Enigma: Physics Encounters Consciousness, Bruce Rosenblum, Fred Kuttner, Oxford University Press, 2nd Edition, 2011.
- R8. Principles of Quantum Computation and Information, Volume I: Basic Concepts, Giuliano Benenti, Giulio Casati, Giuliano Strini, World Scientific Publishing, 2004.
- R9. Quantum Technologies and Industrial Applications: European Roadmap and Strategy Document, Quantum Flagship, K.B. Whaley et al., European Commission, 2020.
- R10. Department of Science & Technology (DST), Government of India, National Mission on Quantum Technologies & Applications – Official Reports and Whitepapers, MeitY/DST Publications, 2020 onward..

Web Resources:

- 1. IBM Quantum Experience and Qiskit Documentation (<https://qiskit.org/>).

2305551	B.Tech. CSE, V-SEMESTER ARTIFICIAL INTELLIGENCE LAB	L	T	P	C
		0	0	3	1.5

Pre-requisite: Knowledge in Computer Programming, Background in linear algebra, data structures and algorithms, and probability.

Course Outcomes:

On successful completion of the course, student will be able to

- CO1. Implement BFS, DFS, Best-First, A*, and AO* algorithms for pathfinding and optimization.
- CO2. Apply heuristic techniques to solve Water-Jug, 8-Puzzle, and 8-Queens problems.
- CO3. Develop intelligent agents using adversarial search with Alpha-Beta pruning and game-playing strategies
- CO4. Build and query knowledge-based systems using unification and rule-based reasoning
- CO5. Evaluate AI solutions using performance metrics like nodes, cutoffs, and optimality
- CO6. Make decisions as an individual or as team member to manage tasks and also engage in Independent and life-long learning with ability to adapt to new and technological changes

List of Exercises/ List of Experiments:

List of Experiments

1. Write a Program to Implement Breadth First Search using Python.
2. Write a program to implement Best First Searching Algorithm
3. Write a Program to Implement Depth First Search using Python.
4. Write a program to implement the Heuristic Search
5. Write a python program to implement A* and AO* algorithm. (Ex: find the shortest path)
6. Write a Program to Implement Water-Jug problem using Python.
7. Write a Program to Implement Alpha-Beta Pruning using Python.
8. Write a Program to implement 8-Queens Problem using Python.
9. Write a program to schedule a meeting among a 5 busy people using Default Reasoning the output should give the time, place and day of the meeting.
10. Write a program to implement the Unification algorithm
11. Develop a knowledge base system consisting of facts and rules about some specialized knowledge domain
12. Write a program to implement 8 puzzle programs using different heuristics. Using it play the game Tic-Tac-Toe at the end the game the program should display the no. of nodes generated, cutoff values at each stage in the form of a table.

Text Books:

- T1. PrateekJoshi, Artificial Intelligence with Python, Packt Publishing, 2017.
- T2. Xiao, Perry. Artificial intelligence programming with Python: from zero to hero. John Wiley & Sons, 2022.

Reference Books:

- R1. Stuart J. Russell and Peter Norvig, Artificial Intelligence A Modern Approach, Fourth Edition, Pearson, 2020
- R2. Martin C. Brown (Author), —Python: The Complete Reference McGraw Hill Education, Fourth edition, 2018
- R3. R. Nageswara Rao , —Core Python Programming McGraw Hill Education India Pvt Ltd 2018.

Online Learning Resources:

1. https://onlinecourses.nptel.ac.in/noc19_cs40/preview
2. https://onlinecourses.nptel.ac.in/noc19_cs41/preview

2305552	B.Tech. CSE, V-SEMESTER COMPUTER NETWORKS AND INTERNET PROTOCOLS LAB	L	T	P	C
		0	0	3	15

Pre-Requisites: -

Course Outcomes:

On successful completion of the course, student will be able to

- CO1.** Implement and analyze various data link layer framing techniques such as character stuffing, bit stuffing, and cyclic redundancy check (CRC) codes.
- CO2.** Develop and simulate flow control and error recovery mechanisms like sliding window and Go-Back-N protocols.
- CO3.** Apply graph algorithms like Dijkstra's and Distance Vector to compute shortest paths and routing tables in a network.
- CO4.** Design and evaluate broadcast trees and frame sorting mechanisms to optimize data transmission in subnets and buffer systems.
- CO5.** Demonstrate the implementation of cryptographic techniques for securing data through encryption and decryption.
- CO6.** Simulate and evaluate network performance under various conditions using NS2, including packet loss, congestion, throughput, and data rate comparisons.

List of Experiments:

1. Implement the data link layer framing methods such as character, character-stuffing and bit stuffing.
2. Write a program to compute CRC code for the polynomials CRC-12, CRC-16 and CRC CCIP
3. Develop a simple data link layer that performs the flow control using the sliding window protocol, and loss recovery using the Go-Back-N mechanism.
4. Implement Dijkstra's algorithm to compute the shortest path through a network
5. Take an example subnet of hosts and obtain a broadcast tree for the subnet.
6. Implement distance vector routing algorithm for obtaining routing tables at each node.
7. Implement data encryption and data decryption
8. Write a program for congestion control using Leaky bucket algorithm.
9. Write a program for frame sorting technique used in buffers.
10. Programs using Wireshark
 - i. Packet Capture Using Wire shark
 - ii. Starting Wire shark
 - iii. Viewing Captured Traffic
 - iv. Analysis and Statistics & Filters.
11. How to run Nmap scan
12. Operating System Detection using Nmap
13. Do the following using NS2 Simulator
 - i. NS2 Simulator-Introduction

- ii. Simulate to Find the Number of Packets Dropped
- iii. Simulate to Find the Number of Packets Dropped by TCP/UDP
- iv. Simulate to Find the Number of Packets Dropped due to Congestion
- v. Simulate to Compare Data Rate& Throughput.
- vi. Simulate to Plot Congestion for Different Source/Destination
- vii. Simulate to Determine the Performance with respect to transmission of Packets

Text Books:

- T1. Andrew S.Tanenbaum, David j.wetherall, Computer Networks, 6th Edition, PEARSON.
- T2. James F.Kurose, Keith W. Ross, Computer Networking: A Top-Down 6th edition, Pearson, 2019.
- T3. Computer Networks: A Systems Approach-Bruce Davie, VMware-Larry Peterson, Princeton University-2019.

Reference Books:

- R1. Computer Networks–B. K. MathanNagan, T. Mahalakshmi- Charulatha Publications PrivateLimited-2019.
- R2. Computer Networks-Dr.Amol V. Dhumane Nitin N. Sakhare-Nirali Prakashan Publishers-2024.
- R3. Data Communications and Networking with TCPIP Protocol Suite-Behrouz A. Forouzan- McGraw Hill-6th Edition

2305553	B.Tech. CSE, V-SEMESTER TINKERING LAB (Common to CSE, CSE (AIML), CSE (DS), AI&ML)	L	T	P	C
		0	0	2	1

Pre-Requisites: Computer Networks, Internet of Things.

Course Outcomes:

On successful completion of the course, student will be able to

- CO1. Demonstrate the ability to design and build basic electronic circuits using breadboards.
- CO2. Implement sensor-based automation systems using Arduino and basic electronic components.
- CO3. Develop and test microcontroller-based applications using simulation tools and physical hardware.
- CO4. Apply appropriate Tools and Techniques to understand and analyze the problems following professional ethics with focus on societal and environmental aspects.
- CO5. Work as a team and communicate results in an effective way.
- CO6. Make decisions as an individual or as team member to manage tasks and also engage in independent and life-long learning with ability to adapt to new and technological changes.

List of Exercises/ List of Experiments:

Minimum Ten experiments are to be conducted.

1. Make your own parallel and series circuits using breadboard for any application of your choice.
2. Demonstrate a traffic light circuit using breadboard.
3. Build and demonstrate automatic Street Light using LDR.
4. Simulate the Arduino LED blinking activity in Tinkercad.
5. Build and demonstrate an Arduino LED blinking activity using Arduino IDE.
6. Analog Output Generation Using PWM in Arduino.
7. Digital Input Using Push Button with Arduino
8. Displaying messages on LCD
9. Obstacle Detection System using Ultrasonic Sensor
10. Touch-Activated LED Control Using Arduino
11. Temperature and Humidity Monitoring Using DHT11 Sensor
12. Motion Detection System Using PIR Sensor
13. Fire Alert System Using Flame Sensor

14. Bluetooth-Based Door Lock System
15. Smart Dustbin Using Ultrasonic Sensor
16. RFID-Based Attendance System
17. Temperature-Controlled Chemical Reactor
18. Facial Recognition Dustbin
19. Contactless Attendance via Face Recognition

Text books:

- T1. Getting Started with Arduino, 3rd Edition, Massimo Banzi and Michael Shiloh
- T2. Adrian McEwen, Hakin Cassimally “Designing the Internet of Things” Wiley India.
- T3. Arshdeep Bahga, Vijay Madisetti “ Internet of Things(A hands on approach)” 1st Edition, VPI publications, 2014.

Reference Books:

- R1. Raj Kamal, “Internet of Things”, McGraw Hill, 1st Edition, 2016.
- R2. Internet of Things, Surya Durbha, Jyothi Joglekar, Oxford Higher Education.
- R3. The Internet of Things, Michael Miller, Pearson.
- R4. The Internet of Things, Samuel Greengard, The MIT Press Ltd.

Web Resources:

1. <https://aim.gov.in/pdf/equipment-manual-pdf.pdf>
2. <https://atl.aim.gov.in/ATL-Equipment-Manual/>
3. <https://aim.gov.in/pdf/Level-1.pdf>
4. <https://aim.gov.in/pdf/Level-2.pdf>
5. <https://aim.gov.in/pdf/Level-3.pdf>

B.Tech. (R23UG) Course Structure

VI-SEMESTER

[illegible]

B.Tech (R23UG) VI Semester Syllabus

2333401	B.Tech., V-SEMESTER MACHINE LEARNING	L	T	P	C
		3	0	0	3

Pre-Requisites: Linear Algebra and Calculus, Vector Calculus, Statistical methods for Data Science, Python Programming.

Course Outcomes:

On successful completion of the course, student will be able to

- CO1.** Analyze various machine learning paradigms, data types, and the complete ML workflow to select appropriate models for given problems.
- CO2.** Apply proximity and distance measures to implement and evaluate nearest neighbor classifiers and regression algorithms on diverse datasets.
- CO3.** Analyze decision tree and Bayesian classifiers, including their trade-offs, and evaluate their performance for classification and regression tasks.
- CO4.** Apply linear discriminant techniques and kernel-based methods to design classifiers, and create multilayer perceptron models using backpropagation.
- CO5.** Evaluate different clustering algorithms and design clustering solutions using hard, soft, fuzzy, and spectral clustering methods on complex datasets.

SYLLABUS:

UNIT- I: INTRODUCTION TO MACHINE LEARNING (08 Periods)

Evolution of Machine Learning, Paradigms for ML, Learning by Rote, Learning by Induction, Reinforcement Learning, Types of Data, Matching, Stages in Machine Learning, Data Acquisition, Feature Engineering, Data Representation, Model Selection, Model Learning, Model Evaluation, Model Prediction, Search and Learning, Data Sets.

UNIT- II: NEAREST NEIGHBOR-BASED MODELS (10 Periods)

Introduction to Proximity Measures, Distance Measures, Non-Metric Similarity Functions, Proximity between Binary Patterns, Different Classification Algorithms Based on the Distance Measures, K-Nearest Neighbor Classifier, Radius Distance Nearest Neighbor Algorithm, KNN Regression, Performance of Classifiers, Performance of Regression Algorithms.

UNIT- III: MODELS BASED ON DECISION TREES (09 Periods)

Decision Trees for Classification, Impurity Measures, Properties, Regression Based on Decision Trees, Bias–Variance Trade-off, Random Forests for Classification and Regression.

The Bayes Classifier: Introduction to the Bayes Classifier, Bayes' Rule and Inference, The Bayes Classifier and its Optimality, Multi-Class Classification | Class Conditional Independence and Naive Bayes Classifier (NBC).

UNIT- IV: LINEAR DISCRIMINANTS FOR MACHINE LEARNING (09 Periods)

Introduction to Linear Discriminants, Linear Discriminants for Classification, Perceptron Classifier, Perceptron Learning Algorithm, Support Vector Machines, Linearly Non-Separable Case, Non-linear SVM, Kernel Trick, Logistic Regression, Linear Regression, Multi-Layer Perceptrons (MLPs), Backpropagation for Training an MLP.

UNIT- V: CLUSTERING

(09 Periods)

Introduction to Clustering, Partitioning of Data, Matrix Factorization | Clustering of Patterns, Divisive Clustering, Agglomerative Clustering, Partitional Clustering, K-Means Clustering, Soft Partitioning, Soft Clustering, Fuzzy C-Means Clustering, Rough Clustering, Rough K-Means Clustering Algorithm, Expectation Maximization-Based Clustering, Spectral Clustering.

Total Periods: 45

Text Books:

T1. Machine Learning Theory and Practice, M N Murthy, V S Ananthanarayana, Universities Press (India), 2024

Reference Books:

R1. Machine Learning, Tom M. Mitchell, McGraw-Hill Publication, 2017.

R2. Machine Learning in Action, Peter Harrington, DreamTech.

R3. Introduction to Data Mining, Pang-Ning Tan, Michel Stenbach, Vipin Kumar, 7th Edition, 2019.

Web Resources:

1. <https://nptel.ac.in/courses/106106139>
2. <https://nptel.ac.in/courses/106105152>
3. <https://www.deeplearning.ai/machine-learning-yearning/>
4. <https://www.cse.huji.ac.il/~shais/UnderstandingMachineLearning/index.html>

2305601	B.Tech. CSE, VI-SEMESTER CRYPTOGRAPHY & NETWORK SECURITY	L	T	P	C
		3	0	0	3

Pre-Requisites: Computer Networks , Discrete mathematics , Number Theory , Introduction to programming and Data Structures and Algorithms.

Course Outcomes:

On successful completion of the course, student will be able to

- CO1.** Identify information security goals, classical encryption techniques and acquire fundamental knowledge on the concepts off in it fields and number theory.
- CO2.** Compare and apply different encryption and decryption techniques to solve problems related to confidentiality and authentication.
- CO3.** Apply the knowledge of cryptographic checksums and evaluate the performance of different message digest algorithms for verifying the integrity of varying message sizes.
- CO4.** Demonstrate the ability to apply **user authentication principles** including **Kerberos** for secure authentication.
- CO5.** Gain proficiency in securing web communications using **TLS** and **HTTPS**, manage secure remote access with **SSH**, and design **firewall policies**.

SYLLABUS:

UNIT-I

(09 Periods)

Computer and Network Security Concepts: Computer Security Concepts, The OSI Security Architecture, Security Attacks, Security Services, Security Mechanisms, A Model for Network Security, **Classical Encryption Techniques:** Symmetric Cipher Model, Substitution Techniques, Transposition Techniques, Steganography, Block Ciphers: Traditional Block Cipher Structure, The Data Encryption Standard, Advanced Encryption Standard: AES Structure, AES Transformation Functions

UNIT-II

(09 Periods)

Number Theory:

The Euclidean Algorithm, Modular Arithmetic, Fermat's and Euler's Theorems, The Chinese Remainder Theorem, Discrete Logarithms, Finite Fields: Finite Fields of the Form $GF(p)$, Finite Fields of the Form $GF(2^n)$.**Public Key Cryptography:** Principles, Public Key Cryptography Algorithms, RSA Algorithm, Diffie Hellman Key Exchange, Elliptic Curve Cryptography.

UNIT-III

(09 Periods)

Cryptographic Hash Functions: Application of Cryp to graphic Hash Functions, Requirements & Security, Secure Hash Algorithm, Message Authentication Functions, Requirements & Security, HMAC &CMAC. **Digital Signatures:** NIST Digital Signature Algorithm, Distribution of Public Keys, X.509 Certificates, Public- Key Infrastructure

UNIT-IV

(09 Periods)

User Authentication: Remote User Authentication Principles, Kerberos. Electronic Mail Security: Pretty Good Privacy (PGP) And S/MIME.

IP Security: IP Security Overview, IP Security Policy, Encapsulating Security Payload, Combining Security Associations, Internet Key Exchange.

UNIT-V

(08 Periods)

Transport Level Security: Web Security Requirements, Transport Layer Security (TLS), HTTPS, Secure Shell (SSH)

Fire walls: Fire wall Characteristics and Access Policy, Types of Fire walls, Fire wall Location and Configurations.

Total Periods: 44

Text books:

T1. Cryptography and Network Security – William Stallings, Pearson Education, 8th Edition.

T2. Cryptography, Network Security and Cyber Laws–Bernard Menezes, Cengage Learning, 2010 edition.

Reference Books:

R1. Cryptography and Network Security-Behrouz A Forouzan, Debdeep Mukhopadhyaya, McGraw Hill, 3rd Edition, 2015.

R2. Network Security Illustrated, Jason Albanese and Wes Sonnenreich, MGH Publishers, 2003.

Web Resources:

1. <https://nptel.ac.in/courses/106/105/106105031/lecture>

2. <https://nptel.ac.in/courses/106/105/106105162/lecture> by Dr. Sourav Mukhopadhyay IIT Kharagpur [Video Lecture]

3. <https://www.mitel.com/articles/web-communication-cryptography-and-network-security-web-articles-by-mitel-power-connections>

2305602	B. Tech CSE, VI SEMESTER SOFTWARE TESTING METHODOLOGIES (Professional Elective-II)	L	T	P	C
		3	0	0	3

Pre-Requisites: Software Engineering

Course Outcomes:

On successful completion of the course, student will be able to

CO1. Know the basic concepts of software testing and its essentials.

CO2. Identify and analyze various types of bugs in software systems.

CO3. Apply corrective measures after understanding the consequences of identified bugs.

CO4. Use of program's control flow as a structural model is the corner stone of testing.

CO5. Performing functional testing using control flow and transaction flow graphs.

SYLLABUS:

UNIT-I: INTRODUCTION TO TESTING

(10 Periods)

Introduction: Purpose of testing, Dichotomies, model for testing, consequences of bugs, taxonomy of bugs, Flow graphs and Path testing: Basics concepts of path testing, predicates, path predicates and achievable paths, path sensitizing, path instrumentation, application of path testing.

UNIT-II: INTRODUCTION TO TRANSACTION FLOW TESTING

(10 Periods)

Transaction Flow Testing: transaction flows, transaction flow testing techniques. Dataflow testing: Basics of dataflow testing, strategies in dataflow testing, application of dataflow testing.

UNIT-III: INTRODUCTION TO DOMAIN TESTING

(09 Periods)

Domain Testing: domains and paths, Nice & ugly domains, domain testing, domains and interfaces testing, domain and interface testing, domains and testability.

UNIT-IV: PATHS, PATH PRODUCTS AND REGULAR EXPRESSIONS

(08 Periods)

Paths, Path products and Regular expressions: path products & path expression, reduction procedure, applications, regular expressions & flow anomaly detection. Logic Based Testing: over view, decision tables, path expressions, kv charts, specifications.

UNIT-V: STATE, STATE GRAPHS AND TRANSITION TESTING

(08 Periods)

State, State Graphs and Transition testing: state graphs, good & bad state graphs, state testing, Testability tips. Graph Matrices and Application: Motivational overview, matrix of graph, relations, power of a matrix, node reduction algorithm, building tools

Text Books:

- T1. Software Testing techniques – Boris Beizer, Dreamtech, second edition.
- T2. Software Testing Tools – Dr. K.V.K.K. Prasad, Dreamtech.

References Books:

- R1. The craft of software testing – Brian Marick, Pearson Education.
- R2. Software Testing Techniques – SPD(Oreille)
- R3. Software Testing in the Real World – Edward Kit, Pearson.
- R4. Effective methods of Software Testing, Perry, John Wiley.
- R5. Art of Software Testing – Meyers, John Wiley.

Web Resources:

- 1. https://onlinecourses.swayam2.ac.in/cec25_cs08/preview
- 2. <https://www.coursera.org/learn/introduction-to-software-engineering>

2305603	B. Tech CSE, VI SEMESTER DATA WAREHOUSING AND DATA MINING	L	T	P	C
		3	0	0	3

Course Outcomes:

On successful completion of the course, student will be able to

- CO1.** Design a Data warehouse system and perform business analysis with OLAP tools.
- CO2.** Apply suitable pre-processing and visualization techniques for data analysis.
- CO3.** Apply frequent pattern and association rule mining techniques for data analysis.
- CO4.** Design appropriate classification and clustering techniques for data analysis.
- CO5.** Infer knowledge from raw data.

SYLLABUS:

UNIT- I: (09 Periods)

Basic Concepts – Data Warehousing Components – Building a Data Warehouse – Database Architectures for Parallel Processing – Parallel DBMS Vendors – Multidimensional Data Model – Data Warehouse Schemas for Decision Support, Concept Hierarchies -Characteristics of OLAP Systems – Typical OLAP Operations, OLAP and OLTP.

UNIT- II: (09 Periods)

Introduction to Data Mining Systems – Knowledge Discovery Process – Data Mining Techniques – Issues – applications- Data Objects and attribute types, Statistical description of data, Data Preprocessing – Cleaning, Integration, Reduction, Transformation and discretization, Data Visualization, Data similarity and dissimilarity measures.

UNIT- III: (08 Periods)

Mining Frequent Patterns, Associations and Correlations – Mining Methods- Pattern Evaluation Method – Pattern Mining in Multilevel, Multi Dimensional Space – Constraint Based Frequent Pattern Mining, Classification using Frequent Patterns.

UNIT- IV: (09 Periods)

Decision Tree Induction – Bayesian Classification – Rule Based Classification – Classification by Back Propagation – Support Vector Machines — Lazy Learners – Model Evaluation and Selection- Techniques to improve Classification Accuracy. Clustering Techniques – Cluster analysis- Partitioning Methods – Hierarchical Methods – Density Based Methods – Grid Based Methods – Evaluation of clustering – Clustering high dimensional data- Clustering with constraints, Outlier analysis-outlier detection methods.

UNIT- V: WEKA TOOL (08 Periods)

Datasets – Introduction, Iris plants database, Breast cancer database, Auto imports database – Introduction to WEKA, The Explorer – Getting started, Exploring the explorer, Learning algorithms, Clustering algorithms, Association–rule learners.

Total Periods: 43

Text Book:

- T1. Jiawei Han and Micheline Kamber, —Data Mining Concepts and Techniques, Third Edition, Elsevier, 2012.

Reference Books:

- R1. Alex Berson and Stephen J. Smith, —Data Warehousing, Data Mining & OLAP, Tata McGraw – Hill Edition, 35th Reprint 2016.
- R2. K.P. Soman, Shyam Diwakar and V. Ajay, —Insight into Data Mining Theory and Practice, Eastern Economy Edition, Prentice Hall of India, 2006.
- R3. Ian H.Witten and Eibe Frank, —Data Mining: Practical Machine Learning Tools and Techniques, Elsevier, Second Edition.

2305604	B. Tech CSE, VI SEMESTER EMBEDDED SYSTEMS DESIGN (Professional Elective –II)	L	T	P	C
		3	0	0	3

Pre-Requisites: NIL

Course Outcomes:

On successful completion of the course, student will be able to

- CO1.** Classify embedded systems based on their purpose, generation, and complexity.
- CO2.** Identify and select appropriate hardware components for an embedded system design.
- CO3.** Differentiate and implement various communication protocols like I2C, SPI, and CAN.
- CO4.** Develop firmware using assembly and high-level programming languages.
- CO5.** Analyze and apply RTOS-based task scheduling and synchronization techniques

SYLLABUS:

UNIT-I: Introduction to Embedded Systems

(9 Periods)

History of embedded systems, Classification of embedded systems based on generation and complexity, Purpose of embedded systems, The embedded system design process-requirements, specification, architecture design, designing hardware and software, components, system integration, Applications of embedded systems, and characteristics of embedded systems.

UNIT-II: Typical Embedded System

(9 Periods)

Core of the embedded system-general purpose and domain specific processors, ASICs, PLDs, COTs; Memory-ROM, RAM, memory according to the type of interface, memory shadowing, memory selection for embedded systems, Sensors, actuators, I/O components: seven segment LED, relay, piezo buzzer, push button switch, other subsystems: reset circuit, brownout protection circuit, oscillator circuit real time clock, watchdog timer.

UNIT-III: Communication Interface

(9 Periods)

Onboard communication interfaces-I2C, SPI, CAN, parallel interface; External communication interfaces-RS232 and RS485, USB, infrared, Bluetooth, Wi-Fi, ZigBe, GPRS, GSM.

UNIT-IV: Embedded Firmware Design and Development

(8 Periods)

Embedded firmware design approaches-super loop-based approach, operating system-based approach; embedded firmware development languages-assembly language-based development, high level language-based development.

UNIT-V: RTOS based Embedded System Design

(9 Periods)

Operating system basics, types of operating systems, tasks, process and threads, multiprocessing and multitasking, task scheduling: non-pre-emptive and pre-emptive scheduling; task communication shared memory, message passing, Remote Procedure Call and Sockets, Task Synchronization: Task Communication/ Synchronization Issues, Task Synchronization Techniques.

Total Periods: 44

Text Books:

T1.Embedded Systems- An integrated approach - Lyla b das, Pearson education 2012.

T2.Computers as Components –Wayne Wolf, Morgan Kaufmann (second edition).

References Books:

R1. Embedded System Design -Frank Vahid, Tony Grivargis, John Wiley.

R2. .Introduction to Embedded Systems - Shibu KV, Mc Graw Hill Education.

R3. Embedded Systems – Raj Kamal, TMH.

2332505	B.Tech. CSE, VI-SEMESTER INTERNET OF THINGS (Professional Elective –II)	L	T	P	C
		3	0	0	3

Pre-Requisites: Computer Networks

Course Outcomes:

On successful completion of the course, student will be able to

CO1. Understand general concepts of Internet of Things.

CO2. Apply design concept to IoT solutions

CO3. Analyse various M2M and IoT architectures

CO4. Evaluate design issues in IoT applications

CO5. Create IoT solutions using sensors, actuators and Devices

SYLLABUS:

UNIT-I: INTRODUCTION TO IOT (08 Periods)

Definition and Characteristics of IoT, physical design of IoT, IoT protocols, IoT communication models, IoT Communication APIs, Communication protocols, Embedded Systems, IoT Levels and Templates.

UNIT-II: PROTOTYPING IOT OBJECTS USING MICROPROCESSOR /MICROCONTROLLER (09 Periods)

Working principles of sensors and actuators, setting up the board – Programming for IoT, Reading from Sensors, Communication: communication through Bluetooth, Wi-Fi.

UNIT-III: IOT ARCHITECTURE AND PROTOCOL (09 Periods)

Architecture Reference Model- Introduction, Reference Model and architecture, IoT reference Model, Protocols- 6LowPAN, RPL, CoAP, MQTT, IoT frameworks- Thing Speak.

UNIT-IV: DEVICE DISCOVERY AND CLOUD SERVICES FOR IOT (09 Periods)

Device discovery capabilities- Registering a device, Deregister a device, Introduction to Cloud Storage models and communication APIs Web-Server, Web server for IoT.

UNIT-V: UAV IoT (10 Periods)

Introduction to Unmanned Aerial Vehicles/Drones, Drone Types. Applications: Defense, Civil, Environmental Monitoring; UAV elements and sensors- Arms, motors, Electronic Speed Controller(ESC), GPS, IMU, Ultra sonic sensors; UAV Software –Arudpilot, Mission Planner, Internet of Drones(IoD)- Case study FlytBase.

Total Periods: 45

Text Books:

- T1. Internet of Things (A Hands-on-Approach), Vijay Madisetti and ArshdeepBahga, VPT, 1st Edition, 2014.
- T2. Handbook of unmanned aerial vehicles, K Valavanis; George J Vachtsevanos, New York, Springer, Boston, Massachusetts : Credo Reference, 2016.

Reference Books:

- R1. From Machine-to-Machine to the Internet of Things: Introduction to a New Age of Intelligence, Jan Holler, VlasiosTsiatsis, Catherine Mulligan, Stefan Avesand, Stamatis Karnouskos, David Boyle, Academic Press, 1st Edition, 2014.
- R2. Internet of Things: A Hands-On Approach, ArshdeepBahga, Vijay Madisetti, Universities Press, 2014.
- R3. The Internet of Things, Enabling technologies and use cases, Pethuru Raj, Anupama C. Raman, CRC Press.
- R4. Rethinking the Internet of Things: A Scalable Approach to Connecting Everything, Francis daCosta, , Apress Publications, 1st Edition, 2013.
- R5. Getting Started with the Internet of Things, CunoPfister, O'Reilly Media, 2011.

Web Resources:

- 1. <https://www.arduino.cc/>
- 2. <https://www.raspberrypi.org/>
- 3. <https://nptel.ac.in/courses/106105166/5>
- 4. <https://nptel.ac.in/courses/108108098/4>

2305605	III B. Tech CSE, VI SEMESTER SOFTWARE PROJECT MANAGEMENT (Professional Elective-III)	L	T	P	C
		3	0	0	3

Pre-Requisites: Software Engineering

Course Outcomes:

On successful completion of the course, student will be able to

C01. Describe the fundamentals of Project Management.

C02. Recognize and use Project Scheduling Techniques

C03. Familiarize with Project Control Mechanisms.

C04. Understand Team Management.

C05. Recognize the importance of Project Documentation and Evaluation.

SYLLABUS:

UNIT-I: (12 Periods)

Conventional Software Management: The water fall model, conventional software Management
Performance Evolution of Software Economics: software Economics. Pragmatic Software Cost Estimation
Improving Software Economics: Reducing Software Product Size, Improving Software Processes,
Improving Team Effectiveness, Improving Automation.

UNIT-II: (8 Periods)

The old way and the new: The principles of conventional software Engineering, principles of modern software management, transitioning to an iterative process. Lifecycle phases: Engineering and production stages, inception, Elaboration, construction, transition phases.

Artifacts of the process: The artifact sets, Management artifacts, Engineering artifacts, programmatic artifacts.

UNIT-III: (10 Periods)

Work Flows of the process: Software process workflows, Inter Trans work flows. Check points of the Process: Major Milestones, Minor Milestones, Periodic status assessments. Iterative Process Planning: work breakdown structures, planning guidelines, cost and schedule estimating, Iteration planning process, Pragmatic planning.

UNIT-IV: (08 Periods)

Process Automation: Automation Building Blocks, The Project Environment. Project Control and Process instrumentation: The seven core Metrics, Management indicators, quality indicators Tailoring the Process: Process discriminants. Managing people and organizing teams.

UNIT-V: (07 Periods)

Project Organizations and Responsibilities: Line - of-Business Organizations, Project Organizations, evolution of Organizations. Future Software Project Management: modern Project Profiles, Next generation Software economics, modern process transitions.

Case Study: The Command Center Processing and Display System-Replacement(CCPDS-R)

Total Periods: --45

Text Books:

T1. Software Project Management, Walker Royce, Pearson Education, 2012.

T2. Bob Hughes, Mike Cotterell and Rajib Mall—Software Project Management, 6th Edition, Mc Graw Hill Edition, 2017.

Reference Books:

R1. Pankaj Jalote,—Software Project Management in practice, 5th Edition, Pearson Education, 2017.

R2. Murali K. Chemuturi, Thomas M. Cagley Jr. Mastering Software Project Management: Best Practices, Tools and Techniques, J. Ross Publishing, 2010.

R3. Sanjay Mohapatra,—Software Project Management, Cengage Learning, 2011

Web Resources:

<http://nptel.ac.in/courses/106101061/29>

2332601	B.Tech. CSE, VI-SEMESTER NATURAL LANGUAGE PROCESSING (Professional Elective-III)	L	T	P	C
		3	0	0	3

Pre-Requisites: Data Mining

Course Outcomes:

On successful completion of the course, student will be able to

- CO1.** Understand the various NLP Applications and Organization of Natural language, able to learn and implement realistic applications using Python.
- CO2.** Apply the various Parsing techniques, Bayes Rule, Shannongame, Entropy and Cross Entropy
- CO3.** Understand the fundamentals of CFG and parsers and mechanism in ATN's.
- CO4.** Apply Semantic Interpretation and Language Modelling.
- CO5.** Apply the Machine Translation and multilingual Information Retrieval systems and Automatic Summarization

SYLLABUS:

UNIT-I: INTRODUCTION TO NATURAL LANGUAGE (07 Periods)

The Study of Language, Applications of NLP, Evaluating Language Understanding Systems, Different Levels of Language Analysis, Representations and Understanding, Organization of Natural language Understanding Systems, Linguistic Background: An outline of English Syntax.

UNIT-II: GRAMMARS AND PARSING (09 Periods)

Grammars and Parsing – Top – Down and Bottom-Up Parsers, Transition Network Grammars, Feature Systems and Augmented Grammars, Morphological Analysis and the Lexicon, Parsing with Features, Augmented Transition Networks, Bayes Rule, Shannongame, Entropy and Cross Entropy.

UNIT-III: GRAMMARS FOR NATURAL LANGUAGE (06 Periods)

Grammars for Natural Language, Movement Phenomenon Language, Gap Threading, Human Preferences in Parsing, Shift Reduce Parsers, Deterministic Parsers.

UNIT-IV: SEMANTIC INTERPRETATION (11 Periods)

Semantic & Logical form, Word senses & ambiguity, The basic logical form language, Encoding ambiguity in the logical Form, Verbs & States in logical form, The microroles, Speech acts & embedded sentences, Defining semantics structure model theory.

Language Modeling

Introduction, nGram Models, Language model Evaluation, Parameter Estimation, Language Model Adaptation, Types of Language Models, Language Specific Modelling Problems, Multilingual and Cross lingual Language Modelling.

UNIT-V: Machine Translation**(12 Periods)**

Survey: Introduction, Problems of Machine Translation, Is Machine Translation Possible, Brief History, Possible Approaches, Current Status. Anusarakaor Language Accessor: Background, Cutting the GordianKnot, The Problem, Structure of Anusaraka System, User Interface, Linguistic Area, Giving up Agreementin Anusarsaka Output, Language Bridges.

Multilingual Information Retrieval

Introduction, Document Pre-processing, Monolingual Information Retrieval, CLIR, MLIR, Evaluation in Information Retrieval, Tools, Software and Resources.

Multilingual Automatic Summarization: Introduction, Approach esto Summarization, Evaluation, How to Builda Summarizer, Competitions and Datasets.

Total Periods: 45**Text Books:**

- T1. James Allen, Natural Language Understanding, 2nd Edition, 2003, Pearson Education.
- T2. Multilingual Natural Language Processing Applications: From Theory To Practice- Daniel M.Bikel and ImedZitouni, Pearson Publications.
- T3. Natural Language Processing, A paninian perspective, Akshar Bharathi, Vineetchaitanya, Prentice – Hall of India.

Reference Books:

- R1. Charniack, Eugene, Statistical Language Learning, MITPress,1993.
- R2. Jurafsky, Dan and Martin, James, Speech and Language Processing, 2nd Edition, Prentice Hall,2008.
- R3. Manning, Christopher and Hen rich, Schutze, Foundations of Statistical Natural Language Processing, MIT Press,1999.

Web Resources:

- 1. <https://nptel.ac.in/courses/106/105/106105158/>
- 2. <http://www.nptelvideos.in/2012/11/natural-language-processing.html>.

2305606	B.Tech VI Semester DISTRIBUTED OPERATING SYSTEMS (Professional Elective-III)	L	T	P	C
		3	0	0	3

Pre-Requisites: Operating System

Course Outcomes

CO1: Understand the design approaches of advanced operating systems

CO2: Analyze the design issues of distributed operating systems.

CO3: Evaluate design issues of multi-processor operating systems.

CO4: Identify the requirements Distributed File System and Distributed Shared Memory.

CO5: Formulate the solutions to schedule the real time applications.

SYLLABUS:

UNIT - I: Architectures of Distributed Systems (08 Periods)

System Architecture Types, Distributed Operating Systems, Issues in Distributed Operating Systems, Communication Primitives. Theoretical Foundations: Inherent Limitations of a Distributed System, Lamport's Logical Clocks, Vector Clocks, Causal Ordering of Messages, Termination Detection.

UNIT – II: Distributed Mutual Exclusion (08 Periods)

The Classification of Mutual Exclusion Algorithms, Non-Token –Based Algorithms: Lamport's Algorithm, The Ricart-Agrawala Algorithm, Maekawa's Algorithm, Token- Based Algorithms: Suzuki-Kasami's Broadcast Algorithm, Singhal's Heuristic Algorithm, Raymond's Heuristic Algorithm.

UNIT – III: Distributed Deadlock Detection (09 periods)

Preliminaries, Deadlock Handling Strategies in Distributed Systems, Issues in Deadlock Detection and Resolution, Control Organizations for Distributed Deadlock Detection, Centralized- Deadlock – Detection Algorithms, Distributed Deadlock Detection Algorithms, Hierarchical Deadlock Detection Algorithms

UNIT – IV: Multiprocessor System Architectures (08 periods)

Introduction, Motivation for multiprocessor Systems, Basic Multiprocessor System Architectures Multi Processor Operating Systems: Introduction, Structures of Multiprocessor Operating Systems, Operating Design Issues, Threads, Process Synchronization, Processor Scheduling. Distributed File Systems: Architecture, Mechanisms for Building Distributed File Systems, Design Issues

UNIT – V: Distributed Scheduling**(09 periods)**

Issues in Load Distributing, Components of a Load Distributed Algorithm, Stability, Load Distributing Algorithms, Requirements for Load Distributing, Task Migration, Issues in task Migration Distributed Shared Memory: Architecture and Motivation, Algorithms for Implementing DSM, Memory Coherence, Coherence Protocols, Design Issues

TEXT BOOK:

T1. Advanced Concepts in Operating Systems, Mukesh Singhal, Niranjana G. Shivaratri, Tata Mc Graw- Hill Edition 2001

REFERENCE BOOK:

R1. Distributed Systems: Andrew S. Tanenbaum, Maarten Van Steen, Pearson Prentice Hall, Edition – 2, 2007

2305607	B.Tech. CSE, VI-SEMESTER METAVERSE (PROFESSIONAL ELECTIVE –III)	L	T	P	C
		3	0	0	3

Pre-Requisites: NIL

Course Outcomes:

On successful completion of the course, student will be able to

- CO1. Understand the characteristics, and interdisciplinary nature of the Metaverse, the opportunities and risks it presents.
- CO2. Analyze Metaverse layers, the technologies used in creating them, as well as design theories and practices for Metaverse.
- CO3. Examine and discuss Metaverse platforms, applications and the latest technological developments in this area.
- CO4. Identify cyber security issues, understand cybercrime, and discuss the open challenges.
- CO5. Building Metaverse Applications

SYLLABUS:

UNIT-I:

(12 Periods)

Metaverse fundamentals: Metaverse evolution, Metaverse importance and characteristics, the interdisciplinary nature of the Metaverse, Metaverse opportunities and risks, Computer-mediated communication (social presence theory, social information processing theory, media richness theory, cyborg theory), Avatar-mediated communication.

UNIT-II:

(08 Periods)

The seven layers of Metaverse: Experience Discovery, Creator economy, Spatial computing, Decentralization, Human interface, Infrastructure

Metaverse Technologies part I: AR/VR/MR/XR, 3D reconstruction, Game engines, Smart glasses, wearables, haptic devices, headsets and headwear.

UNIT III:

(10 Periods)

Metaverse technologies part II: Blockchain, smart contracts, tokens, NFTs, Cryptography, Artificial Intelligence (AI), Internet of Things (IoT), Edge computing and 5G, 6G. Design theories and practices: Social presence and co-presence, Motion sickness and cybersickness, Uncanny valley, Sense of self-location, sense of agency and sense of body ownership, Universal simulation principle, Prototyping, Evaluation techniques (qualitative and quantitative).

UNIT IV:

(08 Periods)

Tools and technologies for Metaverse UX and UI: Tools and services for avatar systems, Spatial user

interface design, Cross-platform user experience design, Multimodal user interface, Technologies and devices for human computer interaction in Metaverse, Metaverse platforms: Decentraland, SANDBOX, Roblox, Axie Infinity, uHive, Hyper Nation, Nakamoto (NAKA), Metahero (HERO), Star Atlas (ATLAS), Bloktopia (BLOK), Stageverse, Spatial, PalkaCity, Viverse, Sorare, Illuvium, Upland, Second Life, Sansar, Sensorium Galaxy

UNIT V:

(07 Periods)

Metaverse applications - part I: Gaming and entertainment, Travel and tourism, Education and learning, Remote working, Commerce and business, Metaverse applications - part II: Real estate, Banking and Finance, Healthcare, Social media, Fashion, Metaverse and cyber security: Cyber security concerns in Metaverse: Social engineering attacks, Data theft, Decentralization vs vulnerabilities, Cyber security risks in Metaverse: process, people, technology, Metaverse and cybercrime: Scam and theft, Rug pull, Money manipulation and wash trading, Money laundering, Metaverse challenges and open issues: Persistency, Interoperability and scalability, Maturity, Regulation, Usefulness and ease-of-use, Privacy and data security, Content creation, NFTs and creator economy, Social, legal and ethical issues in the Metaverse

Total Periods: 45

Text Books:

T1. The Metaverse, Terry Winters, Independently published, 2021, ISBN: 979-8450959283

Reference Books:

R1. Ball, M., 2022, —The Metaverse and How It Will Revolutionize Everything—, Liveright, ISBN: 978-1324092032

R2. Damar, M. (2021). Metaverse shape of your life for future: A bibliometric snapshot. Journal of Metaverse, 1(1), 1–8.

R3. Day, J. (2022) Metaverse will see cyber warfare attacks unlike anything before: ‘Massively elevated’, February 28.

R4. <https://www.express.co.uk/news/science/1570844/metaverse-news-cyber-warfare-attacks-virtual-worlds-russia-china-spt>.

R5. Polyviou, A., Sharma K., Pappas, I.O.(2023). Training in the metaverse: Employing physiological data to improve how we build metaverses for businesses. The next generation internet: The role of metaverses, AR, VR, MR, and digital twins, Temple University Institute for Business and Information Technology Link: <https://ibit.temple.edu/nextgenerationinternet>

R6. QuHarrison T. , Keeney, S., 2022, —The Metaverse Handbook: Innovating for the Internet's NextTectonic Shift, Wiley, ISBN: 978-1119892526

R7. The mistocleous, M., Christodoulou, K., & Katelaris, L. (2023). An Educational Metaverse Experiment: The first on-chain and in- Metaverse academic course. Information Systems. EMCIS2022. Lecture Notes in Business Information Processing, Springer, Cham.

R8. Stephenson, N., 1992, —Snow Crash, ISBN: 978-055338

2333451	B.Tech. CSE, V-SEMESTER MACHINE LEARNING LAB	L	T	P	C
		0	0	3	1.5

Pre-Requisites: : Linear Algebra and Calculus, Vector Calculus, Statistical methods for Data Science, Python Programming.

Course Outcomes:

On successful completion of the course, student will be able to

- CO1.** Apply statistical measures (central tendency, dispersion) and essential data pre-processing techniques for effective data preparation.
- CO2.** Implement and evaluate supervised learning algorithms such as KNN, Decision Trees, Random Forest, Naïve Bayes, Support Vector Machines, and Logistic Regression for classification and regression problems.
- CO3.** Develop predictive models using simple linear regression and Multi-Layer Perceptron (MLP) neural networks for solving regression and classification tasks.
- CO4.** Apply unsupervised learning techniques including K-means, Fuzzy C-Means, and Expectation-Maximization for clustering, and evaluate their performance using appropriate metrics.
- CO5.** Integrate data pre-processing, model selection, and performance evaluation to design complete end-to- end machine learning solutions for real-world applications.

List of Exercises/List of Experiments:

Minimum Ten experiments are to be conducted.

1. Compute Central Tendency Measures: Mean, Median, Mode Measure of Dispersion: Variance, Standard Deviation.
2. Apply the following Pre-processing techniques for a given dataset.
 - a. Attribute selection
 - b. Handling Missing Values
 - c. Discretization
 - d. Elimination of Outliers
3. Apply KNN algorithm for classification and regression.
4. Demonstrate decision tree algorithm for a classification problem and perform parameter tuning for better results.
5. Demonstrate decision tree algorithm for a regression problem.
6. Apply Random Forest algorithm for classification and regression

7. Demonstrate Naïve Bayes Classification algorithm.
8. Apply Support Vector algorithm for classification.
9. Demonstrate simple linear regression algorithm for a regression problem.
10. Apply Logistic regression algorithm for a classification problem.
11. Demonstrate Multi-layer Perceptron algorithm for a classification problem.
12. Implement the K-means algorithm and apply it to the data you selected. Evaluate performance by measuring the sum of the Euclidean distance of each example from its class center. Test the performance of the algorithm as a function of the parameters K.
13. Demonstrate the use of Fuzzy C-Means Clustering.
14. Demonstrate the use of Expectation Maximization based clustering algorithm.

Software Required: Python/R/Weka

Reference Books:

- R1. Python: The Complete Reference, Martin C. Brown, McGraw Hill Education, Fourth edition, 2018.
- R2. Core Python Programming, R. NageswaraRao , Dreamtech Press India Pvt Ltd 2018.
- R3. Machine Learning, Tom M. Mitchell, McGraw-Hill Publication, 2017
- R4. Machine Learning in Action, Peter Harrington, DreamTech

Web Resources:

1. Machine Learning A-Z (Python & R in Data Science Course) | Udemy
2. Machine Learning | Coursera

2305651	B.Tech. CSE, VI-SEMESTER Cryptography and Network Security Lab	L	T	P	Credits
		0	0	3	1.5

Course Outcomes

- CO1.** Apply classical encryption techniques such as Caesar Cipher, Substitution Cipher, and Hill Cipher to encrypt and decrypt messages using Java.
- CO2.** Implement and analyze symmetric encryption algorithms such as DES, Blowfish, and Rijndael (AES) using C or Java.
- CO3.** Design and implement public-key cryptographic systems like RSA and Diffie-Hellman Key Exchange to establish secure communication.
- CO4.** Use cryptographic hash functions (SHA-1, MD5) to compute and verify message digests, ensuring message integrity.
- CO5.** Utilize Java Cryptography Architecture (JCA) and tools like Java KeyTool for secure key generation and encryption in real-world applications.
- CO6.** Demonstrate the ability to use web-based technologies (HTML and JavaScript) for implementing client-side cryptographic protocols.

List of Experiments:

1. Write a C program that contains a string (char pointer) with a value `__Hello world^`. The program should XOR each character in this string with 0 and displays the result.
2. Write a C program that contains a string (char pointer) with a value `__Hello world^`. The program should AND or and XOR each character in this string with 127 and display the result.
3. Write a Java program to perform encryption and decryption using the following algorithms
 - a. Ceaser cipher b. Substitution cipher c. Hill Cipher
4. Write a C/JAVA program to implement the DES algorithm logic.
5. Write a C/JAVA program to implement the Blowfish algorithm logic.
6. Write a C/JAVA program to implement the Rijndael algorithm logic.
7. Write the RC4 logic in Java Using Java cryptography; encrypt the text `—Hello world` using Blowfish. Create your own key using Java key tool.
8. Write a Java program to implement RSA algorithm.

9. Implement the Diffie-Hellman Key Exchange mechanism using HTML and JavaScript.
10. Calculate the message digest of a text using the SHA-1 algorithm in JAVA.
11. Calculate the message digest of a text using the MD5 algorithm in JAVA.

Projects

1. Secure Chat Application Using RSA and AES (Java)
2. File Encryption and Decryption Tool
3. Web-Based Diffie-Hellman Key Exchange Demonstrator
4. Digital Signature Generator and Verifier
5. Password Strength Evaluator and Hash Generator

References :

1. Cryptography and Network Security: Principles and Practice by William Stallings , Pearson Education

2339653	FULL STACK DEVELOPMENT – II (Skill Enhancement Course)	L	T	P	C
		0	1	2	2

Course Outcomes:

- CO1: Building fast and interactive UIs
- CO2: Applying Declarative approach for developing web apps
- CO3: Understanding ES6 features to embrace modern JavaScript
- CO4: Building reliable APIs with Express. Js
- CO5: Create and manage Git repositories, track changes, and push code to GitHub.
- CO6: Collaborate effectively in teams using GitHub tools such as Issues, Pull Requests, Code Reviews, and shared repositories.

Experiments covering the Topics:

- Introduction to DOM (Document Object Model), Ecma Script (ES6) standards and features like Arrow functions, Spread operator, Rest operator, Type coercion, Type hoisting, String literals, Array and Object Destructuring.
- Basics of React.js like React Components, JSX, Conditional rendering Differences between Real DOM and Virtual DOM.
- Important React.js concepts like React hooks, Props, React forms, Fetch API, Iterative rendering using JavaScript map() function.
- JavaScript runtime environment node.js and its uses, Express.js and Routing, Micro-Services architecture and MVC architecture, database connectivity using (My SQL)
- Introduction to My SQL, setting up MySQL and configuring, Databases, My SQL queries, subqueries, creating My SQL driver for database connectivity to Express.js server.
- Introduction to Git and GitHub and upload project& team collaboration

Sample Experiments:

1. Introduction to Modern JavaScript and DOM

- a. Write a JavaScript program to link JavaScript file with the HTML page
- b. Write a JavaScript program to select the elements in HTML page using selectors
- c. Write a JavaScript program to implement the event listeners
- d. Write a JavaScript program to handle the click events for the HTML button elements
- e. Write a JavaScript program to With three types of functions
 - i. Function declaration
 - ii. Function definition
 - iii. Arrow functions

2. Basics of React.js

- a. Write a React program to implement a counter button using react class components
- b. Write a React program to implement a counter button using react functional components
- c. Write a React program to handle the button click events in functional component
- d. Write a React program to conditionally render a component in the browser
- e. Write a React program to display text using String literals

3. Important concepts of React.js

- a. Write a React program to implement a counter button using React use State hook
- b. Write a React program to fetch the data from an API using React use Effect hook
- c. Write a React program with two react components sharing data using Props.
- d. Write a React program to implement the forms in react
- e. Write a React program to implement the iterative rendering using map() function.

4. Introduction to Git and GitHub

a. Setup

- Install Git on local machine.
- Configure Git (user name, email).
- Create GitHub account and generate a personal access token.

b. Basic Git Workflow

- Create a local repository using git init
- Create and add files → git add .
- Commit files → git commit -m "Initial commit"
- Connect to GitHub remote → git remote add origin <repo_url>
- Push to GitHub → git push -u origin main

c. Branching and Collaboration

- Create a branch → git checkout -b feature1
- Merge branch to main → git merge feature1
- Resolve merge conflicts (guided)

5. Upload React Project to GitHub

- Create a new React app using npx create-react-app myapp
- Initialize a git repo and push to GitHub
- Use .gitignore to exclude node_modules
- Create multiple branches: feature/navbar, feature/form
- Practice merge and pull requests (can use GitHub GUI)

6. Introduction to Node.js and Express.js

- a. Write a program to implement the `'hello world'` message in the route through the browser using Express
- b. Write a program to develop a small website with multiple routes using Express.js
- c. Write a program to print the `'hello world'` in the browser console using Express.js
- d. Write a program to implement the CRUD operations using Express.js
- e. Write a program to establish the connection between API and

Database using Express – My SQL driver

7. Introduction to My SQL

COMPUTER SCIENCE AND ENGINEERING

B.Tech. (R23UG) Course Structure**VII-SEMESTER**

S.No.	COURSE CODE	COURSE TITLE	CC	L	T	P	IM	EM	CR
1	2339609	Deep Learning	PCC	3	0	0	30	70	3
2	2332604	Cloud Computing	PCC	3	0	0	30	70	3
3	2332605	Software Architecture and Design Patterns	PEC-IV	3	0	0	30	70	3
	2305701	Cyber Security							
	2305702	Mobile Adhoc Networks							
	2332603	Computer Vision							
4	2305703	Agile methodologies	PEC-V	3	0	0	30	70	3
	2305704	Block chain Technology							
	2305705	Cyber Physical Systems							
	2305706	DevOps							
5		Open Elective Course –III	OEC-III	3	0	0	30	70	3
6		Open Elective Course –IV	OEC-IV	3	0	0	30	70	3
7	2339751	Prompt Engineering	SC	0	1	2	30	70	2
8	2305752	Industry Internship	PROJ	-	-	-	-	50	2
9	23HS701	Gender Sensitization	MC	2	0	0	30	-	-
Total				20	01	02	240	540	22

B.Tech (R23UG) VII Semester Syllabus

2339609	B.Tech. CSE, VII-SEMESTER DEEP LEARNING	L	T	P	C
		3	0	0	3

Pre-Requisites: Machine Learning

Course Outcomes:

On successful completion of the course, student will be able to

- CO1. Demonstrate the mathematical foundation of neural network
- CO2. Describe the machine learning basics
- CO3. Differentiate architecture of deep neural network
- CO4. Build a convolutional neural network
- CO5. Build and train RNN and LSTMs

SYLLABUS:

UNIT-I

(12 Periods)

Linear Algebra: Scalars, Vectors, Matrices and Tensors, Matrix operations, types of matrices, Norms, Eigen decomposition, Singular Value Decomposition, Principal Components Analysis. Probability and Information Theory: Random Variables, Probability Distributions, Marginal Probability, Conditional Probability, Expectation, Variance and Covariance, Bayes' Rule, Information Theory. Numerical Computation: Overflow and Underflow, Gradient-Based Optimization, Constrained Optimization, Linear Least Squares.

UNIT-II

(8 Periods)

Machine Learning: Basics and Under fitting, Hyper parameters and Validation Sets, Estimators, Bias and Variance, Maximum Likelihood, Bayesian Statistics, Supervised and Unsupervised Learning, Stochastic Gradient Descent, Challenges Motivating Deep Learning. Deep Feed forward Networks: Learning XOR, Gradient-Based Learning, Hidden Units, Architecture Design, Back-Propagation and other Differentiation Algorithms.

UNIT-III

(10 Periods)

Regularization for Deep Learning: Parameter Norm Penalties, Norm Penalties as Constrained Optimization, Regularization and Under-Constrained Problems, Dataset Augmentation, Noise Robustness, Semi-Supervised Learning, Multi-Task Learning, Early Stopping, Parameter Tying and Parameter Sharing, Sparse Representations, Bagging and Other Ensemble Methods, Dropout, Adversarial Training, Tangent Distance, Tangent Prop and Manifold Tangent Classifier. Optimization for Training Deep Models: Pure Optimization,

Challenges in Neural Network Optimization, Basic Algorithms, Parameter Initialization Strategies, Algorithms with Adaptive Learning Rates, Approximate Second-Order Methods, Optimization Strategies and Meta-Algorithms.

UNIT-IV

(8 Periods)

Convolutional Networks: The Convolution Operation, Pooling, Convolution, Basic Convolution Functions, Structured Outputs, Data Types, Efficient Convolution Algorithms, Random or Unsupervised Features, Basis for Convolutional Networks.

UNIT-V

(7 Periods)

Sequence Modeling: Recurrent and Recursive Nets: Unfolding Computational Graphs, Recurrent Neural Networks, Bidirectional RNNs, Encoder-Decoder Sequence-to-Sequence Architectures, Deep Recurrent Networks, Recursive Neural Networks, Echo State Networks, LSTM, Gated RNNs, Optimization for Long-Term Dependencies, Auto encoders, Deep Generative Models.

Total Periods: 45

Text Books:

- T1. Ian Good fellow, Yoshua Bengio, Aaron Courville, —Deep LearningI, MIT Press,2016.
- T2. Josh Patterson and Adam Gibson, —Deep learning: A practitioner's approachI, O'Reilly Media, First Edition,2017.

Reference Books:

- R1. Fundamentals of Deep Learning, Designing next-generation machine intelligence algorithms, Nikhil Buduma, O'Reilly, Shroff Publishers,2019.
- R2. Deep learning Cook Book, Practical recipes to get started Quickly, Douwe Osinga, O'Reilly, Shroff Publishers 2019

Web Resources:

1. <https://keras.io/datasets/>
2. <http://deeplearning.net/tutorial/deeplearning.pdf>
3. <https://arxiv.org/pdf/1404.7828v4.pdf>
4. <https://www.cse.iitm.ac.in/~miteshk/CS7015.html>
5. <https://www.deeplearningbook.org>
6. <https://nptel.ac.in/courses/106105215>

2332604	B.Tech. CSE, VII-SEMESTER CLOUD COMPUTING	L	T	P	C
		3	0	0	3

Pre-Requisites: Computer Networks, Operating System, Programming in Python, Database management systems, Web Technologies

Course Outcomes:

On successful completion of the course, students will be able to

- CO1.** Explain the fundamental concepts, characteristics, service models, and core technologies of cloud computing.
- CO2.** Ability to design applications for Cloud environment
- CO3.** Design and develop applications using python
- CO4.** Analyze big data using clustering and classification techniques, and evaluate cloud application performance through benchmarking tools.
- CO5.** Assess cloud security practices, migration strategies, and the adoption of cloud solutions in industries like healthcare, education, and manufacturing.

SYLLABUS:

UNIT – I: BASICS OF CLOUD COMPUTING (12 Periods)

Introduction to cloud computing: Introduction, Characteristics of cloud computing, Cloud Models, Cloud Services Examples, Cloud Based services and applications

Cloud concepts and Technologies: Virtualization, Load balancing, Scalability and Elasticity, Deployment, services level Agreements, Billing.

Cloud Services and Platforms: Compute Services, Storage Services, Identity and Access Management services

UNIT – II: HADOOP AND PYTHON (08 Periods)

Hadoop Map Reduce: Apache Hadoop, Hadoop Map Reduce Job Execution

Cloud Application Design: Reference Architecture for Cloud Applications, Cloud Application Design Methodologies, Data Storage Approaches.

Python Basics: Introduction, Installing Python, Python data Types, Control flow, Function, File handling, Classes.

UNIT – III: PYTHON FOR CLOUD COMPUTING (09 Periods)

Python for Cloud: Python for Amazon web services, Python for Google Cloud Platform, Python for windows Azure, Python for Map Reduce, Python packages of Interest, Designing a RESTful web API.

Cloud Application Development in Python: Image Processing APP, Map Reduce App

**UNIT – IV: BIG DATA, MULTIMEDIA AND TUNING
Periods)**

(08

Big Data Analytics: Introduction, Clustering Big Data, Classification of Big data

Multimedia Cloud: Introduction, Case Study: Live video Streaming App

Cloud Application Benchmarking and Tuning: Introduction, Work load Characteristics, Application Performance Metrics, Bench marking Tools, Load Testing

**UNIT – V: APPLICATIONS AND ISSUES IN CLOUD
Periods)**

(08

Cloud Security: Introduction, Authentication, Authorization, Identity Access Management, Data Security

Cloud for Industry, Health care & Education: Cloud Computing for Healthcare, Cloud Computing for Manufacturing Industry, Cloud computing for Education.

Migrating in to a Cloud: Introduction, Broad Approaches to migrating into the cloud, the seven– step model of migration in to a cloud.

Organizational readiness and Change Management in The Cloud Age: Introduction, Basic concepts of Organizational Readiness, Drivers for changes: A frame work to comprehend the competitive environment, common change management models, change management maturity models.

**Total
Periods:45**

Text Books:

- T1. Cloud computing A hands - on Approach |By Arshdeep Bahga, Vijay Madisetti, Universities Press, 2016
- T2. Cloud Computing Principles and Paradigms: By RajKumar Buyya, James Broberg, Andrzej Goscinski, Wiley, 2016.

Reference Books:

- R1. Masterin g Cloud Computing by Rajkumar Buyya, Christian Vecchiola, S Thamarai Selvi, TMH
- R2. Cloud computing AHands-On Approach by Arshdeep Bahga and Vijay Madisetti.
- R3. Cloud Computing: A Practical Approach, Anthony T.Velte, To by J.Velte, Robert Elsenpeter, Tata Mc Graw Hill, rp 2011.
- R4. Enterprise Cloud Computing, Gautam Shroff, Cambridge University Press, 2010.
- R5. Cloud Application Architectures: Building Applications and Infrastructure in the Cloud, George Reese,O_Reilly, SPD, rp 2011.
- R6. Essentials of Cloud Computing by K.Chandrasekaran. CRC Press.

Web Resources:

- 1. https://onlinecourses.nptel.ac.in/noc25_cs107/preview
- 2. <https://www.coursera.org/specializations/cloud-computing>
- 3. <https://www.geeksforgeeks.org/cloud-computing/cloud-computing/>

2332605	B.Tech. CSE, VII-SEMESTER SOFTWARE ARCHITECTURE AND DESIGN PATTERNS Professional Elective-IV	L	T	P	C
		3	0	0	3

Pre-Requisites: SOFTWARE ENGINEERING

Course Outcomes:

On successful completion of the course, student will be able to

CO1. To highlight the evolution of patterns.

CO2. To learn how to add functionality to designs while minimizing complexity.

CO3. To learn what designs patterns really are or not.

CO4. To know about specific design patterns.

CO5. To learn how to use design patterns to keep code quality without over design.

SYLLABUS:

UNIT-I: ENVISIONING ARCHITECTURE (12 Periods)

Envisioning Architecture: The Architecture Business Cycle, What is Software Architecture, Architectural patterns, reference models, reference architectures, architectural structures and views. Creating an Architecture: Quality Attributes, Achieving qualities, Architectural styles and patterns, designing the Architecture, Documenting software architectures, Reconstructing Software Architecture.

UNIT-II: ANALYZING ARCHITECTURE (08 Periods)

Analyzing Architectures: Architecture Evaluation, Architecture design decision making, ATAM, CBAM. Moving from one system to many: Software Product Lines, Building systems from off the shelf components, Software architecture in future.

UNIT-III: PATTERNS (10 Periods)

Patterns: Pattern Description, Organizing catalogs, role in solving design problems, Selection and usage. Creational and Structural patterns: Abstract factory, builder, factory method, prototype, singleton, adapter, bridge, composite, façade, flyweight.

UNIT-IV: BEHAVIOURAL PATTERNS (08 Periods)

Behavioral patterns: Chain of responsibility, command, Interpreter, iterator, mediator, memento, observer, state, strategy. template method, visitor

**UNIT-V:
(07 Periods)**

Case Studies: The World Wide Web - a case study in interoperability, Air Traffic Control – a case study in designing for high availability, Celsius Tech – a case study in product line development.

Total

Periods: 45

Text Books:

T1. Software Architecture in Practice, second edition, Len Bass, Paul Clements & Rick Kazman, Pearson Education, 2003.

T2. Design Patterns, Erich Gamma, Pearson Education.

Reference Books:

R1. Beyond Software architecture, Luke Hohmann, Addison wesley, 2003.

R2. Software architecture, David M. Dikel, David Kane and James R. Wilson, Prentice Hall PTR, 2001.

R3. Software Design, David Budgen, second edition, Pearson education, 2003.

R4. Head First Design patterns, Eric Freeman & Elisabeth Freeman, O'REILLY, 2007.

R5. Design Patterns in Java, Steven John Metsker & William C. Wake, Pearson education, 2006 6.
J2EE Patterns, Deepak Alur, John Crupi & Dan Malks, Pearson education, 2003.

2305701	B.Tech VII Semester CYBER SECURITY Professional Elective-IV	L	T	P	C
		3	0	0	3

Prerequisite: Computer Networks and Operating Systems.

Course Outcomes:

After completion of the course, students will be able to

CO1: Classify the cybercrimes and understand the Indian ITA 2000

CO2: Analyse the vulnerabilities in any computing system and find the solutions

CO3: Predict the security threats of the future

CO4: Investigate the protection mechanisms

CO5: Design security solutions for organizations

UNIT I: Introduction to Cybercrime (8 Periods)

Introduction, Cybercrime, and Information Security, Who are Cybercriminals, Classifications of Cybercrimes, And Cybercrime: The legal Perspectives and Indian Perspective, Cybercrime and the Indian ITA 2000, A Global Perspective on Cybercrimes.

UNIT II: Cyber Offenses: How Criminals Plan Them (9 Periods)

Introduction, How Criminals plan the Attacks, Social Engineering, Cyber stalking, Cyber cafe and Cybercrimes, Botnets: The Fuel for Cybercrime, Attack Vector, Cloud Computing.

UNIT III: Cybercrime: Mobile and Wireless Devices (9 Periods)

Introduction, Proliferation of Mobile and Wireless Devices, Trends in Mobility, Credit card Frauds in Mobile and Wireless Computing Era, Security Challenges Posed by Mobile Devices, Registry Settings for Mobile Devices, Authentication service Security, Attacks on Mobile/Cell Phones,

Mobile Devices:Security Implications for Organizations, Organizational Measures for Handling Mobile, Organizational Security Policies an Measures in Mobile Computing Era, Laptops.

UNIT IV: Tools and Methods Used in Cybercrime (8 Periods)

Introduction, Proxy Servers and Anonymizers, Phishing, Password Cracking, Keyloggers and Spywares, Virus and Worms, Trojan Horse and Backdoors, Steganography, DoS and DDoS attacks, SQL Injection, Buffer Overflow.

UNIT V: Cyber Security: Organizational Implications

(9

Periods)

Introduction, Cost of Cybercrimes and IPR issues, Web threats for Organizations, Security and Privacy Implications, Social media marketing: Security Risks and Perils for Organizations, Social Computing and the associated challenges for Organizations.

Total Periods:43

Text books:

T1. Sunit Belapure and Nina Godbole, "Cyber Security: Understanding Cyber Crimes, Computer

Forensics and Legal Perspectives", Wiley India, 1st Edition, 2011.

T2. R.K. Chaubey, "Cyber Crime and Cyber Law", Allahabad Law Agency, 2016.

Reference Books:

R1. Jonathan Clough, "Principles of Cybercrime", Cambridge University Press, 2nd Edition, 2015.

R2. William Stallings, "Computer Security: Principles and Practice", Pearson, 4th Edition, 2018.

R3. Thomas R. Peltier, "Information Security Policies, Procedures, and b Standards", Auerbach Publications, 2004.

Online Learning Resources:

1. <http://nptel.ac.in/courses/106105031/40>
2. <http://nptel.ac.in/courses/106105031/39>
3. <http://nptel.ac.in/courses/106105031/38>

2305702	MOBILE ADHOC NETWORKS (Professional Elective-IV) B.Tech VII Semester (CSE)	L	T	P	C
		3	0	0	3

Pre-Requisites: Computer Networks

Course Outcomes:

On successful completion of the course, student will be able to

CO1. Describe the unique issues in ad-hoc/sensor networks.

CO2. Describe current technology trends for the implementation and deployment of wireless

Ad-hoc/sensor networks.

CO3. Discuss the challenges in designing MAC, routing and transport protocols for wireless

Ad-hoc/sensor networks.

CO4. Discuss the challenges in designing routing and transport protocols for wireless

Ad-hoc/sensor networks.

CO5. Comprehend the various sensor network Platforms, tools and applications.

SYLLABUS:

UNIT-I (10 Periods)

Introduction to Ad Hoc Networks: Characteristics of MANETs, Applications of MANETs and challenges of MANETs -Routing in MANETs: Criteria for classification, Taxonomy of MANET routing algorithms, Topology based routing algorithms, Position based routing algorithms, Other routing algorithms.

UNIT-II (8 Periods)

Data Transmission: Broadcast storm problem, Broadcasting, Multicasting and Geocasting - TCP over Ad Hoc: TCP protocol overview, TCP and MANETs, Solutions for TCP over Ad hoc

UNIT-III (8 Periods)

Basics of Wireless, Sensors and Applications: Applications, Classification of sensor networks, Architecture of sensor network, Physical layer, MAC layer, Link layer.

UNIT-IV (8 Periods)

Data Retrieval in Sensor Networks: Routing layer, Transport layer, High-level application layer support, Adapting to the inherent dynamic nature of WSNs, Sensor Networks and

mobile robots-Security: Security in Ad-Hoc networks, Key management, Secure routing, Cooperation in MANETs, Intrusion Detection systems.

UNIT-V:

(9

Periods)

Sensor Network Platforms and Tools: Sensor Network Hardware, Berkeley motes, Sensor Network Programming Challenges, Node-Level Software Platforms -Operating System: Tiny OS -Imperative Language: nesC, Data flow style language: Tiny GALS, Node Level Simulators, ns- 2 and its sensor network extension.

Total

Periods: 43

Text Books:

- T3. Ad Hoc and Sensor Networks –Theory and Applications, Carlos Corderio Dharma P. Aggarwal, World Scientific Publications, March 2006
- T4. Wireless Sensor Networks: An Information Processing Approach, Feng Zhao, Leonidas Guibas, Elsevier Science

Reference Books:

- R1. Ad Hoc Wireless Networks: Architectures and Protocols, C. Siva Ram Murthy, B. S. Manoj, Pearson Education.
- R2. Wireless Sensor Networks: Technology, Protocols, and Applications, Kazem Sohraby, Daniel Minoli, Taieb Znati, Wiley.
- R3. Wireless Sensor and Actuator Networks: Algorithms and Protocols for Scalable Coordination and Data Communication, Amiya Nayak, Ivan Stojmenovic, Wiley.

Web Resources:

- 1. <https://nptel.ac.in/courses/106105160>
- 2. <https://www.nsnam.org/>
- 3. <https://github.com/contiki-os/contiki/wiki/Cooja-Simulator>

2332603	B.Tech. CSE, VII-SEMESTER COMPUTER VISION (Professional Elective-V)	L	T	P	C
		3	0	0	3

Pre-Requisites: Computer Graphics

Course Outcomes:

On successful completion of this course, the students will be able to

- CO1.** Identify basic concepts, terminology, theories, models and methods in the field of computer vision. (L2)
- CO2.** Describe known principles of human visual system (L1).
- CO3.** Describe basic methods of computer vision related to multi-scale representation, edge detection. (L2)
- CO4.** Describe basic primitives, stereos, object and object recognition. (L2)
- CO5.** Suggest a design of a computer vision system for a specific problem (L4)

UNIT-I: LINEAR FILTERS (9 Periods)

Introduction to Computer Vision, Linear Filters and Convolution, Shift Invariant Linear Systems, Spatial Frequency and Fourier Transforms, Sampling and Aliasing Filters as Templates, Technique: Normalized Correlation and Finding Patterns, Technique: Scale and Image Pyramids.

UNIT-II: EDGE DETECTION (10 Periods)

Noise- Additive Stationary Gaussian Noise, Why Finite Differences Respond to Noise, Estimating Derivatives – Derivative of Gaussian Filters, Why Smoothing Helps, Choosing a Smoothing Filter, Why Smooth with a Gaussian? Detecting Edges-Using the Laplacian to Detect Edges, Gradient-Based Edge Detectors, Technique: Orientation Representations and Corners.

UNIT-III: TEXTURE (9 Periods)

Representing Texture –Extracting Image Structure with Filter Banks, Representing Texture using the Statistics of Filter Outputs, Analysis (and Synthesis) Using Oriented Pyramids – The Laplacian Pyramid, Filters in the Spatial Frequency Domain, Oriented Pyramids, Application: Synthesizing Textures for Rendering, Homogeneity, Synthesis by Sampling Local Models, Shape from Texture, Shape from Texture for Planes.

UNIT-IV: SEGMENTATION BY CLUSTERING (9 Periods)

What is Segmentation, Human Vision: Grouping and Gestalt, Applications: Shot Boundary Detection and Background Subtraction. Image Segmentation by Clustering Pixels, Segmentation by Graph-Theoretic Clustering. The Hough Transform, Fitting Lines, Fitting Curves

UNIT-V: RECOGNIZATION BY RELATIONS BETWEEN TEMPLATES (8 Periods)

Finding Objects by Voting on Relations between Templates, Relational Reasoning Using Probabilistic Models and Search, Using Classifiers to Prune Search, Hidden Markov Models, Application: HMM and Sign Language Understanding, Finding People with HMM.

**Total
Periods: 45**

Text Books:

T1. David A. Forsyth, Jean Ponce, Computer Vision – A modern Approach, PHI, 2003

Reference Books:

- R1. Geometric Computing with Clifford Algebras: Theoretical Foundations and Applications in Computer Vision and Robotics, Springer; 1st edition, 2001 by Sommer.
- R2. Digital Image Processing and Computer Vision, 1/e, by Sonka.
- R3. Computer Vision and Applications: Concise Edition (With CD) by Jack Academy Press, 2000.

2305703	B.Tech. CSE, VII-SEMESTER AGILE METHODOLOGIES (Professional Elective-V)	L	T	P	C
		3	0	0	3

Pre-Requisites: Software Engineering

Course Outcomes:

On successful completion of the course, students will be able to

- CO1.** Explain the principles, values, and structure of Agile methodologies and differentiate them from traditional software development approaches.
- CO2.** Analyze and apply various Agile frameworks such as SCRUM, XP, Crystal, and Lean Production in software development lifecycle.
- CO3.** Demonstrate knowledge of Agile-based knowledge management strategies and evaluate the role of story cards in knowledge sharing.
- CO4.** Apply Agile practices in requirements engineering, including requirements abstraction, prioritization, and modeling in dynamic environments.
- CO5.** Evaluate Agile approaches to software quality assurance, including Test-Driven Development and metrics in Feature-Driven Development.

SYLLABUS:

UNIT - 1: AGILE METHODOLOGY (10 Periods)

Theories for Agile Management – Agile Software Development – Traditional Model vs. Agile Model - Classification of Agile Methods – Agile Manifesto and Principles – Agile Project Management – Agile Team Interactions – Ethics in Agile Teams - Agility in Design, Testing – Agile Documentations – Agile Drivers, Capabilities and Values

UNIT - II: AGILE PROCESSES (09 Periods)

Lean Production - SCRUM, Crystal, Feature Driven Development- Adaptive Software Development - Extreme Programming: Method Overview – Lifecycle – Work Products, Roles and Practices.

UNIT - III: AGILITY AND KNOWLEDGE MANAGEMENT (11 Periods)

Agile Information Systems – Agile Decision Making - Earl_S Schools of KM – Institutional Knowledge Evolution Cycle – Development, Acquisition, Refinement, Distribution,

Deployment, Leveraging – KM in Software Engineering – Managing Software Knowledge – Challenges of Migrating to Agile Methodologies – Agile Knowledge Sharing – Role of Story-Cards – Story-Card Maturity Model (SMM).

UNIT - IV: AGILITY AND REQUIREMENTS ENGINEERING (08 Periods)

Impact of Agile Processes in RE–Current Agile Practices – Variance – Overview of RE Using Agile Managing Unstable Requirements – Requirements Elicitation – Agile Requirements Abstraction Model – Requirements Management in Agile Environment, Agile Requirements Prioritization – Agile Requirements Modeling and Generation – Concurrency in Agile Requirements Generation.

UNIT - V: AGILITY AND QUALITY ASSURANCE (07 Periods)

Agile Product Development – Agile Metrics – Feature Driven Development (FDD) – Financial and Production Metrics in FDD – Agile Approach to Quality Assurance - Test Driven Development – Agile Approach in Global Software Development.

Total Periods:45

Text Books:

- T1. David J. Anderson and Eli Schragenheim, —Agile Management for Software Engineering: Applying the Theory of Constraints for Business Results, Prentice Hall, 2003.
- T2. Hazza and Dubinsky, —Agile Software Engineering, Series: Undergraduate Topics in Computer Science, Springer, 2009.

Reference Books:

- R1. Craig Larman, —Agile and Iterative Development: A Manager's Guide, Addison-Wesley, 2004.
- R2. Kevin C. Desouza, —Agile Information Systems: Conceptualization, Construction, and Management, Butterworth-Heinemann, 2007.

Web Resources:

4. <https://www.nptelvideos.com/video.php;lx ,lc ?id=904>
5. <https://www.coursera.org/learn/agile-development-and-scrum>
6. <https://www.udemy.com/course/scrum-master-training/?couponCode=LETSLEARNNOW>

2305704	B.Tech CSE, VII- Semester BLOCKCHAIN TECHNOLOGY (Professional Elective-V)	L	T	P	C
		3	0	0	3

Pre-Requisites: Python and Java Script

Course Outcomes:

After completion of the course, students will be able to

CO1: Describe the fundamentals and evolution of blockchain technology and distinguish between centralized and decentralized applications.

CO2: Explain core blockchain concepts such as cryptographic hashing, consensus, mining, smart contracts, and peer-to-peer networks.

CO3: Analyze and design blockchain architectures using reference models and enterprise use cases; evaluate cryptographic tokens and deployment strategies.

CO4: Implement decentralized applications (DApps) and smart contracts using Ethereum tools and frameworks (Truffle, Ganache, Metamask, etc.).

CO5: Develop blockchain applications using Hyperledger Fabric and explore advanced blockchain technologies including IPFS, oracles, and blockchain-integrated AI/ML.

SYLLABUS:

UNIT- I: INTRODUCTION

(09 Periods)

Introduction, Scenarios, Challenges Articulated, Block chain, Block chain Characteristics, Opportunities Using Block chain, History of Block chain. Evolution of Block chain: Evolution of Computer Applications, Centralized Applications, Decentralized Applications, Stages in Block chain Evolution, Consortia, Forks, Public Block chain Environments, Type of Players in Block chain Ecosystem, Players in Market.

UNIT-II: BLOCKCHAIN CONCEPTS

(09 Periods)

Block chain Concepts: Introduction, Changing of Blocks, Hashing, Merkle-Tree, Consensus, Mining and Finalizing Blocks, Currency aka tokens, security on block chain, data storage on block chain, wallets, coding on block chain: smart contracts, peer-to-peer network, types of block chain nodes, risk associated with block chain solutions, life cycle of block chain transaction.

UNIT – III: ARCHITECTING BLOCKCHAIN SOLUTIONS

(09 Periods)

Architecting Block chain solutions: Introduction, Obstacles for Use of Block chain ,Block chain Relevance Evaluation Frame work, Block chain Solutions Reference Architecture, Types of Block chain Applications. Cryptographic Tokens, Typical Solution Architecture for Enterprise Use Cases, Types of Block chain Solutions , Architecture Considerations

,Architecture with Block chain Platforms, Approach for Designing Block chain Applications.

UNIT-IV: ETHEREUM BLOCKCHAIN IMPLEMENTATION (08 Periods)

EthereumBlockchainImplementation: Introduction, TunaFishTrackingUseCase, Ethereum Ecosystem, Ethereum Development, Ethereum Tool Stack, Ethereum Virtual Machine, Smart Contract Programming ,Integrated Development Environment ,Truffle Framework ,Ganache ,Unit Testing, Ethereum Accounts, My Ether Wallet, Ethereum Networks / Environments , Infura, Ether scan, Ethereum Clients ,Decentralized Application ,Meta mask ,Tuna Fish Use Case Implementation ,Open Zeppelin Contracts.

UNIT-V: HYPER LEDGER BLOCK CHAIN IMPLEMENTATION (10 Periods)

Hyper ledger Block chain Implementation: Introduction, Use Case–Car Owner ship Tracking, Hyper ledger Fabric, Hyper ledger Fabric Transaction Flow, FabCar Use Case Implementation, Invoking Chain code Functions Using Client Application.

Advanced Concepts in Block chain: Introduction, Inter Planetary File System (IPFS), Zero Knowledge Proofs, Oracles, Self-Sovereign Identity, Block chain with IoT and AI/ML Quantum Computing and Block chain, Initial Coin Offering, Block chain Cloud Offerings, Block chain and its Future Potential.

Total

Periods: 45

Text Books:

- T1. Ambadas, Arshad Sarfarz Ariff, Sham— Block chain for Enterprise Application Developers II, Wiley , 2020.
- T2. Andreas M.Antonopoulos,—Mastering Bitcoin: Programming the Open Block chain II,O'Reilly,2017

Reference Books:

- R1. Block chain: A Practical Guide to Developing Business, Law, and Technology Solutions, Joseph Bambara, Paul R. Allen, Mc Graw Hill.
- R2. Blockchain: Blue print for a New Economy, Melanie Swan,O'Reilly

Web Resources:

<https://github.com/blockchainedindia/resources>

2305705	B.Tech. CSE, VII-SEMESTER CYBER PHYSICAL SYSTEMS (Professional Elective-V)	L	T	P	C
		3	0	0	3

Pre-Requisites: NIL

Course Outcomes:

On successful completion of the course, student will be able to

CEO1. Understand the core principles behind CPS

CEO2. Identify Security mechanisms of Cyber physical systems

CEO3. Understand Synchronization in Distributed Cyber-Physical Systems

CEO4. To Understand the Scheduling for Cyber-Physical Systems

CEO5. To understand the various Cyber-Physical System models

SYLLABUS:

UNIT-I: SYMBOLIC SYNTHESIS FOR CYBER-PHYSICAL SYSTEMS (12 Periods)

Introduction and Motivation, Basic Techniques - Preliminaries, Problem Definition, Solving the Synthesis Problem, Construction of Symbolic Models, Advanced Techniques: Construction of Symbolic Models, Continuous-Time Controllers, Software Tools.

UNIT-II: SECURITY OF CYBER-PHYSICAL SYSTEMS (08 Periods)

Introduction and Motivation, Basic Techniques - Cyber Security Requirements, Attack Model, Countermeasures, Advanced Techniques: System Theoretic Approaches

UNIT III: (10 Periods)

Synchronization in Distributed Cyber-Physical Systems: Challenges in Cyber-Physical Systems, A Complexity-Reducing Technique for Synchronization, Formal Software Engineering, Distributed Consensus Algorithms, Synchronous Lockstep Executions, Time-Triggered Architecture, Related Technology, Advanced Techniques

UNIT IV: REAL-TIME SCHEDULING FOR CYBER-PHYSICAL SYSTEMS (08 Periods)

Introduction and Motivation, Basic Techniques - Scheduling with Fixed Timing Parameters, Memory Effects, Multiprocessor/Multicore Scheduling, Accommodating Variability and

Uncertainty

UNIT V: CAPSTONE PROJECT AND CASE STUDIES

(07 Periods)

Model Integration in Cyber-Physical Systems

Introduction and Motivation, Causality, Semantic Domains for Time, Interaction Models for Computational Processes, Semantics of CPS DSMLs, Advanced Techniques, For Spec, The Syntax of CyPhyML, Formalization of Semantics, Formalization of Language Integration.

Total

Periods: 45

Text Books:

T1. Raj Raj kumar, Dion is io De Niz, and Mark Klein, Cyber-Physical Systems, Addison-Wesley Professional.

T2. Rajeev Alur, Principles of Cyber-Physical Systems, MIT Press, 2015

Web Resources:

1. https://onlinecourses.nptel.ac.in/noc25_cs116/preview

2305706	B.Tech. CSE, VII-SEMESTER DEVOPS (Professional Elective -V)	L	T	P	C
		3	0	0	3

Pre-Requisites: Software Engineering and Agile Methodologies.

Course Outcomes:

On successful completion of the course, student will be able to

- CO1. Analyze the historical evolution, foundational principles, cultural components, and practical significance of DevOps to build strategic justification for its adoption within organizations.
- CO2. Apply the Business Model Canvas to software delivery workflows and evaluate DevOps optimization plays for improving efficiency across delivery pipelines.
- CO3. Apply DevOps practices to foster innovation through microservices architectures, API ecosystems, and platform-based development, and design integrated DevOps strategies to support these technologies.
- CO4. Evaluate strategies for scaling DevOps across enterprises by analyzing cultural, structural, and procedural factors, including team models, tool standardization, and security considerations.
- CO5. Create transformation strategies to lead successful DevOps adoption using pilot initiatives, collaboration frameworks, and structured roadmaps that align with organizational goals.

SYLLABUS:

UNIT- I: DEVOPS AND ADOPTING DEVOPS

(09 Periods)

DevOps: An Overview, DevOps: Origins, DevOps: Roots, DevOps: Practices DevOps: Culture.
Adopting DevOps: Developing the Playbook. Developing a Business Case for a DevOps:
 Developing the Business Case.

UNIT- II: DEVELOPING A BUSINESS CASE FOR A DEVOPS TRANSFORMATION

(09 Periods)

Completing the Business Model Canvas, Customer Segments, Value Segments, Value Propositions, Channels, Customer Relationships, Revenue Streams, Key Resources, Key Activities, Key Partnerships, Cost Structures. DevOps Plays for Optimizing the delivery Pipeline: DevOps as an optimization Exercise, Core Themes, The DevOps Plays, Specializing Core Plays.

UNIT- III: DEVOPS PLAYS FOR DRIVING INNOVATION

(09 Periods)

Optimize to Innovate, The Uber Syndrome, Innovation and the Role of Technology, Core Themes, play: Build a DevOps Platform, play: Deliver Microservices Architectures, play: DevOps an API Economy, play: Organizing for Innovation.

UNIT- IV: SCALING DEVOPS FOR THE ENTERPRISE

(09 Periods)

Core Themes, play: DevOps Center of Competency, play: Developing Culture of Innovation at Scale, play: Developing a Culture of continuous Improvement, play: Team Models for DevOps, play: Standardization of Tools and Process, play: Security Considerations for DevOps, Play: DevOps and Outsourcing.

UNIT- V: LEADING DEVOPS ADOPTION IN THE ENTERPRISE**(09 Periods)**

Play: DevOps as a transformation Exercise, play: Developing a Culture of Collaboration and Trust, play: DevOps Thinking for the Line of Business, play: starting with Pilot Projects, Play: Rearing Unicorns on an Aircrafts Carrier. Appendix Case Study: Example DevOps Adoption Roadmap Organization Background, Roadmap Structure, Adoption Roadmap.

Total Periods: 45**Text Books:**

T1. The DevOps Adoption Playbook, Sanjeev Sharma, Published by John Wiley & Sons, Inc.2017.

Reference Books:

R1. DevOps for Dummies, Sanjeev Sharma & Bernie Coyne, Published by John Wiley & Sons, Inc.

R2. DevOps for Developers, Michael Huttermann, Apress publishers,2012.

Web Resources:

1. Learning DevOps with Terraform Infrastructure Automation Course | Udemy

2339751	B.Tech, CSE VII- SEMESTER PROMPT ENGINEERING (Skill Enhancement Course)	L	T	P	C
		0	1	2	2

Prerequisites :AI

Course Out comes:

- CO1. Explore the fundamentals and evolution of prompt engineering.
- CO2. Gaining the ability to craft effective closed-ended, open-ended, and role-based prompts.
- CO3. Learning to probe and stress-test AI models for bias and robustness.
- CO4. Applying prompt optimization techniques and performance evaluation methods.
- CO5. Mitigating bias and promoting ethical prompting practices in NLP/ML systems.

SYLLABUS:

Module 1: Introduction to Prompt Engineering

Lesson 1: Foundations of Prompt Engineering

- Overview of prompt engineering and its significance in NLP and ML.
- Historical context and evolution of prompt-based approaches.

Module 2: Types of Prompts and Their Applications

Lesson 2: Closed-Ended Prompts

- Understanding and creating prompts for specific answers.
- Applications in question- answering systems.

Lesson 3: Open-Ended Prompts

- Crafting prompts for creative responses.
- Applications in language generation models.

Module 3: Strategies for Effective Prompting

Lesson 4: Probing Prompts

- Designing prompts to reveal model biases.
- Ethical considerations in using probing prompts.

Lesson 5: Adversarial Prompts

- Creating prompts to stress-test models.
- Enhancing robustness through adversarial prompting.

Module 4: Fine-Tuning and Optimizing with Prompts

Lesson 6: Fine-Tuning Models with Prompts

- Techniques for incorporating prompts during model training.
- Balancing prompt influence and generalization.

Lesson 7: Optimizing Prompt Selection

- Methods for selecting optimal prompts for specific tasks.
- Customizing prompts based on model behavior.

Module 5: Evaluation and Bias Mitigation

Lesson 8: Evaluating Prompt Performance

- Metrics and methodologies for assessing model performance with prompts.
- Interpreting and analyzing results.

Lesson 9: Bias Mitigation in Prompt Engineering

- Strategies to identify and address biases introduced by prompts.
- Ensuring fairness and inclusivity in prompt-based models.

Module 6: Real-World Applications and Case Studies

Lesson 10: Case Studies in Prompt Engineering

- Exploration of successful implementations and challenges in real-world scenarios.
- Guest lectures from industry experts sharing their experiences.

Text books:

1. "Prompt Engineering in Action" – *Danny D. Sullivan*
2. "The Art of Prompt Engineering with Chat GPT: A Hands-On Guide" – *Nathan Hunter*.

Reference Books:

1. "Prompt Engineering in Practice" – *Michael F. Lewis*
2. "Mastering AI Prompt Engineering: The Ultimate Guide for Chat GPT Users" – *Adriano Damiao*
3. "Writing AI Prompts For Dummies" – *Stephanie Diamond and Jeffrey Allan*
4. "Prompt Engineering Guide" (Online Resource) – *prompting*

Online Resource link :

1. <https://www.udemy.com/course/understanding-prompt-engineering/?couponCode=LETSLEARNNOW>

VIII-SEMESTER

S.No.	COURSE CODE	COURSE TITLE	CC	L	T	P	IM	EM	CR
1	2305851	Internship	INTERN	-	-	-	-	100	4
2	2305852	Project Work	PROJ	-	-	-	60	140	8
Total				-	-	-	60	240	12